



SACS

Wave Response

Version 24.00

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1 INTRODUCTION

1.1 OVERVIEW

The Wave Response program module is designed to compute the dynamic responses of a structure subjected to wave action including forces due to water particle velocities and accelerations.

The program uses the dynamic characteristics calculated by Dynpac and hydrodynamic properties along with wave kinematics calculated by Seastate program module.

1.2 PROGRAM FEATURES

Wave Response requires a SACS model file, Seastate input, and dynamic mode shape and mass files in addition to the Wave Response input file. The program is designed to produce an equivalent static load file that may be solved directly or a modal response file used by the Combine program to calculate the internal loads for subsequent stress analysis.

Some of the main features and capabilities of the Wave Response program module are:

1. Ability to use a full structural model for use in Wave Response analysis.
2. The final steady state analysis can be obtained without a time history integration.
3. The effects of structural compliance can be included such that the damping effects of the fluid are automatically included.
4. The effects of buoyancy can be considered for floating structures.
5. The nonlinearities of the wave forces are represented directly.
6. Ability to plot wave characteristics such as surface profile, hydrodynamic forces, base shear and over turning moment along with structural overall characteristics such as modal coordinates, velocities and accelerations.
7. Plots joint and member results for user selected joints and/or members.
8. Generates base shear and over turning moment transfer function and plots.
9. Response due to Pierson-Moskowitz or Jonswap spectra may be determined in addition to user defined surface history.
10. Ability to create equivalent static loads, including both inertia loads as well as hydrodynamic loads, to be used for static analysis.
11. Obtains generalized modal forces from fully expanded six degree of freedom mode shapes.
12. Various output load case selection criteria including time of maximum or maximum minus minimum base shear or overturning moment and modal dynamic minus the modal static results.
13. Output of modal static and modal dynamic responses facilitates the calculation of Dynamic Amplification Factors.
14. Supports Airy, Stokes, Stream Function, Cnoidal and Solitary wave theories.
15. Supports multiple seeds when generating surface profiles or inputting user defined profiles.
16. Time history integration or Fourier decomposition methods for random wave analysis
17. Perform time-history analysis of combined seismic forces, wave loads, and wind turbine forces.
18. Include nonlinear soil-pile interaction in the time history analysis.

1.3 PROGRAM STRUCTURE

The Wave Response program can be run in two basic modes, deterministic wave mode or the random wave mode. In either procedure, the structural compliance effects can be determined by an iterative procedure and all Seastate override capabilities are supported.

1.3.1 Deterministic Wave Response

In the deterministic procedure, the steady state response of the structure is calculated due to the passage of an infinite wave train composed of a single repeatable wave. An iterative process precludes the necessity of performing a time history integration procedure by calculating the steady state response directly and iterating to correct for any nonlinearities. Any type of wave theory available in the Seastate program, including Airy, Stokes, Stream Function, Cnoidal and Solitary may be used.

1.3.2 Random Wave Analysis

In the random wave procedure, either a time history of the wave surface profile or a wave spectrum is specified. The program reproduces the surface profile or the spectrum by linear addition of AIRY waves. The total wave kinematics are determined by adding the velocities and accelerations from the various components.

1.3.3 Stress Recovery

The Wave Response program allows for stress recovery using either the equivalent static load method, an enhanced modal acceleration method or a modal acceleration method.

Note: Although the program supports three methods, the equivalent static load method has been implemented as a complete replacement of the modal acceleration methods and is the recommended procedure for stress recovery.

1.3.3.1 Equivalent Static Load Method

One of the most useful features of the Wave Response program is the ability to generate equivalent static loads. This method develops equivalent static loading used to represent the forces on the structure due to fluid motion, including the relative motion between the structure and the fluid and inertia.

Equivalent static loads consist of both inertia and hydrodynamic loading. The inertia portion of the loading is determined from modal accelerations, while the hydrodynamic portion of the loading represents the actual fluid forces resulting from both fluid and structural motion.

Because high frequency modes react statically, the only significant dynamic amplification occurs in low frequency overall structural modes. Therefore, all significant structural inertia load may be obtained from the low frequency overall structural modes. Furthermore, because the significant portion of the hydrodynamic loading is also not affected by local structural motion, described in the high frequency modes, the total loading on the structure may be represented by equivalent static loading defined by the inertia loads, determined from the low frequency overall structural modes, and the applied hydrodynamic forces (including the effects of structural motion). These equivalent static loads may then be solved directly to obtain stresses.

Note: The equivalent static load technique eliminates the need to use a large number of modes by adding the inertia loading resulting from only those modes with significant dynamic response to the actual hydrodynamic loading. Hydrodynamic loading is not restricted to the modal DOFs.

1.3.3.2 Enhanced Modal Acceleration Method

Although the equivalent static load method is recommended for stress recovery, the Wave Response program supports an enhanced modal acceleration technique.

The enhanced modal acceleration method determines dynamic stress by subtracting the modal static stress, i.e. stress resulting from generalized forces only, from the total stress resulting from modal responses. The resulting dynamic stresses may then be combined with static stresses determined from a separate static analysis.

1.3.3.3 Modal Acceleration Method

The Wave Response program allows for stress recovery using the modal acceleration technique. This technique requires that sufficient number of modes be used so that all dynamic and static response is included in the solution. In general, the equivalent static load method and the enhanced modal acceleration method are preferred techniques.

2 ANALYSIS PROCEDURE

The Wave Response program is generally used to generate loading for fatigue or extreme wave analyses or to determine dynamic amplification factors. It may be used for deterministic wave response analysis in addition to random wave analysis. Typically, wave response analyses may be classified as one of the following:

- A. Dynamic Fatigue Analysis
- B. Deterministic Extreme Wave Analysis
- C. Random Wave Analysis
- D. Dynamic Amplification Factors

The Wave Response program requires a SACS model file, Seastate input (specified in either a separate file or in the model file) and a Wave Response input file.

2.1 BASIC ANALYSIS OPTIONS

Regardless of the analysis type, basic analysis options must be specified in the Wave Response input file.

2.1.1 Units

The units are entered in columns 8-9 on the WROPT line. Enter 'EN', 'ME' or 'MN' for English, Metric with kilogram force or Metric with kilonewton force, respectively.

2.1.2 Maximum Number of Iterations

The maximum number of iterations allowed is designated in columns 46-50 of the WROPT line. If no iterations are to be performed (single pass analysis), enter '-1'.

Note: For fatigue analysis, API RP-2A recommends that the relative velocity formulation of Morison's equation be omitted. If the relative velocity formulation is to be omitted, no iterations should be performed.

2.1.3 Damping

Damping factors can have a profound effect on analysis results. For single pass analyses, the damping value input by the user should include all sources of damping including fluid damping. For iterative analyses, because fluid damping is automatically accounted for, the damping value should not include any damping due to the fluid.

If all modes have the same damping, the overall damping as a percent of critical is input in columns 11-15 on the DAMP line and columns 16-80 should be left blank. The following shows total critical damping of 3.0% for all modes:

	1	2	3	4	5	6	7	8
1	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
2	WROPT	MEPSL	MAXSES	15	-1			
3	DAMP	3.0						
4	END							

If the damping value is different for various modes, the damping value for each mode must be specified in the appropriate columns. Damping values must be specified for each mode and must be expressed as a

nonzero positive number. The sample below shows various damping values for the 15 modes to be included in the analysis.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	MEPSL	MAXSES		15	-1		
2	DAMP							
3	DAMP	3.0	3.0	3.0	3.0	2.8	2.8	2.8
4	DAMP	2.5	2.5	2.5	2.5	2.5		
5	END							

Also, radiation damping can be accounted for in the analysis (time history only) by entering 'R' in column 5 of the above DAMP cards. The analysis uses formulation by Cook (1978) to calculate the damping which is added to structural damping entered in the DAMP card. Note: This is damping is only computed for the first mode since the formulation does not provide accurate results for higher modes.

2.1.3.1 Mode Selection

By default, the response of all modes is considered in the wave response analysis. If the response of some modes is to be ignored, the number of modes to consider should be stipulated in columns 41-45 on the WROPT line. When the number of modes 'n' is specified, the program assumes that the first 'n' modes are to be considered unless mode numbers are designated using the MODSEL input line. For example, the following designates that modes 1-10 and modes 16-20 are to be considered in the analysis.

Note: The number of modes specified on the MODSEL line must be equal to the number of modes designated on the WROPT line.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	MEPSL	ALL	ES	15	-1		
2	DAMP	3.0						
3	MODSEL	1	2	3	4	5	6	7
		8	9	10	16	17	18	19
		20						

2.2 DYNAMIC FATIGUE ANALYSIS

The Wave Response program can be used to for dynamic fatigue analysis. The program has specialized capabilities that allow the user to readily generate global transfer function plots and to generate the required loading for the fatigue transfer functions.

2.2.1 Creating Global Transfer Function Plots

The Wave Response program can be used to create global base shear and/or overturning moment transfer function plots which may be employed to define frequencies to be used in a detailed analysis. Generating transfer function plots requires that hydrodynamic and wave information be specified in the Seastate input file (or the SACS model file) while analysis and plot options are designated in a Wave Response input file.

2.2.1.1 Hydrodynamic and Wave Data

Hydrodynamic data is specified in the Seastate input or SACS model file. The wave data required may be generated automatically from basic wave parameters or may be input directly by the user.

2.2.1.1.1 Generating Wave Data Automatically

The wave data required for the global transfer function plots may be generated automatically by the Seastate program using the GNTRF input line. This feature creates a series of constant steepness waves in descending period order based on parameters specified by the user.

Typically, 20 to 60 waves of constant steepness* should be used with more waves concentrated near the natural period of the structure. If period step size or steepness is not constant for all waves, multiple GNTRF lines may be utilized. One GNTRF line should be used for each desired series of waves.

Note: For some spectral analyses, it may be required to use one steepness for the larger waves and another steepness for the smaller waves.

The total number of waves to be created in the series is designated in columns 11-13. The beginning period corresponds to the period of the first wave in the series and is specified in columns 21-26. The height of each wave is determined from the steepness stipulated in columns 14-20, while the wave period step size in columns 27-32 is used to determine the period spacing between waves. The wave direction, the number of crest positions and the wave type are specified in columns 46-51, 52-54 and 55-58, respectively.

For example, the following Seastate input generates two series of 12 waves each. The first GNTRF line creates twelve waves beginning with a period of 10.5 seconds through 5.0 seconds spaced 0.5 seconds apart while the second creates waves from 4.75 seconds through 2.0 seconds spaced 0.25 seconds apart.

Note: The DYN option must be specified on the LDOPT line in order to use this feature. Also, the transfer function type input in columns 9-10 of the GNTRF line is ignored for dynamic analysis.

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	* 000 Degree transfer function							
3	LDOPT	+Z	1.03	7.85	-50.00	50.00	ME DYN	NPNP
4	FILE S							
5	CDM							
6	CDM AP							
7	MGROV	0.000	10.00	2.500				
8	MGROV	10.000	45.000	5.000				
9	GRPOV							
10	GRPOVAL	LG2 F						
11	GRPOV	PL2NF		0.001	0.001	0.001		
12	GRPOV	W.BNF 0.010		0.001	0.001	0.001		
13	LOAD							
14	LOADCN	1						
15	GNTRF	12	0.5	10.5	0.5	50.	-50.	18
16	LOADCN	2						
17	GNTRF	12	0.5	4.75	.25	50.	-50.	18
18	END							

Note: Each GNTRF line is preceded by a LOADCN line of which only the first one has a load case name specified.

2.2.1.1.2 Specifying Wave Data Directly

The user may specify the waves used to generate global transfer function plots directly in the Seastate input file or the SACS model file.

Waves used to generate the transfer function plots must be specified in descending period order for a particular direction. Generally, 20 to 60 waves of constant steepness* should be used with more waves concentrated near the natural period of the structure.

Note: Depending on the type of analysis, more than one wave steepness may be used.

The following sample input shows a set of eleven waves with a steepness of 1/15 used to generate a transfer function for the 0 degree wave direction. Metric units are used and the natural period of the structure is 2.0 seconds. The relationship between wave height and wave period was determined from the following:

$$5.12T^2 = Le \quad 1.56T^2 = Lm \quad S = \frac{H}{L}$$

where Le and Lm are the length of an Airy wave in feet and meters, respectively, and S is the wave steepness

Note: The 'AL' wave position option should not be used on the WAVE line when specifying waves to be used by the Wave Response Program.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	* 000 Degree transfer function							
2	LDOPT	+Z	1.03	7.85	-50.00	50.00	ME DYN	NPNP
3	FILE S							
4	CDM							
5	CDM AP							
6	MGROV							
7	MGROV	0.000	10.00	2.500				
8	MGROV	10.000	45.000	5.000				
9	GRPOV							
10	GRPOVAL	LG2 F						
11	GRPOV	PL2NF		0.001	0.001	0.001		
12	GRPOV	W.BNF 0.010		0.001	0.001	0.001		
13	LOAD							
14	LOADCN	1						
15	WAVE							
16	WAVE1.00AIRY	10.3	10.0		D	20.00	18MS10	1 0
17	LOADCN	2						
18	WAVE							
19	WAVE1.00AIRY	6.62	8.00		D	20.00	18MS10	1 0
20	LOADCN	3						
21	WAVE							
22	WAVE1.00AIRY	3.72	6.00		D	20.00	18MS10	1 0
23	LOADCN	4						
24	WAVE							
25	WAVE1.00AIRY	2.59	5.00		D	20.00	18MS10	1 0
26	LOADCN	5						
27	WAVE							
28	WAVE1.00AIRY	1.65	4.00		D	20.00	18MS10	1 0
29	LOADCN	6						
30	WAVE							
31	WAVE1.00AIRY	0.93	3.00		D	20.00	18MS10	1 0
32	LOADCN	7						
33	WAVE							
34	WAVE1.00AIRY	0.65	2.50		D	20.00	18MS10	1 0
35	LOADCN	8						
36	WAVE							
37	WAVE1.00AIRY	0.52	2.25		D	20.00	18MS10	1 0

38	LOADCN	9										
39	WAVE											
40	WAVE1.00AIRY	0.41	2.00		D	20.00	18MS10	1	0			
41	LOADCN	10										
42	WAVE											
43	WAVE1.00AIRY	0.32	1.75		D	20.00	18MS10	1	0			
44	LOADCN	11										
45	WAVE											
46	WAVE1.00AIRY	0.23	1.50		D	20.00	18MS10	1	0			
47												

Note: For this sample, each wave direction is handled as a separate execution. Therefore, only waves of the same direction are input in the Seastate input file. Also, note that wave periods were spaced closer near the natural period of the structure.

2.2.1.2 Wave Response Options

Wave Response options including analysis and plot options are designated in the Wave Response input file.

2.2.1.2.1 Plot Options

Enter 'PSL' in columns 10-12 of the WROPT line to have plots generated. Plot options are designated using the PSEL, PLTTF and TFLCAS input lines.

The type of transfer function, either base shear or overturning moment, is designated using the PLTTF line. Enter 'OM' for a plot of the overturning moment transfer function and/or 'BS' for a plot of the base shear transfer function.

The TFLCAS line is used to specify which waves are to be included in the transfer function plots. The first and last wave of each plot are stipulated. The TFLCAS line may be omitted if all waves are to be plotted.

Enter 'S' in column 69 of the PSEL line to have a tick mark or 'x' placed at each point on the plot corresponding to a frequency of one of the waves used to generate the plot. Otherwise the PSEL line is not required.

The following is a sample Wave Response input file used to generate overturning moment and base shear transfer function plots. As designated on the TFLCAS line, waves 1-11 are to be used to generated the plots.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	MEPSL			15	-1		
2	PSEL						S	
3	PLTTF		OM BS					
4	TFLCAS	1	11					
5	DAMP		3.0					
6	END							

Note: The WSPEC input line may be used to plot response functions. Maximum base shear or Maximum overturning moment should be selected as the load case selection option on the WROPT line.

2.2.2 Generating Loading for Fatigue Analysis

The Wave Response program can be used to generate equivalent static loading utilized in the transfer functions. Generating the equivalent static loading for transfer functions requires that hydrodynamic and wave data be specified in the Seastate input or SACS model file while analysis and load options are designated in a Wave Response input file.

2.2.2.1 Hydrodynamic and Wave Data

Hydrodynamic data must be specified by the user in the Seastate input file. The wave data required to create transfer function loading may be generated automatically from basic wave parameters or may be input directly by the user.

2.2.2.1.1 Selecting Waves and Crest Positions

The user should select a sufficient number of waves to define all of the peaks and valleys in the transfer functions. The global base shear and/or overturning moment plots may be utilized to define the wave frequencies required to generate sufficiently accurate transfer functions. The set of waves chosen may be limited to a maximum height equal to the design wave and a minimum height of 1 foot.

For each wave selected, a sufficient number of wave crest positions should be used to ensure that an accurate representation of the true dynamic forces is obtained and that any fictitious transients are eliminated. Typically, a wave step size between 18-22.5 degrees is adequate.

Note: When using wave spreading, the same wave periods must be used for all of the transfer functions.

2.2.2.1.2 Generating Wave Data Automatically

For each wave direction, 12 to 30 waves of constant steepness* are typically required for an accurate transfer function. If the critical wave periods, i.e. periods corresponding to a peak or valley of the global transfer functions, can be defined as a series of waves with equal period spacing, the GNTRF feature may be used to generate the required wave data. If period step size or steepness is not constant for all waves chosen, multiple GNTRF lines may be utilized to define multiple wave series.

Note: For some spectral analyses, it may be required to use one steepness for the larger waves and another steepness for the smaller waves.

The total number of waves to be created in a series is designated in columns 11-13. The beginning period corresponds to the period of the first wave in the series and is specified in columns 21-26. The height of each wave is determined from the steepness stipulated in columns 14-20, while the wave period step size in columns 27-32 is used to determine the period spacing between waves. The wave direction, the number of crest positions and the wave type are specified in columns 46-51, 52-54 and 55-58, respectively.

The following Seastate input generates three wave series containing a total of 16 waves. The first GNTRF line creates four waves beginning with a period of 12.0 seconds through 6.0 seconds spaced 2.0 seconds apart while the second creates waves with 5.0 and 4.0 second periods, respectively. The final series of waves contains 10 waves beginning with a period of 3.8 seconds through 2.0 seconds spaced 0.20 seconds apart. All waves have a steepness of 1/20 and will be stepped through the structure using 18 positions.

Note: The DYN option must be specified on the LDOPT line in order to use this feature. Also, the transfer function type input in columns 9-10 of the GNTRF line is ignored for dynamic analysis.

	1	2	3	4	5	6	7	8
1	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
2	* 000 Degree transfer function							
3	LDOPT	+Z	1.03	7.85	-50.00	50.00	ME DYN	NPNP
4	FILE S							
5	CDM							
6	CDM AP							

6	MGROV							
7	MGROV	0.000	10.00	2.500				
8	MGROV	10.000	45.000	5.000				
9	GRPOV							
10	GRPOVAL	LG2 F						
11	GRPOV	PLINN			0.001	0.001	0.001	
12	GRPOV	PL2NN			0.001	0.001	0.001	
13	GRPOV	W.BN	0.010		0.001	0.001	0.001	
14	LOAD							
15	LOADCN							
16	GNTRF	4	0.5	12.0	2.0	50.	-50.	18AIRY
17	GNTRF	2	0.5	5.0	1.0	50.	-50.	18AIRY
18	GNTRF	10	0.5	3.8	.20	50.	-50.	18AIRY
19	END							

2.2.2.1.3 Specifying Wave Data Directly

The waves selected to generate the transfer function loading may be input by the user directly. The waves must be specified in descending period order in the Seastate input file with 12 to 20 waves of constant steepness* typically required for an accurate transfer function.

Note: Depending on the type of analysis, more than one wave steepness may be used.

In the following example, eight waves were deemed critical to form an accurate transfer function for the 0 degree wave direction. Each wave is stepped through the structure at 20 degree increments at a total of 18 positions for which a load case will be created.

Note: Although a load case is to be created for every wave position, the 'AL' wave position option should not be used on the WAVE line. The number of positions to be saved as load cases is input in the Wave Response input file. See the ensuing section 'Load Options' for details on selecting wave positions for loading.

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	LOAD							
3	LOADCN	1						
4	WAVE							
5	WAVE1.00AIRY	10.3	10.0		D	20.00	18MS10	1 0
6	LOADCN	2						
7	WAVE							
8	WAVE1.00AIRY	6.62	8.00		D	20.00	18MS10	1 0
9	LOADCN	3						
10	WAVE							
11	WAVE1.00AIRY	3.72	6.00		D	20.00	18MS10	1 0
12	LOADCN	4						
13	WAVE							
14	WAVE1.00AIRY	1.65	4.00		D	20.00	18MS10	1 0
15	LOADCN	5						
16	WAVE							
17	WAVE1.00AIRY	0.93	3.00		D	20.00	18MS10	1 0
18	LOADCN	6						
19	WAVE							
20	WAVE1.00AIRY	0.65	2.50		D	20.00	18MS10	1 0
21	LOADCN	7						
22	WAVE							
23	WAVE1.00AIRY	0.52	2.25		D	20.00	18MS10	1 0
24	LOADCN	8						
25	WAVE							
26	WAVE1.00AIRY	0.32	1.75		D	20.00	18MS10	1 0
27	END							

Note: For this example, each wave direction is handled as a separate execution. Therefore, only waves of the same direction are input in the Seastate input file.

2.2.2.2 Wave Response Options

Wave Response options including analysis, load and plot options are designated in the Wave Response input file.

2.2.2.2.1 Load Options

Load options are designated on the WROPT line. Enter 'ES' or 'US' in columns 19-20 to generate equivalent static loads representing both inertia loading and hydrodynamic loading.

Note: The generated equivalent static loads may be saved to a load file or may be appended to the existing model file. Generating modal response factors is not applicable for fatigue analysis. For Fatigue analysis, the 'US' option generates unsorted equivalent static loads and can save execution time.

The output load case selection criteria is designated in columns 15-18. Enter 'ALL' to have a load case created for each wave position. The following table details the available load case selection options that apply to creating loading for transfer functions:

Option	Description
ALL	Creates a load case for each wave position
MAXM	Creates one load case containing the loading for the wave position yielding the maximum overturning moment
MAXS	Creates one load case containing the loading for the wave position yielding the maximum base shear
M-NM	Creates one load case consisting of the difference between the loading from the position yielding maximum overturning moment and the loading from the position yielding the minimum overturning moment
M-NS	Creates one load case consisting of the difference between the loading from the position yielding maximum base shear and the loading from the position yielding the minimum base shear
MMnM	Creates n load cases corresponding to the wave positions yielding the n highest overturning moments (Note: n must be input as an integer)
MMnS	Creates n load cases corresponding to the wave positions yielding the n highest base shears (Note: n must be input as an integer)
MKnM	Creates various load cases starting with the position yielding the maximum overturning moment then skipping n wave positions
MKnS	Creates various load cases starting with the position yielding the maximum base shear then skipping n wave positions
WAVE	Use the wave crest position specified on the WAVE definition

The following is a sample Wave Response input file used to generate equivalent static loading. A load case was created for each position of each wave. Note that global base shear and overturning moment plots are also desired.

```

      1      2      3      4      5      6      7      8
123456789012345678901234567890123456789012345678901234567890
1  WROPT  MEPSL  ALL  US      15  -1
2  PLTTF
3  TFLCAS  1      8
4  DAMP      3.0
5  END

```

Note: Global transfer function plots may be created in conjunction with the load generation procedure. See the previous section 'Creating Global Transfer Function Plots' for details.

2.3 DETERMINISTIC EXTREME WAVE ANALYSIS

The Wave Response program can be used to generate equivalent static loading or modal response factors for deterministic extreme wave analysis. Generating the equivalent static loading or modal response factors for extreme wave analysis requires that hydrodynamic and wave data be specified in the Seastate input file (or the SACS model file) while analysis and load options are designated in a Wave Response input file.

2.3.1 Hydrodynamic, Wave and Current Data

The hydrodynamic data along with wave and current data used to generate equivalent static loading or modal response factors for dynamic analysis must be input in the Seastate input file directly. Waves may be specified in any order and waves of various directions may be input in the same file.

The following sample shows the input for three extreme waves for which equivalent static loads are to be generated.

```

      1      2      3      4      5      6      7      8
123456789012345678901234567890123456789012345678901234567890
1  * 000 Degree transfer function
2  LDOPT      +Z      1.03      7.85  -50.00      50.00      ME  DYN      NPNP
3  FILE S
4  CDM
5  CDM AP
6  MGROV
7  MGROV      0.000      10.00      2.500
8  MGROV      10.000      45.000      5.000
9  GRPOV
10 GRPOVAL      LG2 F
11 GRPOV      PL2NN      0.001  0.001  0.001
12 GRPOV      W.BN  0.010      0.001  0.001  0.001
13 LOAD
14 LOADCN  1
15 WAVE
16 WAVE1.00AIRY  12.0      13.0      0.00      D      0.00  20.00  18MS10 1 0
17 LOADCN  2
18 WAVE
19 WAVE1.00AIRY  12.0      13.0      45.0      D      0.00  20.00  18MS10 1 0
20 LOADCN  3
21 WAVE
22 WAVE1.00AIRY  12.0      13.0      90.0      D      0.00  20.00  18MS10 1 0
23 END

```

2.3.2 Wave Response Options

Wave Response options including analysis and plot options are designated in the Wave Response input file.

2.3.2.1 Load Options

Load options are designated on the WROPT line. The program can create equivalent static loads representing both inertia loading and hydrodynamic loading or modal response factors. Enter 'ES' in columns 19-20 to generate equivalent static loads or 'MR' for modal response factors.

Note: Equivalent static loads may be saved to an external load file or appended to the model file. Modal response factors are saved in a Combine input file.

The output load case selection criteria is designated in columns 15-18. The following table details the available load case selection options that apply to creating loading for dynamic wave analysis using the equivalent static load method or the modal acceleration method:

Option	Description
ALL	Creates a load case for each wave position
MAXM	Creates one load case containing the loading for the wave position yielding the maximum overturning moment
MAXS	Creates one load case containing the loading for the wave position yielding the maximum base shear
M-NM	Creates one load case consisting of the difference between the loading from the position yielding maximum overturning moment and the loading from the position yielding the minimum overturning moment
M-NS	Creates one load case consisting of the difference between the loading from the position yielding maximum base shear and the loading from the position yielding the minimum base shear
MMnM	Creates n load cases corresponding to the wave positions yielding the n highest overturning moments (Note: n must be input as an integer)
MMnS	Creates n load cases corresponding to the wave positions yielding the n highest base shears (Note: n must be input as an integer)
MKnM	Creates various load cases starting with the position yielding the maximum overturning moment then skipping n wave positions
MKnS	Creates various load cases starting with the position yielding the maximum base shear then skipping n wave positions

When using the enhanced modal acceleration stress recovery method, the static portion must be eliminated from the total response. The Wave Response program has the ability to subtract static loading from the total load thus creating a load case containing only the portion of the load due to dynamics. The following load case options create loading containing only dynamic loads and are applicable only when using the enhanced modal acceleration method:

Option	Description
D-S	Creates a load case with only the dynamic portion of the loading, for each wave position
MDSM	Creates one load case containing only the dynamic portion of the loading for the wave position yielding the maximum overturning moment
MDSS	Creates one load case containing only the dynamic portion of the loading for the wave position yielding the maximum base shear
BDSM	Creates one load case consisting of the difference between dynamic loading from the position yielding maximum overturning moment and the dynamic loading from the position yielding the minimum overturning moment
BDSS	Creates one load case consisting of the difference between dynamic loading from the position yielding maximum base shear and the dynamic loading from the position yielding the minimum base shear
DSnM	Creates n load cases, consisting of dynamic loading only, corresponding to the wave positions yielding the n highest overturning moments (Note: n must be input as an integer)
DSnS	Creates n load cases, consisting of dynamic loading only, corresponding to the wave positions yielding the n highest base shears (Note: n must be input as an integer)
DKnM	Creates various load cases, containing only the dynamic portion of the load, starting with the position yielding the maximum overturning moment then skipping n wave positions

DKnS	Creates various load cases, containing only the dynamic portion of the load, starting with the position yielding the maximum base shear then skipping n wave positions
------	--

The following is a sample Wave Response input file used to generate equivalent static loading for an extreme wave analysis. For each wave, one load case was created corresponding to the crest position yielding the maximum base shear.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	MEPSL	MAXSES		15	-1		
2	PSEL		JO	OM	BS			
3	PSJO	101DX	201DX					
4	DAMP	3.0						
5	END							

2.3.2.2 Plot Options

Various plots may be created in conjunction with the load generation procedure. The program has the ability to create plots containing generalized forces, modal displacements and modal velocities.

Plot features are invoked by specifying 'PSL' in columns 10-12 on the WROPT line while plot contents are designated in columns 7-51 on the PSEL line. The following plot options are applicable to dynamic load generation:

'GF' Generalized Forces	'JO' Joint Displacements (requires PSJO line)
'MC' Modal Coordinates	'MF' Member Forces (requires PSMF line)
'MV' Modal Velocities	'OM' Overturning Moment
'MA' Modal Accelerations	'BS' Base Shear
'HF' Hydrodynamic Forces	

For joint displacement plots, the joints and the degrees of freedom desired are designated using the PSJO line. The member loads desired for the member forces plots are specified on the PSMF lines.

Note: If required, the PSJO line and/or the PSMF lines should follow immediately after the PSEL input line.

The following is a sample Wave Response input file used to generate dynamic wave loading for an extreme wave analysis. Base shear and overturning moment plots are requested on the PSEL line. Joint displacement plots showing the X displacement of joints 101 and 201 are also desired.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	MEPSL	MAXSES		15	-1		
2	PSEL		JO	OM	BS			
3	PSJO	101DX	201DX					
4	DAMP	3.0						
5	END							

2.4 DYNAMIC AMPLIFICATION FACTOR DETERMINATION

The Wave Response program can be used to calculate dynamic amplification factors for deterministic waves specified in the Seastate input file.

2.4.1 Hydrodynamic and Wave Data

The hydrodynamic and wave data for which to calculate dynamic amplification must be specified in the Seastate input file.

In the following example, the dynamic amplification for four waves are to be determined. Each wave is stepped through the structure at 20 degree increments at a total of 18 positions.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	LOAD							
2	LOADCN 1							
3	WAVE							
4	WAVE1.00AIRY	10.3	10.0	0.00	D	0.00	20.00	18MS10 1 0
5	LOADCN 2							
6	WAVE							
7	WAVE1.00AIRY	10.3	10.0	45.0	D	0.00	20.00	18MS10 1 0
8	LOADCN 3							
9	WAVE							
10	WAVE1.00AIRY	10.3	10.0	90.0	D	0.00	20.00	18MS10 1 0
11	LOADCN 4							
12	WAVE							
13	WAVE1.00AIRY	10.3	10.0	180.	D	0.00	20.00	18MS10 1 0
14	END							

2.4.2 Wave Response Options

Wave Response options including analysis are designated in the Wave Response input file. The following is a sample Wave Response input file used to determine dynamic amplification factors.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT ME				15	-1		
2	DAMP	3.0						
3	END							

Note: Only basic analysis options are required. Load options are not required when calculating DAFs and should not be specified.

2.4.3 Interpreting Results

The Wave Response program prints the information required to calculate the dynamic amplification factors in the output listing file. For a particular wave, the DAF may be determined by dividing the maximum dynamic shear in the direction of the wave by the maximum static shear in the same direction. This factor can then be used to factor results or loading obtained from a purely static analysis.

Note: The maximum dynamic and maximum static values are not likely to occur at the same time point due to normal phase shifts that occur in dynamic responses. Also, it should be noted that this approach results in an approximate value since each quantity could have a different DAF.

2.5 RANDOM WAVE ANALYSIS

Random wave analysis may be required when the design wave contains significant energy at frequencies near the natural frequencies of the structure or for structures with low natural frequencies. The Wave Response program may be used for random wave analyses including extreme wave analysis, motion studies, predicting short term statistics, calculating dynamic amplification factors, etc. The program has the following capabilities with regard to random wave analysis:

- A. Generating Loading for Extreme Wave Analysis
- B. Generating Responses
- C. Generating Surface profile plots

For random wave analysis, random linear waves with modified crest kinematics are generated based on wave spectra and parameters input in the Wave Response input file.

Hydrodynamic properties required for the analysis are generated automatically by the Seastate program based on a dummy reference wave specified in the Seastate input or SACS model file.

2.5.1 Hydrodynamic Properties

The hydrodynamic properties are generated as part of the Wave Response execution based on input data in the Seastate input file. In addition to basic input, the Seastate input should contain a dummy reference wave selected to wet the maximum height likely to occur during the random wave. This reference wave is used to generate the hydrodynamic properties of each structural element. It is also used to determine the number of load segments required on each member to accurately describe the loading.

Typically, the largest wave that would occur during the random seastate should be used. The wave direction should be the same as the random wave and the crest position located at the center of the structure. The following is an example of a Seastate input file:

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	* 000 Degree transfer function							
3	LDOPT	+Z	1.03	7.85	-50.00	50.00	ME DYN	NPNP
4	FILE S							
5	CDM							
6	CDM AP							
7	MGROV	0.000	10.00	2.500				
8	MGROV	10.000	45.000	5.000				
9	GRPOV							
10	GRPOV	LG2 F						
11	LOAD							
12	LOADCN	1						
13	WAVE							
14	WAVE1.00AIRY	12.0	13.0		D	2.5	1MS10	1 0
15	END							

2.5.1.1 Aerodynamic Properties

If random wind is included in the random wave analysis, aerodynamic and a reference wind must be specified in the Seastate input or SACS model file.

Reference wind load data is specified in the same load case as the reference wave. The wind direction and mean wind velocity should be input in the reference wind.

The following sample input shows a reference wind with a mean velocity of 100 for the 0 degree direction included with the reference wave.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	* 000 Degree transfer function							
2	LDOPT	+Z	1.03	7.85	-50.00	50.00	ME DYN	NPNP
3	FILE S							
4	CDM							
5	CDM AP							
6	MGROV							
7	MGROV	0.000	10.00	2.500				
8	MGROV	10.000	45.000	5.000				
9	GRPOV							
10	GRPOV	LG2 F						
11	LOAD							
12	LOADCN	1						
13	WAVE							
14	WAVE1.00AIRY	12.0	13.0		D	2.5	1MS10	1 0
15	WIND							
16	WIND D	100.	0.00	AP08				
17	END							

2.5.2 Wave Response Options

Random wave parameters and wave response options including analysis and plot options are designated in the Wave Response input file.

2.5.2.1 Analysis Options

Wave response options are specified on the WVROPT line. The following details the options.

2.5.2.1.1 Random Wave Technique

For random wave analysis, the random wave analysis technique is specified on the WROPT line in columns 68-69.

Enter 'RW' if the wave spectra are to be decomposed into Fourier components. This technique is most suited for structures with a natural frequency that is higher than the wave frequencies.

For structures with a natural frequency lower than the wave frequencies, a time history integration may be required and is designated by 'TH' in columns 68-69.

Since the member distributed load is nonlinear, the time history technique by default calculates wave velocity for each member using a piecewise linear function used to represent the segmentation of the load. For larger structures, a wave velocity grid can be set up to expedite the wave velocity calculation. Enter 'TG' in columns 68-69 if a velocity grid is to be used. By default, velocities are calculated using 10 horizontal grid points and variably spaced vertical grid points determined by the program. The number of grid points to be used may be overridden in columns 7-14 on the THIST line.

The following designates time history integration option using 5 horizontal grid points. The program will determine the appropriate number of vertical grid points.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	WROPT	ME			15	-1		TG
2	DAMP		3.0					
3	WAVTIM		50.	-50.		0.25	1 1	2
4	WSPEC	PM	15.0	12.0				
5	THIST	5					TR	

2.5.2.1.2 Cut-Off Elevation

By default, wave kinematics are calculated from the mudline to the water surface. To reduce execution time, the user may optionally input the elevation below which wave kinematics and fluid damping effects are not to be calculated in columns 70-75. This elevation can be determined based on a horizontal velocity ratio entered in columns 70-75 by specifying 'A' in column 65. The minimum velocity is determined by multiplying the horizontal velocity at the water surface by the ratio entered. The elevation below the surface at which the horizontal velocity is below the minimum velocity is set as the cutoff elevation.

The following sample designates that a random wave analysis is to be performed and that no wave kinematics are to be calculated below elevation 15.0.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	WROPT	ME	ALL		15	-1		RW 15.0
2	DAMP		3.0					
3	WAVTIM		50.	-50.		0.25	1 1	2
4	WSPEC	PM	15.0	12.0				
5	THIST	5					TR	

Note: When using a cut-off elevation, although wave kinematics and fluid damping effects are not included below the cut-off elevation current effects are included below the cut-off elevation.

2.5.2.1.3 Floating Structure Options

For floating structures, the effects of buoyancy change may be considered and the constant term in the Fourier analysis may be omitted by specifying 'BUOY' and 'AO' in columns 61-64 and 66-67, respectively.

2.5.2.2 Plot Options

Various plots may be created in conjunction with the load generation procedure. The program has the ability to create plots containing generalized forces, modal displacements and modal velocities.

Plot features are invoked by specifying 'PSL' in columns 10-12 on the WROPT line while plot contents are designated in columns 7-51 on the PSEL line. The following plot options are applicable to dynamic load generation:

'SP' Surface Profile	'MF' Member Forces (requires PSMF line)
'GF' Generalized Forces	'JO' Joint Displacements (requires PSJO line)
'MC' Modal Coordinates	'MA' Modal Accelerations
'MV' Modal Velocities	'OM' Overturning Moment
'BS' Base Shear	'HF' Hydrodynamic Forces
'FE' Hydrodynamic Forces by Elevation (requires ELVSEL line)	

For joint displacement plots, the joints and the degrees of freedom desired are designated using the PSJO line. The member loads desired for the member forces plots are specified on the PSMF lines. The ELVSEL line is required for the 'FE' option. The elevations for hydrodynamic force summation are entered from the mudline up toward the water surface.

A comparison of the modal static base shear and the total hydrodynamic forces may be printed by selecting the 'CS' option on the PSEL line. This information may be used to determine if a sufficient number of modes have been used.

Note: If required, the PSJO line and/or the PSMF lines should follow immediately after the PSEL input line. The ELVSEL line should follow the WAVTIM line and the WSPEC line if used.

2.5.3 Nonlinear Foundation for Time-history Analysis (Technology Preview)

The nonlinear foundation can be optionally included for the wave response time-history analysis ('TH' or 'TG' in columns 68-69 of the WROPT line) by including the NLFOPT line in the input file. Enter 'PSI' in columns 8-11 to include the nonlinear interaction of soil and piles defined in a Pile-Structure Interaction (PSI) input file. If the NLFOPT line (with the PSI option) is entered, two additional input files are required to run the wave response analysis:

- Pile-Structure Interaction (PSI) input file to provide piles and soil specifications.
- Pilehead Superelement file which is used to extract mode shapes (Dynapc analysis).

These files can be selected in SACS Analysis Generator to create a run file for the wave response analysis.

The following example illustrates the wave response input file with the nonlinear foundation option:

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	ENPSL	ALL	ES	10	-1	MR	TH
2	* Nonlinear foundation defined by the PSI input file							
3	NLFOPT	PSI						
4	DAMP	1.000						
5	WAVTIM	+Z	6.404		100.	0.1	500	WH
6	WSPEC1	JS	6.5617	5.8	1.0	5.	0.764	10.
7	END							

2.5.4 Designating Random Wave Data

Unlike deterministic analysis, the wave data for a random wave analysis is designated in the Wave Response input file.

2.5.4.1 Time and Position Parameters

The time series wave parameters along with the overall random analysis parameters are input on the WAVTIM line.

General wave parameters such as water depth, mudline elevation, wave direction and initial crest position are designated in columns 11-34.

The analysis time increment and the maximum number of Airy wave components are stipulated in columns 42-48 and 49-52, respectively. The Airy wave option is specified in column 59 as follows:

'ST' Standard Airy wave using kinematics calculated at the crest and trough

'VS' Use the kinematics determined at the water line at the crest

'WH' Use kinematics determined at the water line at the surface, decaying these values using normal Airy exponential decay for values below the surface.

'CA' Use Classical Airy wave theory where there is no crest or trough and wave kinematics are calculated at the mean water line

'LS' Use linear stretching above water surface and normal Airy wave exponential decay for values below the surface

A wave kinematics factor can be inputted in columns 68-71 to account for wave spreading and wave profile irregularity. If current is included, enter 'LN' to linearly stretch the current profile to the wave crest position.

The following sample designates 50.0 water depth, mudline elevation -50.0, 0.0 degree wave direction and an analysis time increment of 0.25 seconds. Wave kinematics used in the crest are those calculated at the water surface.

	1	2	3	4	5	6	7	8
8	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
9	WROPT	ME	MAXSES		15	-1	RW	15.0
10	DAMP		3.0					
11	WAVTIM		50.	-50.	0.25	1	1	2
12	WSPEC	PM	15.0	12.0				
13	THIST	5					TR	

2.5.4.2 Wave Kinematics

The Wave Response program has the ability to consider the change in position of the structure due to its own motion. This, however, requires the program to recalculate the wave kinematics for every iteration. Considerable computer time can be saved if the wave kinematics from the first iteration are used for all subsequent iterations. Typically, using kinematics from the first iteration should not cause any significant loss in accuracy since the displacements of the structure are usually very small as compared to the lengths of the significant wave components.

By default, the wave kinematics from the first iteration are used for all subsequent iterations. If wave kinematics are position dependant, i.e. wave kinematics are to be recalculated for each iteration, enter 'PD' in columns 7-8 of the WAVTIM line.

Note: The motion of the structure is always taken in to account in subsequent iterations with regards to the relative water particle velocities and accelerations regardless of the wave kinematics option chosen.

2.5.4.3 Wave Surface Profile

The wave surface profile may be determined from a wave height spectral density function (Wave Spectrum) or may be specified manually by the user.

2.5.4.3.1 Wave Spectrum

When using a wave spectrum, the wave spectral density data is input using the WSPEC line. For Pierson-Moskowitz, Jonswap or Ochi-Hubble, the spectrum type is designated by 'PM', 'JS' or 'OH' in columns 11-12, respectively. The significant wave height and dominant period are input in columns 13-19 and 20-26, respectively. For Ochi-Hubble, the wind driven parameters are entered in columns 34-47. A user-defined spectrum is designated with 'US' in columns 11-12. In this case the significant wave height, dominant time period or wind driven parameters are not input. Instead, the program uses period or frequency values and their corresponding wave spectrum values which are input on immediately following SPEC lines.

By default the time duration is fifty times the dominant period. The time duration may be overridden in columns 48-54. If random amplitude is to be generated in addition to random phase angles, enter 'RA' in columns 55-56. For Jonswap spectra, Gamma and C are taken as 3.3 and 1.525, respectively. The default values for Gamma and C may be overridden in columns 34-40 and 41-47, respectively.

By default, the program assumes that the spectrum is divided such that at least one percent of the area under the spectrum is represented by each wave component. The minimum percent for each wave component may be overridden in columns 27-33. The spectrum may, however, be divided such that different percentages are used in different areas of the spectrum using the PERRNG line.

The surface profile only option may be used to terminate the analysis after generating the surface profile. This feature allows the user to check that the time duration is adequate to represent the spectrum within the desired degree of accuracy and is invoked by 'SP' in columns 57-58.

For a user-defined wave spectrum, the spectrum is defined with SPEC lines immediately following the WSPEC line. Up to 100 wave spectrum points may be input. The wave spectrum is defined by 'P' (period) or 'F' (frequency) in column 5 of the SPEC line. If frequency input is specified, frequencies must be input in ascending order, small to large. If period input is specified, periods must be input in descending order, large to small. A period of zero is never input. Each SPEC line may specify up to four points on the user-defined spectrum. The first point has its period or frequency in columns 9-16 and its wave spectrum value in columns 17-24. The second point has its period or frequency in columns 25-32 and its wave spectrum value in columns 33-40. The third point has its period or frequency in columns 41-48 and its wave spectrum value in columns 49-56. The fourth point has its period or frequency in columns 57-64 and its wave spectrum value in columns 65-72. If the second, third or fourth point period or frequency field is left blank, then that point is ignored.

The following example designates a Pierson-Moskowitz wave spectrum with significant wave height of 15.0 and dominant period of 12.0 seconds.

	1		2		3		4		5		6		7		8	
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	WROPT	ME		MAXSES					15	-1				RW	15.0	
2	DAMP		3.0													
3	WAVTIM		50.	-50.				0.25			1	1	2			
4	WSPEC	PM	15.0	12.0												
5	THIST	5											TR			

2.5.4.3.2 User Defined Profile

The wave surface profile may be defined by the user in an external data file. The number of time points, n, must be specified in columns 1-10 on the first line of the file. The surface profile points for the first time point to the nth time point are specified in groups of six points in columns 1-13, 14-26, 27-39, 40-52, 53-65 and 66-78. Surface profile points correspond to the time T determined from $T = (n-1)DT$ where T is the time and DT is the analysis time interval specified in columns 35-41 on the WAVTIM line. The following sample shows 12 points on the wave surface profile.

	1		2		3		4		5		6		7		8	
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1		12														
2		2.0		3.75		6.0		11.3		14.0		14.25				
3		14.0		12.4		8.0		4.0		2.0		2.15				

2.5.4.4 Multiple Random Seeds

By default, one seed is used when generating the random phase angles of the wave components. A set of sequential equally spaced seeds or a set of user input seeds can be used to generate numerous random surfaces of the random wave.

2.5.4.4.1 Equally Spaced Seeds

Numerous equally spaced sequential seeds can be used by specifying the number of seeds and the first seed value.

When using Pierson Moskowitz, Jonswap or Oci-Hubble spectra, the number of seeds is designated in columns 6-7 and the initial seed value in columns 8-10, respectively, on the WSPEC line defining the spectrum.

The following designates that 20 random surfaces beginning with an initial seed value of 10 are to be created.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WAVTIM	50.	-50.	0.0	0.25	1	1	2
2	WSPEC20	10PM	15.0	12.0				

Note: One wave surface profile is generated for each wave seed.

When inputting user defined profiles, the number of random surfaces to be input is designated in columns 72-74 on the WAVTIM line.

The following designates that 20 user defined surface profiles are to be used.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WAVTIM	50.	-50.		0.25	ST	1	2
							20	

2.5.4.4.2 User Defined Seeds

Up to 99 seeds used to generate surface profiles can be specified by the user on the RNSEED line.

When using Pierson Moskowitz, Jonswap or Oci-Hubble spectra, the number of seeds is designated in columns 6-7 on the WSPEC line. The seeds to be used to generate the surface profiles are input on the RNSEED line.

The following designates that 15 user defined seeds are to be used to generate random wave surface profiles.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WAVTIM	50.	-50.	0.0	0.25	1	1	2
2	WSPEC15	PM	15.0	12.0				
3	RNSEED	10.	11.	12.	13.	14.	20.	21.
							22.	23.
							24.	30.
							31.	32.
							33.	34.

Note: One wave surface profile is generated for each wave seed.

2.5.5 Specifying Random Wind Data

A random wind may be optionally used in the random wave analysis. Aerodynamic and a reference wind must be specified in the Seastate input or SACS model file while the random wind data is entered on the WINDSP line in the Wave Response input file.

Note: Random wind data follows the random wave data and precedes the current data. The wind direction is assumed to coincide with the wave approach direction.

Enter the wind velocity at the reference height of 10 meters (33 feet) used to define the Harris spectrum in columns 11-17. The width of the structure used to calculate the wind coherence function is entered in columns 18-24. This dimension is the width of the structure above the water surface and can be taken as the width normal to the wind approach direction.

Optionally enter the lateral coherence parameter used in the wind coherence function along with the Harris spectrum length and roughness in columns 25-31, 32-39 and 39-45, respectively.

The following applies a random wind with velocity of 100 miles per hour at elevation 33 feet. The width of the structure is taken as the width of the main deck which 55.0 feet. Default values are used for all other input parameters.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	WAVTIM	50.	-50.	0.0	0.25	1	1	2
2	RNSEED	10.	11.	12.	13.	14.	20.	21.
3	WIND	M	100	55.0				

2.5.6 Specifying Current Data

The effects of a steady horizontal current may be included in the random wave analysis. The current at various elevations above the mudline may be specified using CURR input lines. The elevation above the mudline and the current velocity are designated in columns 9-11 and 17-24, respectively. The current direction is entered in columns 25-32 and the blocking factor in columns 41-48 on the first CURR line. The following defines a linearly varying current in the 90.0 degree direction:

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	CURR	0.0	0.5	90.				
2	CURR	50.0	1.0					

2.5.7 Time History Analysis of Combined Seismic, Wave, and Wind Turbine Loads

SACS Wave Response program can perform a combined analysis of wave loads, wind turbine forces, and seismic loads in the time-history simulation (i.e., 'TH' or 'TG' is selected in columns 68-69 on the WROPT line). THLOAD input line can be utilized to import up to two (2) time-history files containing wind turbine forces and seismic forces (or any other force time-history files). Once the THLOAD lines are entered in the Wave Response input file, the files can be selected in SACS Analysis Generator.

SACS Dynamic Response program can *automatically* generate the base-driven earthquake forces by selecting the 'WVR' option on THLOAD in the Dynamic Response Input file. The users are referred to the Dynamic Response user manual for additional information. Input files for other force time-history types can be manually created through SACS Datagen and by selecting the *Force Time History Input File - Wave Response* input type.

The following example illustrates the inputs for two force time-history files. The first input is the Wind Turbine forces given in Flex 5 format, while the second file contains base-driven earthquake load in SACS format.

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	* First Time-History File							
2	THLOAD	MN	FNT05					
3	* Second Time-History File							
4	THLOAD	MN						

Consider the following notes about importing the time-history files:

1. If the Wave Response Analysis Time Increment (entered on the WAVTIM line) does not match the time steps in the imported files, the imported forces will be linearly interpolated.
2. If the time histories of the imported files are shorter than the Wave Response Analysis Time, the imported time history will be padded by the force values at the last time step.
3. If the time histories of the imported files are longer than the Wave Response analysis time, the extra time steps will be ignored.
4. If the imported history starts at the non-zero time, the imported time history will be padded by the force values at the first step between the time zero and the imported history starting time.
5. The force time-history file types entered on THLOAD lines must conform to the file types selected in the SACS Analysis Generator. Otherwise, the Wave Response returns error messages.

2.5.8 Creating the Response File

In addition to plotting response data, response data may be saved to a data file for statistical analysis by the Probability or Global Loading program. The response file contains time histories of the selected parameters. Any plot/data option available on the PSEL line may be saved to the response data file.

To save the data, specify the appropriate selection for the desired data in one of the plot/data selection fields. Enter 'D' in the ensuing option field if the data selected is to be saved to the response file without plotting the data or 'B' if the data is to be saved and plotted. The sample below designates that modal coordinates ('MCB' cols. 13-15), velocities ('MVB' cols. 16-18) and accelerations ('MAB' cols. 19-21) are to be both plotted and saved to the response file, while overturning moment ('OM' cols. 10-11) and base shear ('BS' cols. 7-8) are to be plotted only.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
5	WROPT	MEPSL	MAXSES		15	-1		
6	PSEL		MC MV MA				OM BS	

2.5.8.1 Response File for Global Loading Analysis

The Global Loading module can be used to evaluate loading and response data at various locations in the structure. The response file used by the Global Loading modules requires modal coordinates in order to determine critical load dat. Modal velocities and accelerations are required in the response file if load cases are to be created using the retrieve load method.

When using the 'GLOB' load case selection option however, modal coordinates, velocities and accelerations are automatically stored in the response file for the Global Loading module regardless of whether designated on the PSEL line.

2.5.9 Load Case Options

Load case generation options are designated on the WROPT line. The program can create equivalent static loads representing both inertia loading and hydrodynamic loading or modal response factors.

Note: Equivalent static loads provide and enhanced stress recovery as compared to modal response factors.

Enter 'ES' in columns 19-20 to generate equivalent static loads, 'EI' for equivalent static inertia loads or 'MR' for modal response factors.

Load cases can be optionally created using one of the load case creation procedures. The first procedure creates load cases as part of the random wave analysis while the other creates load cases by retrieving the loading data from a response file generated by a previous random wave analysis.

2.5.9.1 Generating Loading in the Random Wave Analysis

For random analysis, the output load cases may be created as part of the analysis at time points selected automatically by the program, user specified times, or user specified time increments.

Note: The load case creation procedures are mutually exclusive. Only one load case creation procedure may be used.

2.5.9.1.1 Automatic Time Point Selection

The program can create load cases based on output load case selection criteria designated in columns 15-18 on the WROPT line. The following table details the available load case selection options that apply to creating loading for random wave analysis using the equivalent static load method or the modal acceleration method:

Option	Description
ALL	Export loads for all time steps
MAXM	Creates a load case containing the loading for the time point yielding the maximum overturning moment
MAXS	Creates a load case containing the loading for the time point yielding the maximum base shear
GLOB	Creates a load case at time points determined from global loading analysis
MST	Creates 6 load cases in order for Max and Min overturning moment, base shear, and torsion time steps.
WAVE	Creates one
M-NM	Creates 2 load cases for largest overturning moments shear with positive and negative sign.
M-NS	Creates 2 load cases for largest base shear with positive and negative sign.

MMxM	Creates 2x (replace x with a number between 1-9) load cases for largest overturning moments with positive sign and negative sign.
MMxS	Creates 2x (replace x with a number between 1-9) load cases for largest base shear with positive sign and negative sign.
MKxM	Outputs all the time steps starting from 1 and skipping x (replace x with a number between 1-9) time steps along with the time step containing the maximum overturning moment.
MKxS	Outputs all the time steps starting from 1 and skipping x (replace x with a number between 1-9) time steps along with the time step containing the maximum base shear.

The following designates that a load case corresponding to the time point yielding the maximum base shear is to be created.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	MEPSL	MAXSES		15	-1		
2	PSEL	MC	MV	MA	OM	BS		

The 'GLOB' option generates a response file then uses the global loading module to select time points based on the global loading input specified by the user. A Global Loading input file is required when using this option.

The following designates that load cases are to be generated based on the Global Loading options specified by the user in the Global Loading input file.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	MEPSL	GLOBES		15	-1		
2	PSEL	MC	MV	MA	OM	BS		

Note: The PSEL line is optional when using the 'GLOB' option since, a response file containing modal coordinates, velocities and accelerations is automatically created when using this option.

2.5.9.1.2 User Specified Time Points

Load cases can be created at specific time points by inputting the times on the TMSEL input line. Up to twelve time points may be specified and time must be specified in ascending order. Times entered may represent ranges if the range option is invoked by an 'R' in column 79. The following sample specifies that load cases are to be created at 0.0, 5.0, 10.0 and 15.0 seconds.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WAVTIM	50.	-50.	0.0	0.25	1		
2	TMSEL	0.0	5.0	10.0	15.0			
3	DAMP	3.0						

2.5.9.1.3 User Specified Time Intervals

Load cases may be created for specified time increments using the PTSEL line. The first time increment is input in columns 6-10. This value must be an integer corresponding to the analysis time interval number. For example, time 0.0 seconds would correspond to increment number 1. The remaining time points must be entered in ascending order up to a total of twelve time points. The time corresponding to the time increment is determined from $T = (n-1) DT$ where; T is time, n is the increment number and DT is the analysis time increment. The following PTSEL line designates that load cases are to be created at analysis time increments 1, 21, 31 and 41 which correspond to times of 0.0, 5.0, 10.0 and 15.0 seconds.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	WAVTIM	50.	-50.	0.0	0.25	1		
2	PTSEL	1	21	31	41			
3	DAMP	3.0						
4	END							

2.5.9.2 Retrieving Loading from a Response File

Loading can be generated from a response file created by a previous wave response analysis or global loading study. Enter 'RL' in columns 13-14 on the WROPT line to retrieve loading data from a response file.

When retrieving load case data from a response file, load cases are created at time increments specified in a time point selection file by default. Load cases can likewise be created based on user designated times or user specified time increments using the TMSEL, PTSEL and PTSEED lines.

Note: A time point selection file is required if no TMSEL, PTSEL or PTSEED lines are specified. Also, because automatic time point selection options are not applicable when retrieving load data, the 'MAXS', 'MAXM' and 'GLOB' options should not be used to select time points.

2.5.9.2.1 User Specified Time Points

Load cases can be retrieved at specific time points by inputting the times on the TMSEL input line. Up to twelve time points may be specified and time must be specified in ascending order. Times entered may represent ranges if the range option is invoked by an 'R' in column 79.

The following sample specifies that load cases are to be created at 0.0, 5.0, 10.0 and 15.0 seconds. The loading data however will be retrieved from a response file rather than generated by the Wave Response module.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	EN	RL	ES			RW	
2	WAVTIM	50.	-50.	0.0	0.25	1		
3	TMSEL	0.0	5.0	10.0	15.0			
4	DAMP	3.0						
5	END							

2.5.9.2.2 User Specified Time Intervals

Load cases may be retrieved for specified time increments when using a single surface profile using the PTSEL line or using the PTSEED line when multiple seeds are used.

The time increment value must be an integer corresponding to the analysis time interval number. For example, time 0.0 seconds would correspond to increment number 1. The remaining time points must be entered in ascending order up to a total of twelve time points. The time corresponding to the time increment is determined from $T = (n-1) DT$ where; T is time, n is the increment number and DT is the analysis time increment.

The following PTSEL line designates that load cases are to be retrieved at analysis time increments 1, 21, 31 and 41 which correspond to times of 0.0, 5.0, 7.5 and 10.0 seconds.

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	WROPT	EN	RL	ES				RW
3	WAVTIM		50.	-50.	0.0	0.25	1	
4	PTSEL	1	21	31	41			
5	DAMP		3.0					
6	END							

When using multiple seeds, the time increments at which to create load cases are designated using the PTSEED line by entering the time increment number along with the seed number.

The following stipulates that load cases are to be created at time increment 21 for seed 2 and increment 12 for seed 3.

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	WROPT	EN	RL	ES				RW
3	DAMP		3.0					
4	WAVTIM		50.	-50.	0.0	0.25	1	
5	PTSEED	21.	2	12.	3			
6	END							

2.6 RESPONSE FUNCTION STATISTICS

The Wave Response program can be used to calculate and plot statistical parameters of a response function such as RMS base shear and overturning moments. These wave response functions may be employed to determine appropriate transfer functions, wave steepness, etc.

Generating response function data requires that hydrodynamic and wave information be specified in the Seastate input file (or the SACS model file) while analysis and plot options are designated in a Wave Response input file.

2.6.1 Hydrodynamic and Wave Data

Hydrodynamic data is specified in the Seastate input or SACS model file. The wave data required may be generated automatically from basic wave parameters or may be input directly by the user in the same fashion used to generate transfer function plots and stresses used for a dynamic fatigue analysis.

Unlike transfer function generation for fatigue analysis, however, transfer functions for various wave steepness may be required.

For example, the following creates transfer functions for the zero degree direction for 1/20 and 1/25 wave steepness using 25 wave periods.

Note: The DYN option must be specified on the LDOPT line in order to use this feature. Also, the transfer function type input in columns 9-10 of the GNTRF line is ignored for dynamic analysis.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	* 000 Degree transfer function							
2	LDOPT	+Z	1.03	7.85	-50.00	50.00	ME DYN	NPNP
3	FILE S							
4	CDM							
5	CDM AP							
6	MGROV							
7	MGROV	0.000	10.00	2.500				
8	MGROV	10.000	45.000	5.000				
9	GRPOV							
10	GRPOV	LG2 F						
11	LOAD							
12	LOADCN							
13	* PERIODS FOR 000 DEGREE TRANSFER FUNCTION							
14	* 1/20 STEEPNESS							
15	GNTRF	AL 6	.05	10.00	1.00	0.0	18AIRYPF	
16	GNTRF	AL 6	.05	4.75	1.00	0.0	18AIRYPF	
17	GNTRF	AL 11	.05	3.40	1.00	0.0	18AIRYPF	
18	GNTRF	AL 2	.05	2.25	1.00	0.0	18AIRYPF	
19	* 1/25 STEEPNESS							
20	GNTRF	AL 6	.04	10.00	1.00	0.0	18AIRYPF	
21	GNTRF	AL 6	.04	4.75	1.00	0.0	18AIRYPF	
22	GNTRF	AL 11	.04	3.40	1.00	0.0	18AIRYPF	
	18AIRYPF							
23	GNTRF	AL 2	.04	2.25	1.00	0.0	18AIRYPF	
24	END							

2.6.2 Wave Response Options

Wave Response options including analysis and plot options are designated in the Wave Response input file.

2.6.2.1 Plot Options

Enter 'PSL' in columns 10-12 of the WROPT line to have response function plots generated. Plot options are designated using the PLTTF line.

The type of transfer function or response function, either base shear or overturning moment, is designated using the PLTTF line. Enter 'OM' for a plot of the overturning moment transfer function and/or 'BS' for a plot of the base shear transfer function. Enter 'OMD' or 'BSD' for overturning moment and base shear response functions, respectively.

The following is a sample Wave Response input file used to generate overturning moment and base shear response function plots.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	MEPSL			15	-1		
2	PLTTF		OMD	BSD				
3	END							

2.6.2.2 Wave Spectra Data

In order to generate the response function, the wave spectra data must be input in the Wave Response input file using the WSPEC line. One wave spectrum per response function is required.

The following specifies that two response functions for a most probable maximum wave defined by a period of 8.0 and a height of 5.0 are to be generated. The first response function uses the transfer function defined by waves 1-25 and the second uses the transfer function defined by waves 26-50.

1	2	3	4	5	6	7	8
---	---	---	---	---	---	---	---

	12345678901234567890123456789012345678901234567890123456789012345678901234567890
1	WROPT ENPSL ALL 10
2	PLTTF OMDBSD PFS
3	DAMP 2.0
4	WSPEC PM 5.00 8.00 1.0 60. 1 25
5	WSPEC PM 5.00 8.00 1.0 60. 26 50
6	END

3 COMMENTARY

3.1 FLUID FORCE ON A MEMBER

The fluid force on a member is computed using Morison's equation as follows:

$$F = F_D + F_I$$

where F_D is the drag force per unit length and F_I is the inertia force vector per unit length. F_D and F_I are taken as:

$$F_D = C_d \frac{w}{2g} D (U - \dot{z}) |U - \dot{z}|$$
$$F_I = C_m \frac{w}{g} V \frac{\partial U}{\partial t}$$

where C_d and C_m are the drag and inertia coefficients, respectively, w is the weight of water, g is acceleration of gravity, D is the effective diameter and V is displaced volume. U and $\dot{z}$ are taken as the fluid and the structural velocity vector components normal to the member, respectively, while $\partial U / \partial t$ is the local acceleration component of water normal to the member.

Note: For a single pass analysis, $\dot{z}$ is taken as zero.

3.2 DETERMINISTIC THEORETICAL APPROACH

The deterministic approach assumes that an infinite train of a repeatable wave form has passed the structure and that a steady state response has been established. The program analyzes the hydrodynamic forces including the influence of the structure motion on the relative water particle velocities to obtain the applied hydrodynamic forces. Since the hydrodynamic forces are inherently nonlinear, an iterative approach has been adopted.

For the first iteration, the structure is assumed to be at rest ($\dot{z} = 0$). The wave is stepped through the structure such that one full cycle of the wave is completed.

For each wave crest position, the distributed member forces are computed using Morison's equation. Using static equilibrium, the equivalent joint forces are determined. The equivalent joint forces are multiplied by the modal eigenvectors to obtain the modal generalized forces for each wave crest position.

Considering each mode individually and assuming that the generalized force is a repeatable function, the generalized force may be decomposed into various Fourier components, each of which is a sinusoidal function with an associated phase angle. The modal response may then be calculated for each Fourier component of the generalized force. The total response of the mode is then determined by linearly combining the responses due to each component.

The motion of any point on the structure may be determined by summing the responses from all modes. Knowing this motion, the relative fluid velocity, $U - \dot{z}$, can be determined by subtracting the velocity of the structure from the water particle velocity. A new set of member forces are then calculated using Morison's equation.

The process is continued until the difference between the generalized forces for the current iteration and the generalized forces of the previous iteration are within the specified tolerance or the maximum number of iterations has been reached.

During the first iteration, the modal responses of the structure are also calculated assuming no dynamic amplification. These "modal static" values can then be compared to the iterative solution ("modal dynamic") values to determine dynamic amplification factors.

3.3 RANDOM THEORETICAL APPROACH

The random wave iterative approach is the same as that used for the deterministic approach. In the random approach, however, the repeatable wave is a wave train in itself having the statistical properties of the input wave spectrum or the surface profile input by the user. Linear combinations of linear Airy waves are used to simulate the input wave spectrum or wave surface profile. Additionally, the wave kinematics are calculated by the Wave Response program rather than computed by the Seastate program.

3.3.1 Surface Profile

When using a user specified surface profile, a Fourier analysis of the surface profile is performed to determine the Airy wave components that comprise the arbitrary surface profile to be analyzed. After discarding any insignificant components, the arbitrary wave is passed through the structure.

Starting with the Airy wave component having the highest period, the wave kinematics are calculated for each member. The distributed member forces are computed using Morison's equation for each crest position of the wave component. Using static equilibrium, the equivalent joint forces are determined. The equivalent joint forces are multiplied by the modal eigenvectors to obtain the modal generalized forces for each crest position of the component. The surface of the wave component is used as the mean water line of the next Airy wave component.

Considering each mode individually and assuming that the generalized force is a repeatable function, the generalized force may be decomposed into various Fourier components, each of which is a sinusoidal function with an associated phase angle. The modal response may then be calculated for each Fourier component of the generalized force. The total response of the mode is then determined by linearly combining the responses due to each component.

3.3.2 Wave Spectrum

When a wave spectrum is specified, the spectrum itself is analyzed and broken into Airy wave components. Only wave components with periods that are divisible into the analysis time duration are considered as possible components.

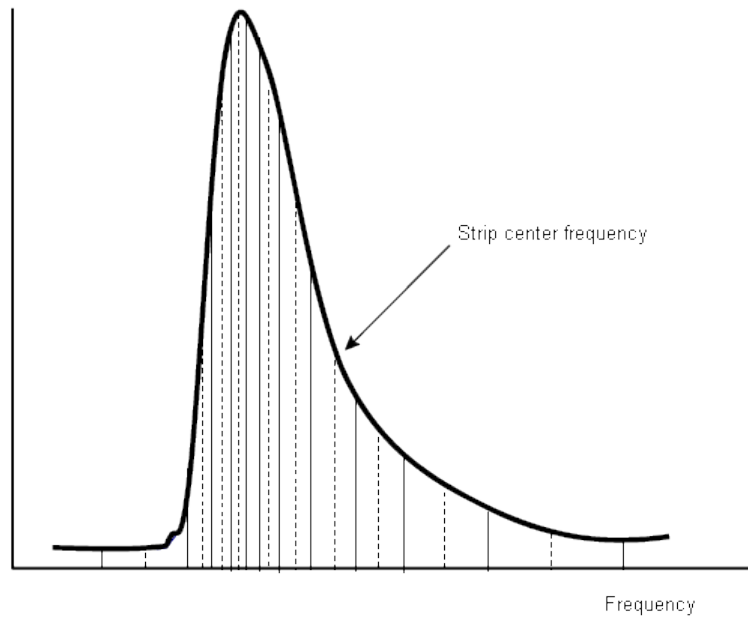
The period of a possible wave component n , is determined from the following equation:

$$T_n = \frac{T_D}{n} \quad n = 1, 2, 3 \dots n_{\max}$$

where T_n is the period, T_D is the analysis time duration and $n_{\max}$ is the last significant component. For example, for an analysis time duration of 1000 seconds, the period of the first possible component is 1000 seconds, the period of the second possible component is 500 seconds, the third is 333 seconds, etc. This

procedure ensures that the analysis will have a repeatable wave train thus eliminating any complications in the subsequent Fourier analyses.

The input spectrum is divided into strips with each strip having a center frequency corresponding to the frequency of one of the possible wave components. The strips are then lumped together so that each lumped strip contains at least the minimum portion of the spectrum allowed (see figure below).



Once the spectrum is divided into components, the analysis procedure is the same as for the user input surface profile.

3.4 Constrained Stream Wave Insertion For Time History Analysis

Waveresponse program can insert a constrained stream wave using the 'WVINS' line at a given time T_{ins} for time history analysis. The users need to provide the crest elevation and, optionally, the wave height for the constrained wave. To create a realistic sea profile, Waveresponse takes the following steps:

1. Create a random background wave from the wave spectrum provided in the 'WSPEC' line.
2. Create a **symmetric** wave surface that attains the max crest elevation at the insertion time T_{ins} and which follows the random wave spectrum statistics.
3. Mix the **symmetric** sea profile with the background profile to create the **modified** profile.
4. Create the **stream wave** profile for the given peak period T . The wave height is either chosen as:
 - The distance between the depth of the closet trough in the modified profile to the provided stream wave crest elevation. If the option for breaking wave height reduction ('A' on column 71 on the WVINS line) is selected, the program ensures the wave height is at most the breaking wave height.
 - Using the wave height provided by the user.
5. Replace the modified profile with the stream profile between $(T_{ins} - T_{i-}, T_{ins} + T_{i+})$ and blends the **stream wave** with the **modified** profile between $(T_{ins} - T_{i-} - T/4, T_{ins} - T_{i-})$ and $(T_{ins} +$

$T_{i+}, T_{ins} + T_{i+} + T/4$). By default $T_{i+} = T_{i-} = T/4$; however, to ensure a realistic sea surface profile, the insertion window can be adjusted automatically by entering 'E' on column 64 of the WVINS line. If this option is selected, the program tries to extend the blending window to create a single constrained wave crest with a well-defined crest.

All the above steps can be viewed in the neutral chart file by entering 'SP' on columns 7-8 of PSEL line.

The stream wave is inserted in the modified profile and blended with it using the following:

$$\eta = f(t) \eta_{stream} + (1 - f(t)) \eta_{modified}$$

Where the wave blending function is defined as:

$$f(t) = \begin{cases} 0 & t < T_{ins} - T_{i-} - T/4 \\ \sin^2 \left(\frac{2\pi(t - T_{ins} + T_{i-})}{T} + \frac{\pi}{2} \right) & T_{ins} - T_{i-} - T/4 \leq t \leq T_{ins} - T_{i-} \\ 1 & T_{ins} - T_{i-} \leq t \leq T_{ins} + T_{i+} \\ \sin^2 \left(\frac{2\pi(T_{ins} + T_{i+} - t)}{T} + \frac{\pi}{2} \right) & T_{ins} + T_{i+} \leq t \leq T_{ins} + T_{i+} + T/4 \\ 0 & t > T_{ins} + T_{i+} + T/4 \end{cases}$$

Figure 1 depicts the blending function.

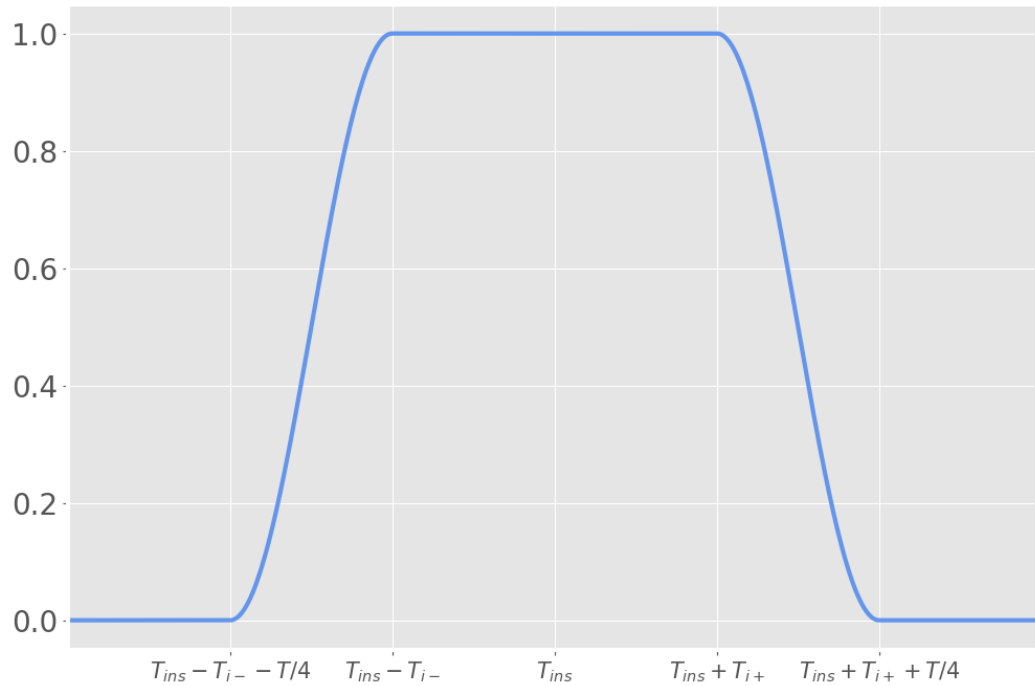


Figure 1- Wave blending function

3.5 SEASTATE DATA

3.5.1 Deterministic Approach

The Wave Response program uses the hydrodynamic capabilities of the Seastate program to generate hydrodynamic properties.

The Seastate program is also used to step the wave through the structure to obtain the complete wave kinematics for an entire wave cycle.

When stepping the wave through the structure, an appropriate number of wave steps should be used to ensure that the higher harmonics contained in the wave forces generated are adequately represented. For example, when a wave is stepped through the structure one complete wave cycle using 18 steps (20 degrees), the fundamental period would be represented by 18 points, the first harmonic by 9 points, the second harmonic by 6, and the third by only 4-1/2 points.

The number of wave steps required is dependant on the primary modes of the structure being analyzed. When the primary modes of the structure are close to the wave period, 16-18 steps should adequately define the forcing function on those modes. However, if the primary modes of the structure have periods closer to the higher harmonics contained in the wave, more wave steps may be necessary.

Note: In general, 16-18 wave steps would be considered the minimum number of steps required in order to obtain reasonably accurate analysis results.

3.5.2 3.4.2 Random Wave Approach

For random wave analysis, the wave kinematics are generated by the Wave response program. Seastate is used to calculate the hydrodynamic properties of each member and to determine whether or not the member is loaded by the wave. For each member loaded by the wave, Seastate also determines the number of load segments required to accurately represent the wave force.

The reference wave to be used by Seastate should wet all members that will be wet at any time during the random wave analysis. In general, a large wave with the crest positioned at the structure center should be used.

Note: Although using a small wave and increasing the water depth may also be used to determine the wetted members, using a larger wave yields more accurate analysis results. When using a larger wave, members are divided into shorter load segments in areas where the highest velocity gradients occur which is typically in the splash zone. For a very small wave with an increased water depth, members in the splash zone will not be adequately subdivided.

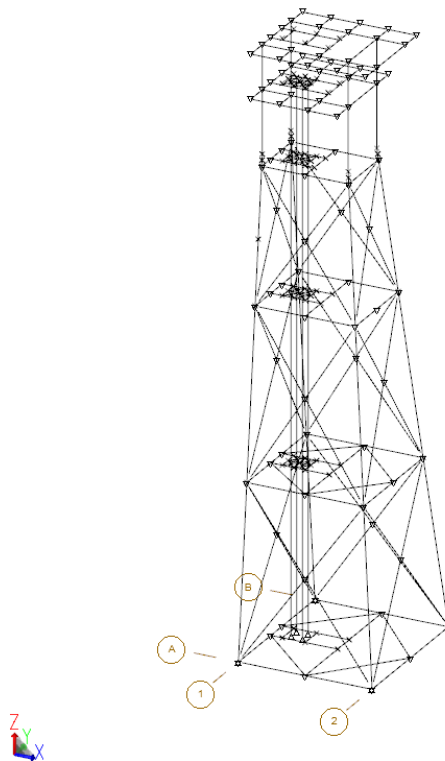
Since wave kinematics are determined by the Wave Response program, fewer wave steps are required for random wave analysis than for deterministic analysis. However, because Seastate is used to determine member load segmentation, a sufficient number of steps must be used to ensure that adequate member load segmentation occurs. Typically, 10 crest positions (36 degrees) is appropriate.

Note: The program selects the wave step yielding the highest degree of member segmentation and the greatest submerged length for each member.

4 SAMPLE PROBLEMS

The structure used for the sample problems is a four pile jacket in 50 meters of water as shown below. A foundation super element generated by PSI is used to represent the nonlinear foundation and fifteen mode shapes were extracted to represent the major deflections of the structure. Four sample problems are used to illustrate the various capabilities of the Wave Response program:

1. The first sample problem generates global base shear and overturning moment transfer function plots typically used to determine critical waves for spectral fatigue analysis transfer functions.
2. Sample Problem 2 illustrates the ability to generate equivalent static loading used to generate the transfer functions for a spectral fatigue analysis.
3. Sample Problem 3 is a deterministic extreme wave analysis. Equivalent static loads are generated.
4. Sample Problem 4 demonstrates the extreme random wave analysis capabilities using a wave spectrum to define the statistical properties of the environment. Equivalent static loads are generated along with a response file used by the Probability program.
5. Sample Problem 5 generates surface profile plots and profile statistics of a random wave using numerous seeds. Time history integration option is used.
6. Sample Problem 6 uses time history integration to generate a response file for the Global Loading program to select critical time points for numerous random wave seeds. It also demonstrates generating load cases at these time points by retrieving loads from the response file.
7. Sample Problem 7 generates response function data for a wave spectrum representing the most probable maximum wave using transfer functions representing various wave steepness.



4.1 SAMPLE PROBLEM 1 – TRANSFER FUNCTION PLOTS

This sample illustrates the ability to plot global base shear and overturning moment transfer function plots. Typically, global transfer function plots are generated in order to check that the specified waves are sufficient to accurately define the transfer function used in a spectral fatigue analysis.

In this sample, the wave data is generated automatically using the GNTRF feature of the Seastate program. Also, Seastate data was specified in a separate input file from the model.

Prior to executing this sample problem, PSI was used to generate a foundation superelement file. Because this is a fatigue analysis, the superelement was created for low deflection levels using small waves similar to those used in the fatigue analysis. The Dynpac program was also executed in order to extract fifteen modes shapes and the generalized mass.

The following is the Seastate input file used in this sample. A detailed description of each of the applicable input lines ensues.

	1	2	3	4	5	6	7	8
1	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
2	* 000 Degree transfer function							
3	* Kilogram Units Steel Test Model							
4	LDOPT	+Z	1.03	7.85	-50.00	50.00	ME DYN	NPNP
5	FILE S							
6	CDM							
7	CDM AP							
8	MGROV							
9	MGROV	0.000	10.00	2.500				
10	MGROV	10.000	45.000	5.000				
11	GRPOV							
12	GRPOV	BL1 F 0.001				5.0 5.0 3.0 3.0		F
13	GRPOV	BL1 F 0.001				3.0 3.0 2.0 2.0		F
14	GRPOV	LG2 F						
15	*****More Group overrides							
16	GRPOV	PL3NN	0.001	0.001	0.001			
17	GRPOV	PL4NN	0.001	0.001	0.001			
18	GRPOV	W.BN	0.010	0.001	0.001	0.001		
19	LOAD							
20	LOADCN	1						
21	GNTRF	12	.05	10.50	0.50		0.0	18
22	LOADCN							
23	GNTRF	12	.05	4.75	0.25		0.0	18
24	END							

The following describes the input lines in the Seastate input file used for Sample Problem 1:

3. The LDOPT line specifies the Seastate options. The 'DYN' option in columns 56-58 designates that Seastate is to be used for dynamic modeling and wave data generation only.

4. The FILE line designates that the model data and Seastate data are specified in separate files. The 'S' option specifies that only loading data in the Seastate input file is to be considered.

19. The first LOADCN line signals the beginning of load case 1.

20. The GNTRF line is used to specify the data used to generate the first set of waves utilized in the transfer function plots as follows:

- a. 12 waves are to be generated as specified by '12' in columns 12-13.

- b. The wave steepness (height/length ratio) is to be 1/20 as designated by '0.05' in columns 14-20.
- c. The period of the first wave of the wave set has a period of 10.5 seconds with each subsequent wave of the set having a period 0.5 seconds lower than the previous wave (i.e. 10.5, 10.0, 9.5, 9.0, etc.).
- d. The wave approach direction is 0.0 degrees as stipulated in columns 46-51.
- e. Each wave is to be stepped through the structure using 18 wave crest positions (20 degree increments).

21. Each GNTRF line must be precede by a LOADCN line.

Note: No load case name is specified on subsequent LOADCN lines.

22. The second GNTRF line is used to specify the data used to generate the second set of waves utilized in the transfer function plots as follows:

- a. 12 waves are to be generated as specified by '12' in columns 12-13.
- b. The wave steepness (height/length ratio) is to be 1/20 as designated by '0.05' in columns 14-20.
- c. The period of the first wave of the wave set has a period of 4.75 seconds with each subsequent wave of the set having a period 0.25 seconds lower than the previous wave (i.e. 4.75, 4.5, 4.25, 4.0, 3.75, 3.5, etc.).
- d. The wave approach direction is 0.0 degrees as stipulated in columns 46-51.
- e. Each wave is to be stepped through the structure using 18 wave crest positions (20 degree increments).

The following is the Wave Response input file used to generate the transfer function plots. A detailed description of each of the applicable input lines ensues.

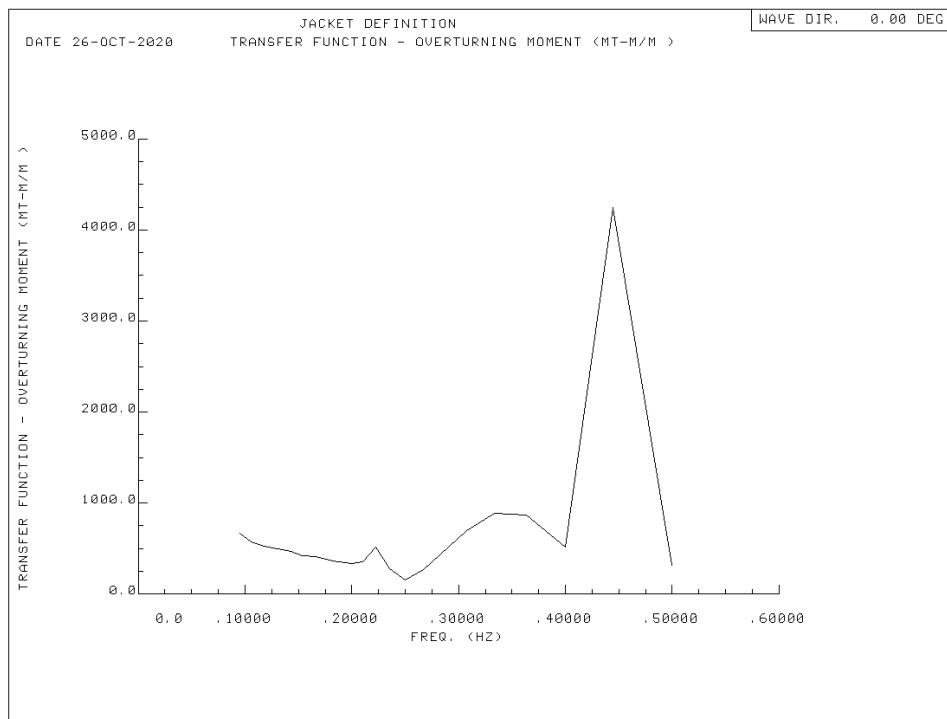
	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	MEPSL	ALL		15	-1		
2	PLTTF			OM	BS			
3	TFLCAS	1	14					
4	DAMP		3.0					
5	END							

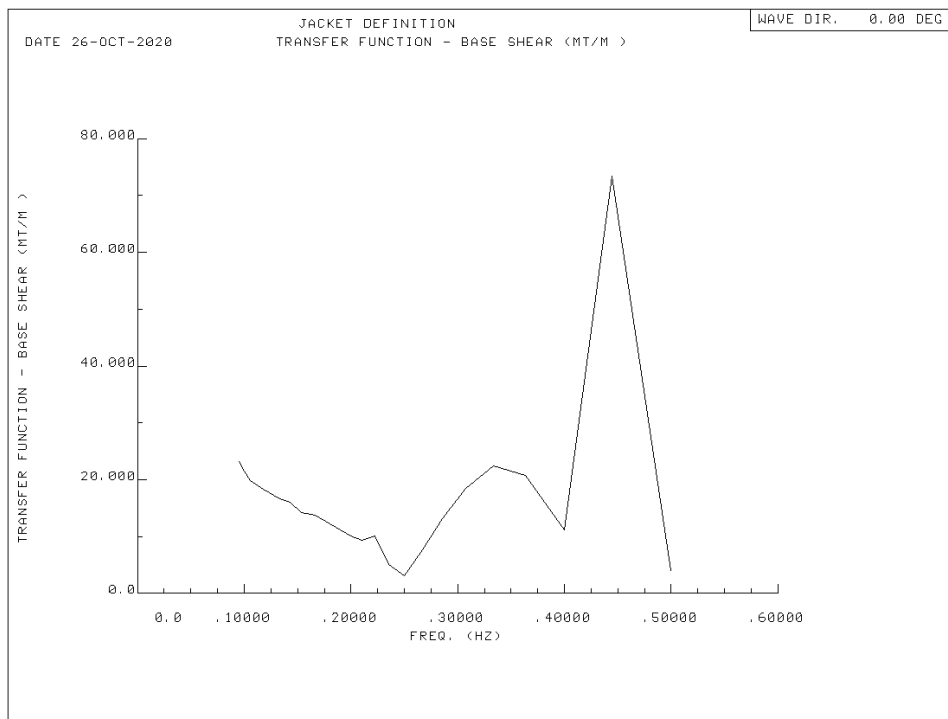
The following describes the input lines in the Wave Response input file used for Sample Problem 1:

1. The WROPT line specifies the Wave Response options as follows:
 - a. Metric kilonewton force units are to be used.
 - b. Plots are to be generated as stipulated by 'PSL' in columns 10-12.
 - c. Fifteen modes are to be considered as designated by '15' in columns 44-45.

-
- d. A single pass analysis is to be performed (i.e. no iterations) as designated by '-1' in columns 49-50.
2. Overturning moment and base shear global transfer function plots are requested on the PLTTF line.
3. Waves 1 through 24 are to be used to plot the transfer functions as designated by '1' and '24' in columns 10 and 13-14 of the TFLCAS line, respectively.
4. Three percent overall critical damping is to be used.

The following are the global overturning moment and base shear transfer function plots created for this sample problem.





4.2 SAMPLE PROBLEM 2 – CREATING EQUIVALENTE STATIC LOADING

Sample Problem 2 illustrates the ability to create the equivalent static loading required in order to generate the transfer functions used in a spectral fatigue analysis. Typically, for each wave direction to be considered, equivalent static loading is generated then solved to create a common solution file. The solution file(s) containing nominal stresses are used by the Fatigue program to create the transfer functions required for spectral fatigue analysis.

This sample details the procedure required to generate the equivalent static loads for one approach direction. The wave data was specified manually by the user in a separate Seastate input file. The waves were selected based on the critical points (i.e. peaks and valleys) on the global transfer function plots generated in Sample Problem 1. The following fourteen waves were selected:

Height(m)	Period	Height(m)	Period	Height(m)	Period
8.55	10.50	1.57	4.50	0.59	2.75
7.76	10.00	1.24	4.00	0.49	2.50
4.36	7.50	0.95	3.50	0.39	2.25
2.79	6.00	0.82	3.25	0.31	2.00
1.75	4.75	0.70	3.00		

Note: The GNTRF feature may be used to generate the wave data automatically if the waves to be used can be expressed as one or more sets of waves with uniform period spacing or if the total number of load conditions created is not a limiting factor. See Sample Problem 1 for sample input using the GNTRF feature.

Airy waves specified in descending period order are used to generate the transfer function loading. The following is the Seastate input file used. A detailed description of each of the applicable input lines ensues.

```

1      1      2      3      4      5      6      7      8
1234567890123456789012345678901234567890123456789012345678901234567890
1      * 000 Degree transfer function
2      LDOPT      +Z      1.03      7.85      -50.00      50.00      ME DYN      NPNP
3      FILE S
4      CDM
5      CDM AP
6      MGROV
7      MGROV      0.000      10.00      2.500
8      MGROV      10.000      45.000      5.000
9      GRPOV
10     GRPOV      BL1 F 0.001      5.0 5.0 3.0 3.0      F
11     GRPOV      BL2 F 0.001      3.0 3.0 2.0 2.0      F
12     *****More Group overrides
13     GRPOV      PL4NN      0.001      0.001      0.001
14     GRPOV      W.BN      0.010      0.001      0.001      0.001
15     LOAD
16     LOADCN      1
17     WAVE
18     WAVE1.00AIRY      8.55      10.50      0.00      D      0.00      20.00      18MS10      1      0
19     LOADCN      2

```

20	WAVE								
21	WAVE1.00AIRY	7.76	10.00	0.00	D	0.00	20.00	18MS10	1 0
22	LOADCN 3								
23	WAVE								
24	WAVE1.00AIRY	4.36	7.50	0.00	D	0.00	20.00	18MS10	1 0
25	LOADCN 4								
26	WAVE								
27	WAVE1.00AIRY	2.79	6.00	0.00	D	0.00	20.00	18MS10	1 0
28	LOADCN 5								
29	WAVE								
30	WAVE1.00AIRY	1.75	4.75	0.00	D	0.00	20.00	18MS10	1 0
31	LOADCN 6								
32	WAVE								
33	WAVE1.00AIRY	1.57	4.50	0.00	D	0.00	20.00	18MS10	1 0
34	LOADCN 7								
35	WAVE								
36	WAVE1.00AIRY	1.24	4.00	0.00	D	0.00	20.00	18MS10	1 0
37	LOADCN 8								
38	WAVE								
39	WAVE1.00AIRY	0.95	3.50	0.00	D	0.00	20.00	18MS10	1 0
40	LOADCN 9								
41	WAVE								
42	WAVE1.00AIRY	0.82	3.25	0.00	D	0.00	20.00	18MS10	1 0
43	LOADCN 10								
44	WAVE								
45	WAVE1.00AIRY	0.70	3.00	0.00	D	0.00	20.00	18MS10	1 0
46	LOADCN 11								
47	WAVE								
48	WAVE1.00AIRY	0.59	2.75	0.00	D	0.00	20.00	18MS10	1 0
49	LOADCN 12								
50	WAVE								
51	WAVE1.00AIRY	0.49	2.50	0.00	D	0.00	20.00	18MS10	1 0
52	LOADCN 13								
53	WAVE								
54	WAVE1.00AIRY	0.39	2.25	0.00	D	0.00	20.00	18MS10	1 0
55	LOADCN 14								
56	WAVE								
57	WAVE1.00AIRY	0.31	2.00	0.00	D	0.00	20.00	18MS10	1 0
58	END								

The following describes the input lines in the Seastate input file used for Sample Problem 2:

2. The LDOPT line specifies the Seastate options. The 'DYN' option in columns 56-58 designates that Seastate is to be used for dynamic modeling and wave data generation only.
 3. The FILE line designates that the model data and Seastate data are specified in separate files. The 'S' option specifies that only loading data in the Seastate input file is to be considered.
 18. The first wave is specified on the WAVE line of load case 1 as follows:
 - a. The wave type, height and period are designated in columns 9-12, 14-18 and 25-30, respectively.
 - b. The wave direction is 0.00 degrees as input in columns 39-44.
 - c. The wave is to be stepped through the structure using a 20 degree increment for 18 crest positions for a total of 360 degrees (one full wave cycle).
- Note: Although the critical wave position criteria is designated in the Wave Response input file, the 'AL' option should not be specified on the WAVE line.*
21. The second wave is designated on the next WAVE input line.

The following is the Wave Response input file used to generate the equivalent static loads and global transfer function plots. A detailed description of each of the applicable input lines ensues.

	1	2	3	4	5	6	7	8
	1234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	MEPSL	ALLES		15	-1		
2	PLTTF			OM	BS			
3	TFLCAS	1	14					
4	DAMP		3.0					
5	END							

The following describes the input lines in the Wave Response input file used for Sample Problem 2:

1. The WROPT line specifies the Wave Response options as follows:
 - a. Metric kilonewton force units are to be used.
 - b. Plots are to be generated as stipulated by 'PSL' in columns 10-12.
 - c. Fifteen modes are to be considered as designated by '15' in columns 44-45.
 - d. A single pass analysis is to be performed (i.e. no iterations) as designated by '-1' in columns 49-50.
 - e. A load case is to be created for all wave positions as stipulated by 'ALL' in columns 15-18.
 - f. 'ES' in columns 19-20 designates that equivalent static loads representing static and inertia loads are to be generated.
2. Overturning moment and base shear global transfer function plots are requested on the PLTTF line.
3. Waves 1 through 14 are to be used to plot the transfer functions as designated by '1' and '14' in columns 10 and 13-14 of the TFLCAS line, respectively.
4. Three percent overall critical damping is to be used.

An equivalent static load case was created for each of the 18 crest position of each wave, therefore yielding a total of 252 load cases for this wave direction. These 252 load cases were solved automatically by selecting the automatic feature in the SACS Executive. The analysis, therefore, resulted in a common solution file consisting of nominal stresses used by the Fatigue program to create the transfer function for this direction.

The following pages contain a portion of the Wave Response output listing file:

** WAVE RESPONSE PARAMETERS **

NO. JOINTS 204
 NO. MEMBERS 401
 NO. PLATES 0
 NO. SHELLS 0
 NO. SOLIDS 0
 MAX. ITERATIONS -1
 ITERATION CONVERGENCE 1.00 PERCENT
 FOURIER SERIES CONVERGENCE . 1.00 PERCENT
 OUTPUT LOAD OPTIONALL TIME POINTS
 EQUIVALENT STATIC LOADS TO BE GENERATED

*** REACTION SUMMARY ABOUT ORIGIN ***

MODE	***** FORCES *****			***** MOMENTS *****		
	X MT	Y MT	Z MT	X MT-M	Y MT-M	Z MT-M
1	-16.37	-0.27	-0.45	42.6	-1959.4	-12.6
2	0.30	-15.23	-0.05	2215.0	22.5	-454.7
3	0.01	7.56	0.70	417.0	-30.1	-2588.5
4	-63.34	1.36	-6.15	597.7	30094.1	19.1
5	-2.51	-74.05	3.27	-31134.7	1345.0	-439.6
6	-2.96	6.23	1.84	3896.2	1841.4	487.2
7	-2.63	30.52	11.01	19925.8	2034.9	828.6
8	0.57	9.85	-13.45	8281.0	-545.3	-843.2
9	6.33	-2.21	-35.38	-859.2	-3783.3	-1222.9
10	5.31	0.86	26.79	1004.9	-5955.7	2059.5
11	-2.06	-0.10	1.41	-98.0	1240.5	-374.8
12	-21.07	4.29	-110.12	1785.0	29466.8	2227.5
13	-1.34	-2.57	-0.12	-4315.8	1445.6	-940.2
14	3.33	2.33	-8.74	1608.6	-3105.4	588.0
15	10.85	16.55	0.49	10162.3	-11713.3	3747.1

* WAVE CASE 1 *

WAVE HEIGHT = 8.55 M WAVE PERIOD = 10.50 SECS WAVE DIRECTION = 0.00 DEGREES

***** MODAL DYNAMIC VERSUS MODAL STATIC COMPARISON OF MUDLINE MOMENTS AND SHEARS *****

STEP	TIME (SECS)	PHASE (DEG)	***** DYNAMIC *****				***** STATIC *****			
			MOMENT (X) MT-M	MOMENT (Y) MT-M	SHEAR (X) MT	SHEAR (Y) MT	MOMENT (X) MT-M	MOMENT (Y) MT-M	SHEAR (X) MT	SHEAR (Y) MT
1	0.00	0.00	4.6	3593.6	120.24	0.12	4.7	3250.7	113.55	0.12
2	0.58	20.00	3.0	2134.3	71.74	0.08	3.1	1966.0	68.68	0.09
3	1.17	40.00	0.6	285.4	10.96	0.02	0.6	348.3	11.96	0.03
4	1.75	60.00	-1.9	-1233.7	-40.08	-0.05	-2.0	-1061.6	-37.68	-0.05
5	2.33	80.00	-4.0	-2032.9	-69.24	-0.13	-4.1	-1849.6	-65.76	-0.13
6	2.92	100.00	-5.6	-2271.0	-80.84	-0.21	-5.5	-2228.5	-79.70	-0.20
7	3.50	120.00	-7.3	-2387.5	-87.38	-0.29	-7.3	-2452.2	-88.54	-0.29
8	4.08	140.00	-9.0	-2619.2	-95.22	-0.35	-9.0	-2633.6	-95.70	-0.36
9	4.67	160.00	-9.8	-2769.5	-98.25	-0.38	-9.7	-2601.8	-94.87	-0.38
10	5.25	180.00	-8.7	-2443.8	-85.10	-0.33	-8.6	-2191.2	-79.96	-0.33
11	5.83	200.00	-5.7	-1492.0	-51.89	-0.20	-5.7	-1352.9	-49.33	-0.20
12	6.42	220.00	-1.7	-194.3	-7.51	-0.05	-1.7	-219.9	-8.10	-0.05
13	7.00	240.00	1.9	988.4	33.68	0.07	2.1	927.1	33.30	0.08
14	7.58	260.00	4.2	1822.9	64.16	0.14	4.2	1786.3	63.76	0.14
15	8.17	280.00	5.1	2451.6	87.64	0.15	4.8	2456.6	87.02	0.15
16	8.75	300.00	5.2	3131.3	111.03	0.13	5.1	3203.6	112.70	0.13
17	9.33	320.00	5.4	3835.1	132.84	0.12	5.5	3749.4	131.37	0.13
18	9.92	340.00	5.4	4145.8	140.40	0.13	5.4	3847.7	134.49	0.12

* WAVE CASE 1 *
 WAVE HEIGHT = 8.55 M WAVE PERIOD = 10.50 SECS WAVE DIRECTION = 0.00 DEGREES

*** MUDLINE MOMENTS AND SHEARS ***

STEP	TIME (SECS)	PHASE (DEG)	MOMENT (X) MT-M	MOMENT (Y) MT-M	BASE MOMENT MT-M	SHEAR (X) MT	SHEAR (Y) MT	BASE SHEAR MT
1	0.00	0.00	4.6	3593.6	3593.6	120.24	0.12	120.24
2	0.58	20.00	3.0	2134.3	2134.3	71.74	0.08	71.74
3	1.17	40.00	0.6	285.4	285.4	10.96	0.02	10.96
4	1.75	60.00	-1.9	-1233.7	1233.7	-40.08	-0.05	40.08
5	2.33	80.00	-4.0	-2032.9	2032.9	-69.24	-0.13	69.24
6	2.92	100.00	-5.6	-2271.0	2271.0	-80.84	-0.21	80.84
7	3.50	120.00	-7.3	-2387.5	2387.5	-87.38	-0.29	87.38
8	4.08	140.00	-9.0	-2619.2	2619.2	-95.22	-0.35	95.22
9	4.67	160.00	-9.8	-2769.5	2769.5	-98.25	-0.38	98.25
10	5.25	180.00	-8.7	-2443.8	2443.8	-85.10	-0.33	85.10
11	5.83	200.00	-5.7	-1492.0	1492.1	-51.89	-0.20	51.89
12	6.42	220.00	-1.7	-194.3	194.3	-7.51	-0.05	7.51
13	7.00	240.00	1.9	988.4	988.4	33.68	0.07	33.68
14	7.58	260.00	4.2	1822.9	1822.9	64.16	0.14	64.16
15	8.17	280.00	5.1	2451.6	2451.6	87.64	0.15	87.64
16	8.75	300.00	5.2	3131.3	3131.3	111.03	0.13	111.03
17	9.33	320.00	5.4	3835.1	3835.1	132.84	0.12	132.84
18	9.92	340.00	5.4	4145.8	4145.8	140.40	0.13	140.40

***** EQUIVALENT STATIC LOAD SUMMATIONS *****

LOAD CASE	***** INERTIA *****			***** HYDRODYNAMICS *****			***** BUOYANCY *****			***** WIND *****		
	FX KG	FY KG	FZ KG	FX KG	FY KG	FZ KG	FX KG	FY KG	FZ KG	FX KG	FY KG	FZ KG
1	56.9	0.0	2.1	1017.7	-0.0	-36.1	0.0	0.0	0.0	0.0	0.0	0.0
2	24.4	0.0	0.9	615.1	0.0	-105.5	0.0	0.0	0.0	0.0	0.0	0.0
3	-21.4	-0.0	-0.8	104.5	0.0	-166.5	0.0	0.0	0.0	0.0	0.0	0.0
4	-45.9	-0.0	-1.7	-353.9	0.1	-134.9	0.0	0.0	0.0	0.0	0.0	0.0
5	-34.7	-0.0	-1.3	-619.5	0.0	-96.3	0.0	0.0	0.0	0.0	0.0	0.0
6	-5.6	0.0	-0.2	-756.1	-0.0	-84.7	0.0	0.0	0.0	0.0	0.0	0.0
7	9.5	0.0	0.2	-839.3	-0.0	-68.4	0.0	0.0	0.0	0.0	0.0	0.0
8	-3.9	0.0	-0.3	-904.0	0.0	-50.1	0.0	0.0	0.0	0.0	0.0	0.0
9	-29.9	0.0	-1.2	-892.7	0.1	-21.5	0.0	0.0	0.0	0.0	0.0	0.0
10	-39.9	-0.0	-1.5	-749.0	0.1	12.9	0.0	0.0	0.0	0.0	0.0	0.0
11	-21.8	0.0	-0.8	-457.8	0.1	49.0	0.0	0.0	0.0	0.0	0.0	0.0
12	8.4	0.0	0.3	-68.3	0.1	79.0	0.0	0.0	0.0	0.0	0.0	0.0
13	23.4	0.0	0.9	320.4	0.0	94.7	0.0	0.0	0.0	0.0	0.0	0.0
14	13.2	-0.0	0.5	603.3	-0.1	100.5	0.0	0.0	0.0	0.0	0.0	0.0
15	-4.8	-0.0	-0.1	813.8	-0.0	98.0	0.0	0.0	0.0	0.0	0.0	0.0
16	-3.9	-0.0	-0.0	1026.2	-0.1	137.5	0.0	0.0	0.0	0.0	0.0	0.0
17	22.8	-0.0	1.0	1184.5	-0.0	101.3	0.0	0.0	0.0	0.0	0.0	0.0
18	53.3	0.0	2.0	1207.3	-0.0	32.8	0.0	0.0	0.0	0.0	0.0	0.0

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***** SUMMATION OF FLUID FORCES *****

POINT      TIME    PHASE    ***** FORCES *****
           SECS    DEGS      X      Y      Z
                        MT      MT      MT

  1      0.00     0.00     81.68    -0.00    -4.27
  2      0.56    20.00     45.70     0.00   -10.31
  3      1.11    40.00      1.99     0.00   -15.55
  4      1.67    60.00    -36.40     0.01   -10.60
  5      2.22    80.00    -58.23     0.00    -8.46
  6      2.78   100.00    -69.83     0.00    -7.30
  7      3.33   120.00    -75.48    -0.00    -5.66
  8      3.89   140.00    -77.19     0.00    -3.87
  9      4.44   160.00    -71.97     0.00    -1.24
 10      5.00   180.00    -57.35    -0.00     1.74
 11      5.56   200.00    -31.99     0.01     4.77
 12      6.11   220.00      0.31     0.01     7.16
 13      6.67   240.00     33.18     0.00     8.40
 14      7.22   260.00     57.05    -0.01     8.85
 15      7.78   280.00     75.01    -0.01     8.48
 16      8.33   300.00     90.96    -0.01    10.65
 17      8.89   320.00    102.60    -0.01     8.19
 18      9.44   340.00    100.93    -0.00     1.90

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***** REACTION SUMMARY FOR ITERATION  0 *****

MOMENT X =      19.69 MT-M   AT TIME =      3.889 SECS AND PHASE ANGLE = 140.00 DEGS
        Y =    2385.57 MT-M   AT TIME =      8.889 SECS AND PHASE ANGLE = 320.00 DEGS
        Z =       9.25 MT-M   AT TIME =      3.889 SECS AND PHASE ANGLE = 140.00 DEGS

SHEAR X =    -116.45 MT   AT TIME =      9.444 SECS AND PHASE ANGLE = 340.00 DEGS
       Y =       0.27 MT   AT TIME =      4.444 SECS AND PHASE ANGLE = 160.00 DEGS
       Z =      -9.83 MT   AT TIME =      8.889 SECS AND PHASE ANGLE = 320.00 DEGS

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**** MAXIMUM MODAL RESPONSES FOR ITERATION  0 ****

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MODE	RESPONSE CM	TIME SECS
1	7.331	9.444
2	-0.163	9.444
3	-0.018	8.889
4	2.690	9.444
5	0.078	9.444
6	0.106	9.444
7	0.047	9.444
8	0.015	8.889
9	-0.218	9.444
10	-0.209	8.889
11	0.114	1.111
12	0.074	8.889
13	-0.076	3.889
14	0.026	5.556
15	0.083	3.889

APPLIED LOAD SUMMARY

LOAD CASE NO. ID	TOTAL FORCE (X) KN	TOTAL FORCE (Y) KN	TOTAL FORCE (Z) KN
1 0001	3.746654E+03	9.109688E-01	5.321604E+01
2 0002	1.785415E+03	6.475003E-01	-6.674426E+01
3 0003	-9.462480E+02	-7.201134E-01	-2.017835E+02
4 0004	-2.587200E+03	-1.360171E+00	-2.098967E+02
5 0005	-2.276026E+03	-4.455052E-01	-1.501700E+02
6 0006	-9.566540E+02	6.501651E-01	-9.192344E+01
7 0007	-2.748742E+02	1.186689E+00	-5.654814E+01
8 0008	-9.906664E+02	1.077944E+00	-6.146021E+01
9 0009	-2.276582E+03	7.593155E-01	-7.133956E+01
10 000A	-2.654883E+03	1.521594E-01	-5.164758E+01
11 000B	-1.495810E+03	2.059803E-02	1.274869E+01
12 000C	3.620682E+02	6.911156E-01	9.331820E+01

13	000D	1.471013E+03	8.300862E-01	1.351895E+02
14	000E	1.221282E+03	-4.889437E-01	1.223302E+02
15	000F	5.059967E+02	-1.545732E+00	9.053220E+01
16	000G	7.215620E+02	-1.531546E+00	1.355347E+02
17	000H	2.177280E+03	-6.933114E-01	1.424015E+02
18	000I	3.719838E+03	1.111557E-01	1.181246E+02
19	000J	3.224058E+03	-3.722586E+00	3.711054E+01
20	000K	1.405523E+03	4.060644E+00	-7.123973E+01
21	000L	-1.027736E+03	5.198527E+00	-1.882955E+02
22	000M	-2.451351E+03	-2.778644E+00	-1.722081E+02
23	000N	-2.148099E+03	-6.922973E+00	-1.334716E+02
24	000O	-9.728761E+02	-2.276334E-02	-8.373519E+01
25	000P	-3.398446E+02	7.755837E+00	-4.972219E+01
26	000Q	-8.864740E+02	4.297652E+00	-4.903834E+01
27	000R	-1.909043E+03	-4.925420E+00	-5.497065E+01
28	000S	-2.166762E+03	-5.537693E+00	-3.709961E+01
29	000T	-1.143397E+03	3.259332E+00	1.894775E+01
30	000U	4.271433E+02	7.208394E+00	8.485068E+01
31	000V	1.337536E+03	-3.338056E-02	1.167609E+02
32	000W	1.101603E+03	-7.008863E+00	1.056826E+02
33	000X	5.202906E+02	-2.882388E+00	8.024809E+01
34	000Y	7.550663E+02	4.364026E+00	1.078973E+02
35	000Z	2.042948E+03	2.469308E+00	1.211714E+02
36	000a	3.319607E+03	-4.674738E+00	9.730127E+01
37	000b	1.396249E+03	2.578352E+00	-6.025933E+00
38	000c	1.232850E+02	-2.298489E+00	-3.790448E+01
39	000d	-1.345736E+03	-2.704706E+00	-7.270948E+01
40	000e	-1.945364E+03	-8.665563E-02	-8.145678E+01
41	000f	-1.332743E+03	7.218290E-01	-6.311930E+01
42	000g	-4.021862E+02	2.309447E-01	-3.312634E+01
43	000h	-2.669993E+02	1.762607E+00	-2.009188E+01
44	000i	-9.433266E+02	3.366834E+00	-2.695901E+01
45	000j	-1.385283E+03	6.418034E-01	-2.794075E+01
46	000k	-8.491095E+02	-4.001489E+00	-4.914407E+00
47	000l	2.250334E+02	-3.465349E+00	2.978403E+01
48	000m	8.825559E+02	2.332646E+00	5.126847E+01
49	000n	8.173346E+02	5.136765E+00	5.065384E+01
50	000o	5.234455E+02	3.453584E-01	4.220708E+01
51	000p	5.486554E+02	-6.024889E+00	3.945088E+01
52	000q	9.650708E+02	-5.510735E+00	4.327463E+01
53	000r	1.526943E+03	1.244683E+00	4.205680E+01
54	000s	1.831278E+03	5.513070E+00	2.300141E+01
55	000t	1.108414E+03	-2.293840E+01	1.646764E+01
56	000u	2.497619E+02	4.725565E+00	-7.140726E+00
57	000v	-1.031444E+03	2.809779E+01	-3.715046E+01
58	000w	-1.715715E+03	2.390033E+01	-5.244340E+01
59	000x	-1.315011E+03	-3.629943E+00	-4.384121E+01
60	000y	-3.444294E+02	-2.701629E+01	-2.217911E+01
61	000z	1.938866E+02	-2.296762E+01	-7.912134E+00
62	0010	-1.914858E+02	4.251307E+00	-1.079585E+01
63	0011	-9.999511E+02	2.725290E+01	-2.116102E+01
64	0012	-1.239415E+03	2.285057E+01	-1.938428E+01
65	0013	-4.291498E+02	-4.693209E+00	2.892282E+00
66	0014	8.950943E+02	-2.797848E+01	3.405149E+01
67	0015	1.692279E+03	-2.376946E+01	5.206451E+01
68	0016	1.421796E+03	3.687749E+00	4.682710E+01
69	0017	5.388876E+02	2.696154E+01	2.800384E+01
70	0018	6.252030E+00	2.282645E+01	1.425187E+01
71	0019	3.111374E+02	-4.440579E+00	1.488611E+01
72	001A	9.882764E+02	-2.742173E+01	2.124757E+01
73	001B	-1.444433E+03	-1.562882E+01	-4.042004E+01
74	001C	-6.157816E+02	-6.974024E+00	-1.582148E+01
75	001D	4.423473E+02	5.869110E+00	1.219549E+01
76	001E	1.283471E+03	1.575878E+01	2.649158E+01
77	001F	1.388425E+03	1.757918E+01	2.739846E+01
78	001G	6.761681E+02	1.174420E+01	1.255224E+01
79	001H	-4.664085E+02	8.938534E-01	-1.774813E+01
80	001I	-1.590369E+03	-1.028561E+01	-4.626136E+01
81	001J	-2.182901E+03	-1.602443E+01	-5.144737E+01
82	001K	-1.712822E+03	-1.479100E+01	-3.276523E+01
83	001L	-2.481039E+02	-8.531307E+00	-2.538776E+00
84	001M	1.373353E+03	2.330818E+00	3.356808E+01
85	001N	2.324631E+03	1.243376E+01	6.253291E+01
86	001O	2.335483E+03	1.632184E+01	6.343390E+01
87	001P	1.466450E+03	1.305490E+01	3.707007E+01
88	001Q	3.557480E+01	2.948071E+00	3.266018E+00
89	001R	-1.252370E+03	-9.929727E+00	-2.514174E+01
90	001S	-1.756706E+03	-1.702881E+01	-4.297926E+01
91	001T	-4.609879E+03	-5.941139E+01	-1.138523E+02
92	001U	-3.955576E+03	-6.568331E+01	-8.944995E+01
93	001V	-1.460448E+03	-3.999550E+01	-2.799646E+01
94	001W	1.682308E+03	5.599946E+00	3.163509E+01
95	001X	3.825873E+03	4.688781E+01	8.762342E+01
96	001Y	4.117114E+03	6.681264E+01	9.870060E+01
97	001Z	2.509222E+03	5.737621E+01	4.755501E+01
98	001a	-4.447350E+02	1.912688E+01	-1.819604E+01
99	001b	-3.292350E+03	-2.784933E+01	-7.171191E+01
100	001c	-4.505477E+03	-5.990783E+01	-1.055427E+02
101	001d	-3.569679E+03	-6.703601E+01	-7.930674E+01
102	001e	-8.928092E+02	-4.236079E+01	-1.188528E+01
103	001f	2.289643E+03	4.052756E+00	5.306193E+01
104	001g	4.446481E+03	4.514461E+01	1.114491E+02
105	001h	4.681848E+03	6.701566E+01	1.113989E+02
106	001i	2.815151E+03	5.790474E+01	6.003133E+01

107	001j	-3.531254E+02	1.893475E+01	-5.402096E+00
108	001k	-3.242267E+03	-2.689085E+01	-7.561417E+01
109	001l	-9.258337E+01	1.451191E+01	-1.034648E+01
110	001m	-7.312316E+02	3.103918E+01	-8.238454E+00
111	001n	-9.624627E+02	3.287502E+01	-1.380337E+01
112	001o	-7.434167E+02	2.142136E+01	-2.498857E+01
113	001p	-3.177282E+02	-4.911711E-01	-3.769645E+00
114	001q	3.245611E+02	-2.374424E+01	2.164261E+01
115	001r	1.074211E+03	-3.475218E+01	2.377508E+01
116	001s	1.429038E+03	-2.841993E+01	2.347095E+01
117	001t	1.130687E+03	-8.980129E+00	2.860029E+01
118	001u	4.539436E+02	1.350717E+01	1.562213E+01
119	001v	-3.009599E+02	2.954792E+01	-1.244564E+01
120	001w	-9.527586E+02	3.302298E+01	-2.293595E+01
121	001x	-1.162005E+03	1.994207E+01	-1.532462E+01
122	001y	-7.101866E+02	-3.437306E+00	-1.316751E+01
123	001z	1.849374E+01	-2.439270E+01	-1.343395E+01
124	0020	4.932078E+02	-3.474879E+01	7.574071E+00
125	0021	6.250822E+02	-2.841064E+01	1.746299E+01
126	0022	4.501953E+02	-8.779175E+00	2.222279E+00
127	0023	-4.446091E+02	-5.476600E-01	-1.285613E+01
128	0024	-1.361669E+01	-3.197826E-01	-3.783140E-01
129	0025	3.903610E+02	-5.226808E-02	1.170213E+01
130	0026	7.151038E+02	5.552699E-01	2.163961E+01
131	0027	9.376034E+02	1.237465E+00	2.890849E+01
132	0028	1.049550E+03	1.980958E+00	3.271727E+01
133	0029	1.057120E+03	2.248891E+00	3.278298E+01
134	002A	9.688612E+02	2.058961E+00	2.928382E+01
135	002B	7.904901E+02	1.266291E+00	2.257783E+01
136	002C	5.243702E+02	2.598829E-01	1.332587E+01
137	002D	1.821028E+02	-7.201557E-01	2.333168E+00
138	002E	-2.110545E+02	-1.278787E+00	-9.437058E+00
139	002F	-6.104007E+02	-1.558227E+00	-2.075583E+01
140	002G	-9.534845E+02	-1.435374E+00	-2.981479E+01
141	002H	-1.178861E+03	-1.294296E+00	-3.529963E+01
142	002I	-1.238579E+03	-9.633807E-01	-3.612260E+01
143	002J	-1.117845E+03	-8.375594E-01	-3.193422E+01
144	002K	-8.354391E+02	-6.207148E-01	-2.399066E+01
145	002L	-6.368659E+02	-1.514578E+00	-1.886136E+01
146	002M	5.098837E+01	-1.834469E+00	7.468907E-01
147	002N	6.889588E+02	-1.440875E+00	1.921825E+01
148	002O	1.193159E+03	-4.430837E-01	3.326500E+01
149	002P	1.494634E+03	1.588089E+00	4.162570E+01
150	002Q	1.562981E+03	3.199779E+00	4.445458E+01
151	002R	1.428414E+03	4.728805E+00	4.353466E+01
152	002S	1.172025E+03	4.711943E+00	3.890893E+01
153	002T	8.840553E+02	3.656183E+00	3.069811E+01
154	002U	5.976262E+02	1.336421E+00	1.839170E+01
155	002V	2.746967E+02	-1.022069E+00	3.153215E+00
156	002W	-1.508582E+02	-2.696490E+00	-1.306380E+01
157	002X	-6.932762E+02	-3.288522E+00	-2.812159E+01
158	002Y	-1.266601E+03	-2.597345E+00	-3.980486E+01
159	002Z	-1.713229E+03	-1.710077E+00	-4.750694E+01
160	002a	-1.885716E+03	-9.280510E-01	-4.990929E+01
161	002b	-1.723042E+03	-7.411767E-01	-4.622675E+01
162	002c	-1.269205E+03	-1.014835E+00	-3.548833E+01
163	002d	-8.469766E+02	-3.215090E+00	-2.451505E+01
164	002e	-7.355274E+01	-2.537225E+00	-1.608659E+00
165	002f	6.900240E+02	-1.172893E+00	1.959870E+01
166	002g	1.347423E+03	1.107768E+00	3.710767E+01
167	002h	1.799726E+03	2.996724E+00	4.772792E+01
168	002i	1.978948E+03	5.393359E+00	5.300525E+01
169	002j	1.882671E+03	5.875362E+00	5.331179E+01
170	002k	1.586379E+03	6.332721E+00	4.993500E+01
171	002l	1.189300E+03	4.582432E+00	4.024481E+01
172	002m	7.159950E+02	2.739061E+00	2.393995E+01
173	002n	2.699780E+02	2.672350E-01	3.512745E+00
174	002o	-2.941879E+02	-1.446745E+00	-1.694814E+01
175	002p	-9.365979E+02	-2.494291E+00	-3.408357E+01
176	002q	-1.566525E+03	-3.656803E+00	-4.727594E+01
177	002r	-2.030615E+03	-3.325138E+00	-5.449692E+01
178	002s	-2.196690E+03	-4.182427E+00	-5.740538E+01
179	002t	-2.013972E+03	-3.423474E+00	-5.355165E+01
180	002u	-1.530395E+03	-3.869131E+00	-4.281928E+01
181	002v	-9.727519E+02	-4.393315E+00	-2.428500E+01
182	002w	-3.642279E+02	-3.835821E+00	-4.007696E+00
183	002x	3.025862E+02	-1.559498E+00	5.290663E+00
184	002y	8.933639E+02	2.917249E+00	1.601480E+01
185	002z	1.356121E+03	5.150488E+00	4.138665E+01
186	0030	1.691823E+03	3.928178E+00	5.421918E+01
187	0031	1.886666E+03	5.824637E+00	5.449537E+01
188	0032	1.865869E+03	7.445386E+00	4.323113E+01
189	0033	1.542967E+03	5.807614E+00	3.091690E+01
190	0034	9.547032E+02	3.777031E+00	2.628547E+01
191	0035	2.348035E+02	2.669862E+00	1.025873E+01
192	0036	-4.599765E+02	1.581011E+00	-5.850683E+00
193	0037	-1.021532E+03	-1.019847E+00	-2.313357E+01
194	0038	-1.421459E+03	-3.527547E+00	-4.300090E+01
195	0039	-1.660444E+03	-4.302025E+00	-4.808425E+01
196	003A	-1.742293E+03	-7.152154E+00	-5.099764E+01
197	003B	-1.667463E+03	-7.766386E+00	-4.807379E+01
198	003C	-1.414220E+03	-5.624391E+00	-3.715351E+01
199	003D	-2.114170E+02	-3.438673E+00	-2.185998E+01
200	003E	-5.293089E+02	-4.298781E+00	-1.614078E+01

201	003F	-7.720366E+02	-3.792460E+00	-6.668974E+00
202	003G	-8.303816E+02	-4.850208E+00	-1.293758E+01
203	003H	-6.651911E+02	-3.462378E+00	-9.244909E+00
204	003I	-3.834747E+02	-2.725544E+00	-1.086189E+01
205	003J	-1.104185E+02	-3.315741E+00	-2.088683E+01
206	003K	7.395023E+01	-1.603050E+00	-5.695681E+00
207	003L	1.722296E+02	3.773828E-01	1.275203E+01
208	003M	2.643603E+02	1.757275E+00	1.182740E+01
209	003N	3.774823E+02	3.281213E+00	1.419448E+01
210	003O	4.802279E+02	5.629331E+00	1.540361E+01
211	003P	5.364385E+02	5.069194E+00	6.529053E+00
212	003Q	5.127890E+02	4.007586E+00	1.264260E+01
213	003R	4.387194E+02	4.677392E+00	1.988987E+01
214	003S	3.502091E+02	2.851148E+00	9.447692E+00
215	003T	2.390826E+02	6.647012E-01	4.438438E+00
216	003U	5.743999E+01	-8.348638E-01	-5.958372E+00
217	003V	1.177985E+04	1.821660E+02	2.794601E+02
218	003W	1.021797E+04	1.800667E+02	2.365318E+02
219	003X	7.433040E+03	1.557983E+02	1.609646E+02
220	003Y	3.713851E+03	1.147734E+02	7.063453E+01
221	003Z	-5.168637E+02	6.082908E+01	-2.830781E+01
222	003a	-4.742931E+03	-1.862503E+00	-1.227261E+02
223	003b	-8.400199E+03	-6.487661E+01	-1.946944E+02
224	003c	-1.099859E+04	-1.200323E+02	-2.459898E+02
225	003d	-1.219943E+04	-1.595485E+02	-2.774291E+02
226	003e	-1.191603E+04	-1.796525E+02	-2.809095E+02
227	003f	-1.016003E+04	-1.780871E+02	-2.353117E+02
228	003g	-7.223578E+03	-1.574748E+02	-1.619684E+02
229	003h	-3.443554E+03	-1.161766E+02	-7.284375E+01
230	003i	7.249671E+02	-6.012915E+01	2.049221E+01
231	003j	4.784749E+03	2.285226E+00	1.156769E+02
232	003k	8.254892E+03	6.432948E+01	1.994968E+02
233	003l	1.073931E+04	1.185203E+02	2.546541E+02
234	003m	1.195653E+04	1.590131E+02	2.804836E+02
235	003n	-1.838614E+02	-3.229894E+00	-3.082726E+00
236	003o	-1.637122E+02	-5.961048E+00	-1.481903E-01
237	003p	-1.082429E+02	-8.130809E+00	3.230385E+00
238	003q	-8.616501E+00	-8.981302E+00	7.877095E+00
239	003r	1.397167E+02	-8.750078E+00	1.037791E+01
240	003s	3.014338E+02	-8.185281E+00	7.635586E+00
241	003t	4.292025E+02	-6.298926E+00	4.118167E+00
242	003u	4.779418E+02	-3.033863E+00	1.396527E+00
243	003v	4.250245E+02	4.672829E-01	-9.832705E-01
244	003w	2.725480E+02	3.382313E+00	-1.661404E+00
245	003x	7.322662E+01	6.228314E+00	2.640011E+00
246	003y	-1.228757E+02	8.211988E+00	4.743197E+00
247	003z	-2.539857E+02	8.999955E+00	1.837105E+00
248	0040	-3.094917E+02	8.993608E+00	-2.658462E+00
249	0041	-3.012618E+02	8.196964E+00	-6.036465E+00
250	0042	-2.566821E+02	6.153078E+00	-9.838659E+00
251	0043	-2.168774E+02	2.534208E+00	-1.261172E+01
252	0044	-1.931193E+02	-6.112117E-01	-8.656649E+00

4.3 SAMPLE PROBLEM 3 – RESPONSE FOR A DETERMINISTIC WAVE FOR THE STEADY STATE CONDITION

Sample Problem 3 is a deterministic extreme wave analysis. The purpose of this example is to obtain the response for a deterministic wave for the steady state condition.

For this sample, a 5.4 second 5 meter Stream Function wave, along with current, was considered for both the 0 degree and 90 degree directions. One equivalent static load case was produced for each wave direction based on the crest position yielding the maximum base shear.

Note: Prior to executing this sample, PSI was executed to obtain the linearized foundation super element. Fifteen mode shapes were also extracted using the Dynpac program.

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	* Extreme Wave Deterministic Analysis							
3	LDOPT	+Z	1.03	7.85	-50.00	50.00	ME DYN	NPNP
4	FILE S							
5	CDM							
6	CDM AP							
7	MGROV							
8	MGROV	0.000	10.00	2.500				
9	MGROV	10.000	45.000	5.000				
10	GRPOV							
11	GRPOV	BL1 F 0.001			5.0	5.0	3.0	3.0
12	GRPOV	BL2 F 0.001			3.0	3.0	2.0	2.0
13	*****More Group overrides							
14	GRPOV	PL4NN	0.001	0.001	0.001			
15	GRPOV	W.BN	0.010	0.001	0.001	0.001		
16	LOAD							
17	LOADCN 1							
18	WAVE							
19	WAVE1.00STRE	5.00		5.40	0.0	D	0.00	20.00 18MS10 1 0
20	CURR							
21	CURR	0.0	0.0	0.00				
22	CURR	50.0	1.0					
23	LOADCN 2							
24	WAVE							
25	WAVE1.00STRE	5.00		5.40	90.0	D	20.00	18MS10 1 0
26	CURR							
27	CURR	0.0	0.0	90.00				
28	CURR	50.0	1.0					
29	END							

The wave, current and hydrodynamic data was specified by the user in a Seastate input file as shown below. A detailed description of each of the applicable input lines ensues.

The following describes the input lines in the Seastate input file used for Sample Problem 3:

2. The LDOPT line specifies the Seastate options. The 'DYN' option in columns 56-58 designates that Seastate is to be used for dynamic modeling and wave data generation only.
3. The FILE line designates that the model data and Seastate data are specified in separate files. The 'S' option specifies that only loading data in the Seastate input file is to be considered.
18. The first wave is specified on the WAVE line of load case 1 as follows:
 - a. The wave type, height and period are designated in columns 9-12, 14-18 and 25-30, respectively.

b. The wave direction is 0.00 degrees as input in columns 39-44.

c. The wave is to be stepped through the structure using a 20 degree increment for 18 crest positions for a total of 360 degrees (one full wave cycle).

Note: Although the critical wave position criteria is designated in the Wave Response input file, the 'AL' option should not be specified on the WAVE line.

20-21. The current data for load case 1 is specified using the CURR input lines.

24. The second wave is designated on the next WAVE input line.

The following is the Wave Response input file used to generate the equivalent static loading and the response plots. One load case corresponding to the wave crest position yielding the maximum base shear was created for each wave analyzed. A detailed description of each of the applicable input lines ensues.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	MEPSL	MAXSES		15	10		
2	PSEL		JO MF OM BS					
3	PSJO	601LDX601LDY601LVX601LVY602LDX602LDY602LVX602LVY						
4	PSMF	501L601LFXB502L602LFXB	A	A	A	A		
5	DAMP	3.0						
6	END							

The following describes the input lines in the Wave Response input file used for Sample Problem 3:

1. The WROPT line specifies the Wave Response options as follows:

a. Metric kilonewton force units are to be used.

b. Plots are to be generated as stipulated by 'PSL' in columns 10-12.

c. Fifteen modes are to be considered as designated by '15' in columns 44-45.

d. A maximum of 10 iterations is allowed as designated by '10' in columns 49-50.

e. A load case is to be created for the wave crest position yielding the maximum base shear as stipulated by 'MAXS' in columns 15-18.

f. 'ES' in columns 19-20 designates that equivalent static loads representing static and inertia loads are to be generated.

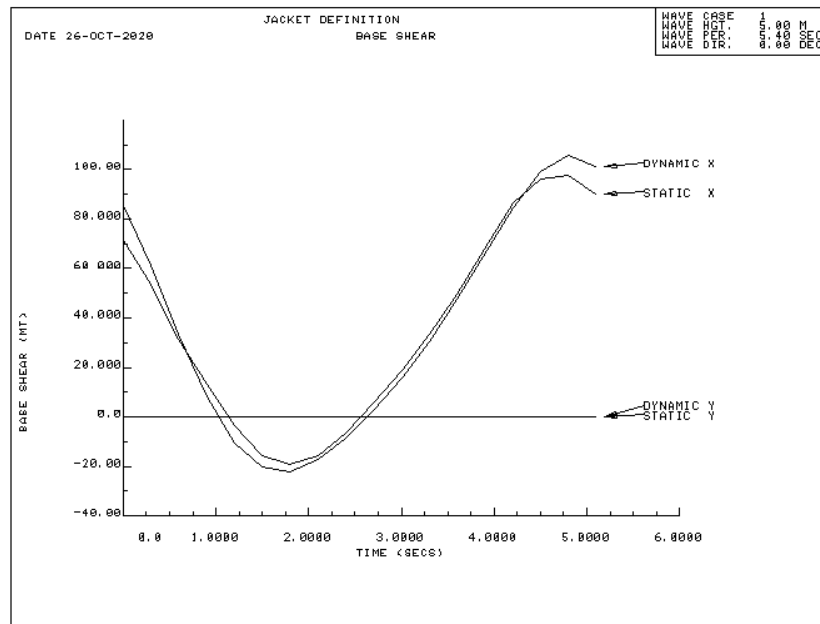
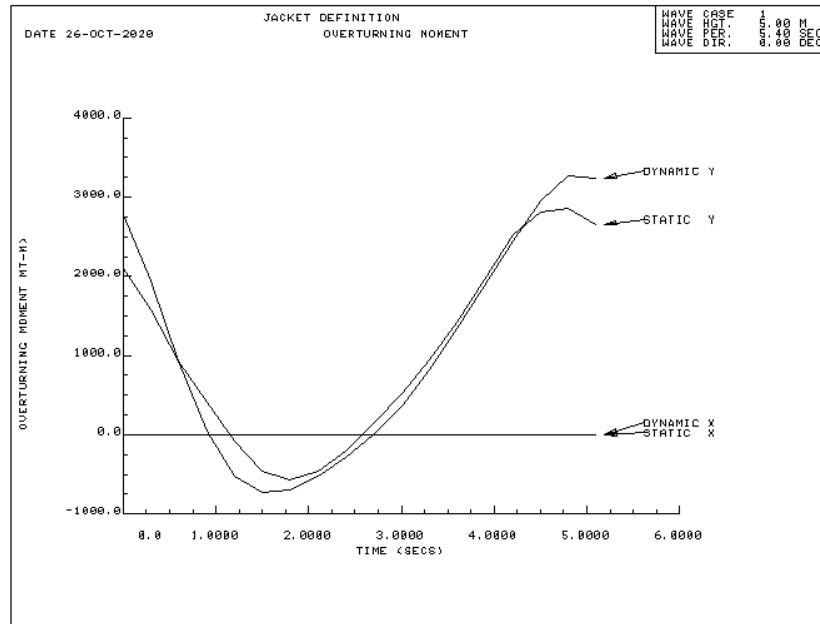
2. Joint displacement, member forces, overturning moment and base shear plots are requested by 'JO', 'MF', 'OM' and 'BS' on the PSEL line.

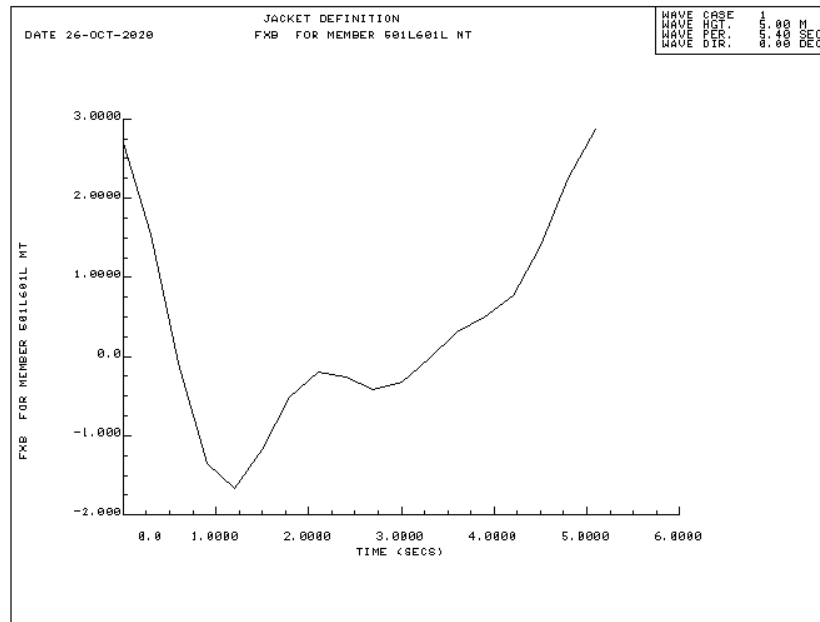
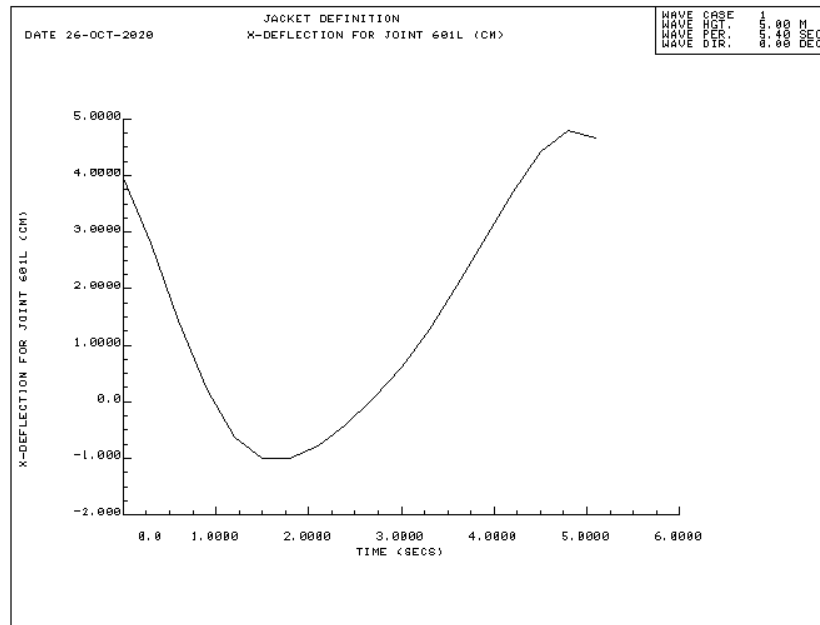
3. The joints and the degree of freedom to be plotted are specified on the PSJO line.

4. Member axial force at the end of members 501-601 and 507-607 are to be plotted as stipulated on the PSMF input line.

5. Three percent overall critical damping is to be used.

The following are some of the plots generated by the Wave Response program for the zero degree direction wave and current (load case 1):





In addition to plots, the Wave Response program creates an output data file the equivalent static load cases. In this sample, the first load case contains equivalent static loads for the 0 degree wave and current, while load case 2 contains the equivalent static loads for the 90 degree wave and current. This data file can be solved automatically as part of the Wave Response analysis using the SACS Executive or may be solved as a separate analysis step.

A portion of the sample problem output is shown on the following pages. Note that during the iteration process, messages show how the iteration process is proceeding. For example, after a particular pass, the

comparison between generalized forces computed from the previous iteration are compared with those from the current pass. An error parameter is listed showing the results of the generalized force comparison for each mode along with an error level for the sum of all the modes. One or more of the modes will typically exhibit error levels higher than either the total iteration error or the allowed error. This is normally due to the fact that these modes have little or no response and a large error between iterations is not significant.

The output also shows the response of mode 1 for the entire wave cycle including the modal displacement, generalized force, and generalized velocity versus time. This type of report is generated for each mode for all the wave cases. The Wave Response program reports the dynamic and the static overturning moment and base shear versus time. This output may be used to determine the Dynamic Amplification Factor by dividing the maximum value of dynamic by the maximum static respective value.

Note: The maximum dynamic overturning moment, or base shear will more than likely not occur at the same time point as the maximum sta

```

***** SUMMATION OF FLUID FORCES *****

POINT      TIME    PHASE      ***** FORCES *****
           SECS    DEGS        X        Y        Z
                           MT        MT        MT

  1         0.00     0.00        61.48     -0.00     -4.60
  2         0.30    20.00        46.17     -0.00     -3.71
  3         0.60    40.00        27.18        0.00     -2.26
  4         0.90    60.00        11.60        0.00     -3.16
  5         1.20    80.00         -2.55        0.00     -3.51
  6         1.50   100.00       -13.20        0.00     -1.89
  7         1.80   120.00       -16.36     -0.00     -0.97
  8         2.10   140.00       -13.45     -0.00     -0.01
  9         2.40   160.00        -5.87     -0.01        0.91
 10         2.70   180.00         4.64     -0.00        1.64
 11         3.00   200.00        16.26     -0.00        2.28
 12         3.30   220.00        28.82     -0.00        2.90
 13         3.60   240.00        43.25     -0.00        3.59
 14         3.90   260.00        58.58     -0.01        4.24
 15         4.20   280.00        74.25     -0.01        4.77
 16         4.50   300.00        82.18     -0.00        3.71
 17         4.80   320.00        83.32        0.00        1.26
 18         5.10   340.00        77.30     -0.00       -2.39

***** REACTION SUMMARY FOR ITERATION    0 *****

MOMENT X =          9.03 MT-M   AT TIME =      5.100 SECS AND PHASE ANGLE = 340.00 DEGS
      Y =       2065.55 MT-M   AT TIME =      4.500 SECS AND PHASE ANGLE = 300.00 DEGS
      Z =         -1.00 MT-M   AT TIME =      3.300 SECS AND PHASE ANGLE = 220.00 DEGS

SHEAR  X =     -105.06 MT   AT TIME =      4.800 SECS AND PHASE ANGLE = 320.00 DEGS
      Y =         0.17 MT   AT TIME =      5.100 SECS AND PHASE ANGLE = 340.00 DEGS
      Z =         -8.03 MT   AT TIME =      4.500 SECS AND PHASE ANGLE = 300.00 DEGS

**** MAXIMUM MODAL RESPONSES FOR ITERATION    0 ****

      MODE      RESPONSE      TIME
              CM              SECS

          1         7.016         4.800
          2        -0.150         4.800
          3        -0.019         4.200
          4         2.393         4.500
          5         0.070         4.800
          6         0.093         4.800

```

7	0.040	4.800
8	0.013	4.500
9	-0.170	4.500
10	-0.157	4.500
11	0.112	0.600
12	0.046	4.500
13	0.038	4.500
14	0.021	4.500
15	-0.023	3.900
ITERATION	1	*** TOTAL RMS DIFFERENCE = 2.818 PERCENT
		ALLOWABLE DIFFERENCE = 1.000 PERCENT

* WAVE CASE 1 *										
WAVE HEIGHT = 5.00 M			WAVE PERIOD = 5.40 SECS			WAVE DIRECTION = 0.00 DEGREES				
***** MODAL DYNAMIC VERSUS MODAL STATIC COMPARISON OF MUDLINE MOMENTS AND SHEARS *****										
***** DYNAMIC *****						***** STATIC *****				
STEP	TIME (SECS)	PHASE (DEG)	MOMENT (X) MT-M	MOMENT (Y) MT-M	SHEAR (X) MT	SHEAR (Y) MT	MOMENT (X) MT-M	MOMENT (Y) MT-M	SHEAR (X) MT	SHEAR (Y) MT
1	0.00	0.00	0.5	2781.1	85.41	-0.12	0.0	2096.7	71.31	-0.12
2	0.30	20.00	0.1	1936.8	61.06	-0.10	-0.3	1580.8	53.42	-0.10
3	0.60	40.00	-1.1	931.4	33.36	-0.05	-0.4	932.0	31.24	-0.06
4	0.90	60.00	-1.5	40.9	7.94	-0.02	-0.7	402.6	13.11	-0.04
5	1.20	80.00	-0.6	-526.3	-10.66	-0.01	-0.6	-84.5	-3.29	-0.01
6	1.50	100.00	1.1	-733.4	-20.44	0.01	-0.3	-456.6	-15.57	0.03
7	1.80	120.00	1.9	-691.7	-22.06	0.01	0.1	-564.9	-19.03	0.04
8	2.10	140.00	0.7	-519.7	-17.30	0.03	0.3	-467.2	-15.52	0.05
9	2.40	160.00	-1.0	-288.3	-8.41	0.06	0.7	-211.3	-6.64	0.03
10	2.70	180.00	-1.4	-2.0	2.79	0.06	0.7	138.1	5.46	0.03
11	3.00	200.00	0.1	365.1	15.64	0.04	1.1	521.6	18.76	0.02
12	3.30	220.00	2.3	816.7	30.34	-0.01	1.2	940.6	33.22	0.01
13	3.60	240.00	3.3	1338.3	47.39	-0.06	1.4	1427.0	49.87	-0.02
14	3.90	260.00	2.4	1891.5	66.14	-0.08	1.3	1959.1	67.89	-0.06
15	4.20	280.00	0.5	2450.2	84.57	-0.11	0.8	2523.2	86.70	-0.11
16	4.50	300.00	-0.8	2958.5	99.25	-0.14	0.3	2812.6	96.29	-0.15
17	4.80	320.00	-0.9	3272.8	105.71	-0.16	-0.3	2867.2	97.76	-0.18
18	5.10	340.00	-0.3	3243.7	101.15	-0.18	-0.4	2657.8	90.29	-0.18

* WAVE CASE 1 *
WAVE HEIGHT = 5.00 M WAVE PERIOD = 5.40 SECS WAVE DIRECTION = 0.00 DEGREES

*** MUDLINE MOMENTS AND SHEARS ***

STEP	TIME (SECS)	PHASE (DEG)	MOMENT (X) MT-M	MOMENT (Y) MT-M	BASE MOMENT MT-M	SHEAR (X) MT	SHEAR (Y) MT	BASE SHEAR MT
1	0.00	0.00	0.5	2781.1	2781.1	85.41	-0.12	85.41
2	0.30	20.00	0.1	1936.8	1936.8	61.06	-0.10	61.06
3	0.60	40.00	-1.1	931.4	931.4	33.36	-0.05	33.36
4	0.90	60.00	-1.5	40.9	41.0	7.94	-0.02	7.94
5	1.20	80.00	-0.6	-526.3	526.3	-10.66	-0.01	10.66
6	1.50	100.00	1.1	-733.4	733.4	-20.44	0.01	20.44
7	1.80	120.00	1.9	-691.7	691.7	-22.06	0.01	22.06
8	2.10	140.00	0.7	-519.7	519.7	-17.30	0.03	17.30
9	2.40	160.00	-1.0	-288.3	288.3	-8.41	0.06	8.41
10	2.70	180.00	-1.4	-2.0	2.4	2.79	0.06	2.79
11	3.00	200.00	0.1	365.1	365.1	15.64	0.04	15.64
12	3.30	220.00	2.3	816.7	816.7	30.34	-0.01	30.34
13	3.60	240.00	3.3	1338.3	1338.3	47.39	-0.06	47.39
14	3.90	260.00	2.4	1891.5	1891.5	66.14	-0.08	66.14
15	4.20	280.00	0.5	2450.2	2450.2	84.57	-0.11	84.57
16	4.50	300.00	-0.8	2958.5	2958.5	99.25	-0.14	99.25
17	4.80	320.00	-0.9	3272.8	3272.8	105.71	-0.16	105.71
18	5.10	340.00	-0.3	3243.7	3243.7	101.15	-0.18	101.15

**** MAXIMUM MOMENT AND SHEAR REPORT ****

MAXIMUM MOMENT 3272.69 MT-M
AT TIME 4.8000 SECS
PHASE 320.00 DEGREES
MAXIMUM SHEAR 105.71 MT
AT TIME 4.8000 SECS
PHASE 320.00 DEGREES

MINIMUM MOMENT -733.37 MT-M
AT TIME 1.5000 SECS
PHASE 100.00 DEGREES
MINIMUM SHEAR -22.06 MT
AT TIME 1.8000 SECS
PHASE 120.00 DEGREES

***** EQUIVALENT STATIC LOAD SUMMATIONS *****

LOAD CASE	FX KG	FY KG	FZ KG	FX KG	FY KG	FZ KG	FX KG	FY KG	FZ KG	FX KG	FY KG	FZ KG
1	77.0	0.1	3.0	814.8	0.0	12.2	0.0	0.0	0.0	0.0	0.0	0.0

4.4 SAMPLE PROBLEM 4 – RANDOM WAVE ANALYSIS

Sample Problem 4 illustrates the random wave analysis capabilities using a wave spectrum to define the wave properties. The statistical properties of the environment were defined using a Pierson-Moskowitz wave spectrum.

Note: When using an input spectrum instead of a random surface profile, the Wave Response program internally generates the time history of the surface profile that will have a spectrum corresponding to the parameters of the input spectrum. Since the Fourier series components are the desired Airy wave components and are taken directly from the input wave spectrum, it is unnecessary for the user to generate the surface profile manually.

The Seastate program is used to generate the hydrodynamic properties of each of the members in the model. Additionally, a dummy reference wave that will wet the parts of the structure that will be wetted during any time of the random wave analysis is used by the Seastate to create the member load segmentation.

Note: The Wave Response program calculates the water particle velocities from the Airy wave components. Therefore, the actual wave specified in the Seastate input file is of little importance as long as it provides an accurate representation of the wetted surface.

The dummy wave and hydrodynamic data was specified by the user in a Seastate input file as shown below. A detailed description of each of the applicable input lines ensues.

	1	2	3	4	5	6	7	8
1	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
2	* Extreme Wave Deterministic Analysis							
3	LDOPT	+Z	1.03	7.85	-50.00	50.00	ME DYN	NPNP
4	FILE S							
5	CDM							
6	CDM AP							
7	MGROV							
8	MGROV	0.000	10.00	2.500				
9	MGROV	10.000	45.000	5.000				
10	GRPOV							
11	GRPOV	BL1 F 0.001				5.0 5.0 3.0 3.0		F
12	GRPOV	BL2 F 0.001				3.0 3.0 2.0 2.0		F
13	*****More Group overrides							
14	GRPOV	PL4NN	0.001	0.001	0.001			
15	GRPOV	W.BN	0.010	0.001	0.001	0.001		
16	LOAD							
17	LOADCN	1						
18	WAVE							
19	WAVE1.00AIRY	6.00		8.0		D	20.00	18MS10 1
20	CURR							
21	END							

The following describes the input lines in the Seastate input file used for Sample Problem 4

2. The LDOPT line specifies the Seastate options. The 'DYN' option in columns 56-58 designates that Seastate is to be used for dynamic modeling and wave data generation only.

3. The FILE line designates that the model data and Seastate data are specified in separate files. The 'S' option specifies that only loading data in the Seastate input file is to be considered.

18. The dummy wave is specified on the WAVE line of load case 1 as follows:

- a. The wave type, height and period are designated in columns 9-12, 14-18 and 25-30, respectively.
- b. The wave direction is 0.00 degrees as input in columns 39-44.
- c. The wave is to be positioned so that the crest is at the center of the structure as designated by '1' in column 68.

Note: The number of crest position steps is of little importance. Typically, positioning the crest at the center of the structure ensures both sufficient wetted surface and the maximum number of member load segments.

The following is the Wave Response input file used to generate the equivalent static loading, response plots and the response file used by the Probability program. One load case corresponding to the time point yielding the maximum base shear was created. A detailed description of each of the applicable input lines ensues.

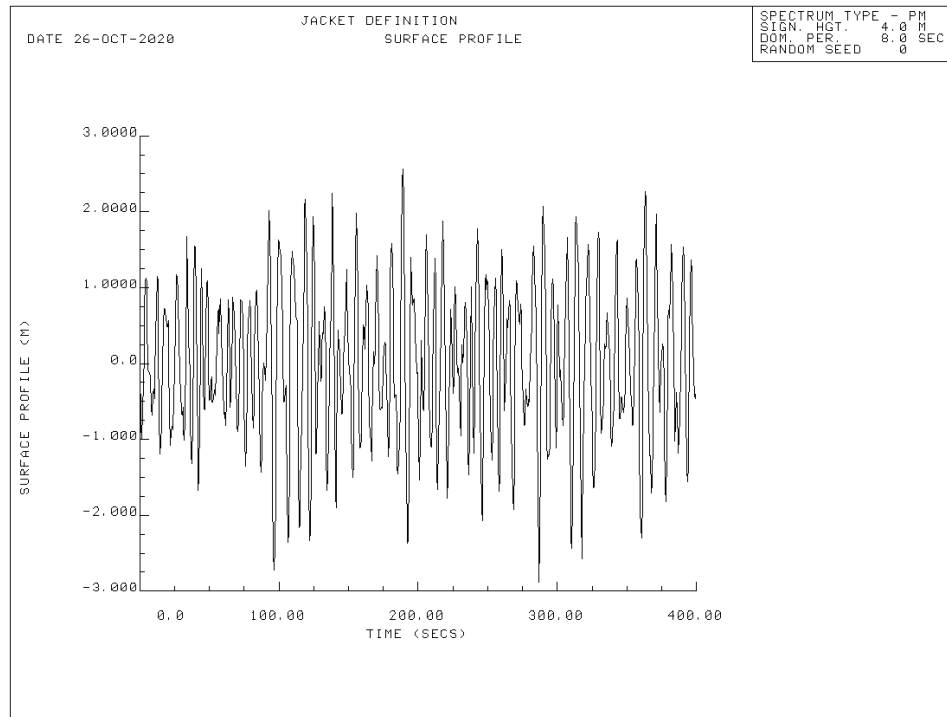
	1	2	3	4	5	6	7	8
1	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
2	WROPT	MEPSL	MAXSES		15	-1		RW
3	PSEL	SPB	GFB	JO	MF	OM	BS	
4	PSJO	601LDX601LDY601LVX601LVY602LDX602LDY602LVX602LVY						
5	PSMF	501L601LFXB502L602LFXB		A		A		A
6	DAMP	3.0						
7	WAVTIM	+Z	50.	-50.		0.5	1.0	1 1 2
8	WSPEC	PM	4.0	8.0				64.0
9	CURR							
10	CURR	50.0	1.0	0.0				
11	END							

The following describes the input lines in the Wave Response input file used for Sample Problem 4:

1. The WROPT line specifies the Wave Response options as follows:
 - a. Metric kilonewton force units are to be used.
 - b. Plots are to be generated as stipulated by 'PSL' in columns 10-12.
 - c. Fifteen modes are to be considered as designated by '15' in columns 44-45.
 - d. A single pass analysis (no iterations) is to be performed as designated by '-1' in columns 49-50.
 - e. A load case is to be created for the time point yielding the maximum base shear as stipulated by 'MAXS' in columns 15-18.
 - f. 'ES' in columns 19-20 designates that equivalent static loads representing static and inertia loads are to be generated.
 - g. A random wave analysis is to be executed by 'RW' in columns 68-69.
2. Plots and response data are requested using the PSEL line as follows:
 - a. The wave surface profile and generalized forces are to be plotted and saved to a data file for use by the Probability program as designated by 'SPB' and 'GFB' in columns 7-9 and 10-12, respectively.

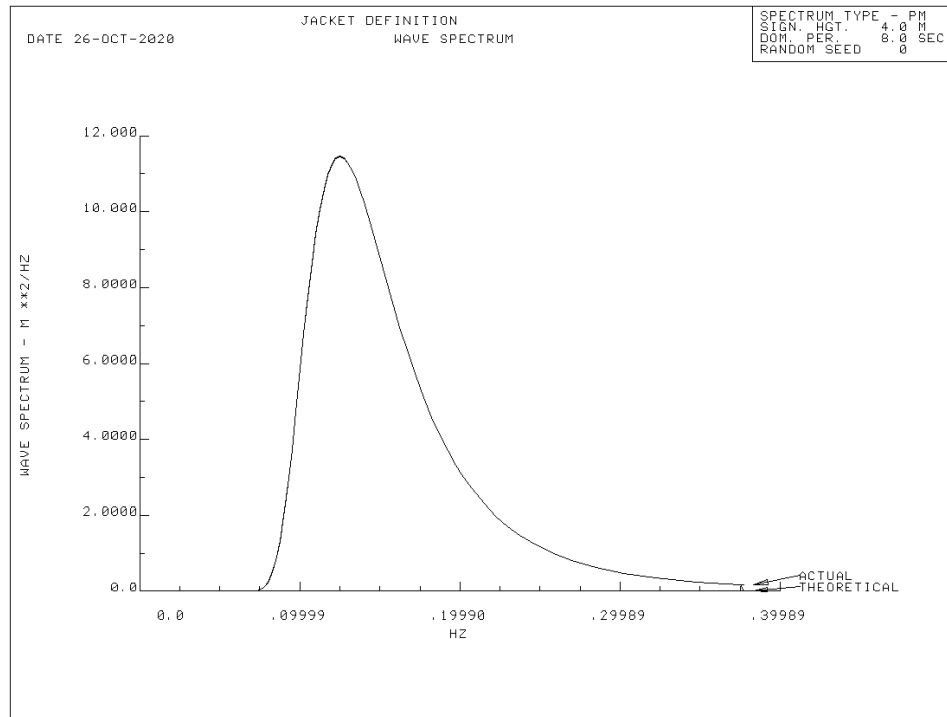
-
- b. Joint displacement, member forces, overturning moment and base shear plots are requested by 'JO', 'MF', 'OM' and 'BS'.
- 3. The joints and the degree of freedom to be plotted are specified on the PSJO line.
 - 4. Member axial force at the end of members 501-601 and 507-607 are to be plotted as stipulated on the PSMF input line.
 - 5. Three percent overall critical damping is to be used.
 - 6. Wave time and position parameters are entered on the WAVTIM line as follows:
 - a. The water depth and mudline elevation are specified in columns 11-16 and 17-22, respectively.
 - b. The analysis time increment is input as 0.5 seconds.
 - c. Standard Airy waves are used with wave kinematics calculated at the mean water line as designated by '1' in column 59.
 - d. Both Fast Fourier transformation options are selected by '1' and '2' in columns 61 and 67, respectively.
 - 7. The wave height spectral density is defined using the WSPEC line as follows:
 - a. Pierson-Moskowitz spectrum is to be used as stipulated by 'PM'.
 - b. The significant wave height is 4.0 meters and the dominant period is 8.0 seconds.
 - c. The time duration is to be 64.0 seconds as designated in columns 48-54.
 - 8. The CURR lines are used to define the current with a linearly varying velocity of 0.0 at the mudline to 1.0 at the water surface (50.0 above the mudline).

Time history plots, equivalent static loading for the time point causing the maximum base shear and a response data file were generated as part of the analysis. The plot of the surface profile generated by the Wave Response program is shown below:

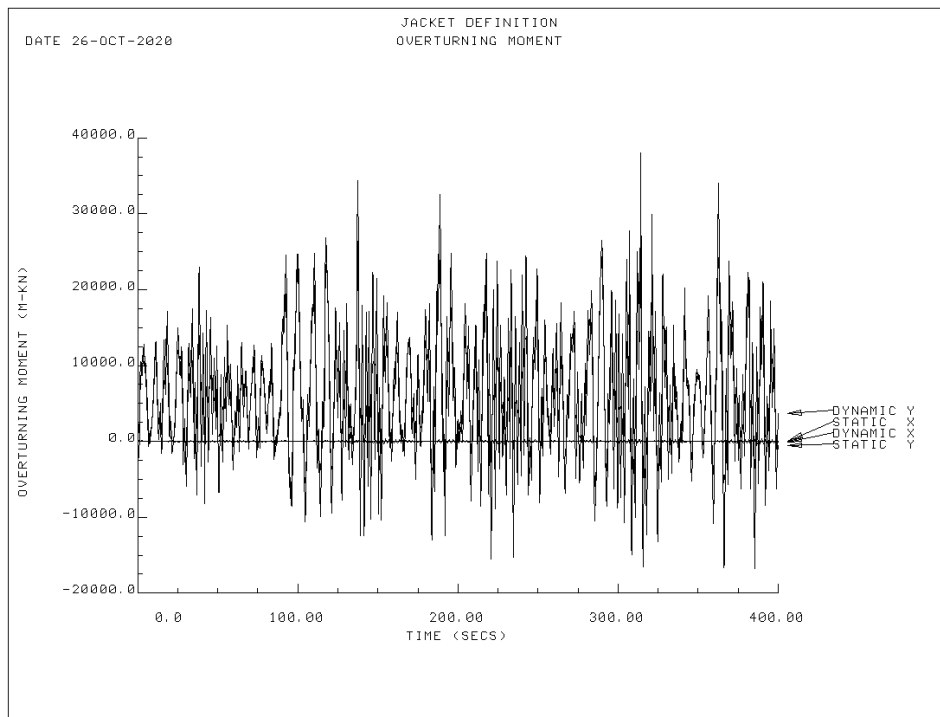


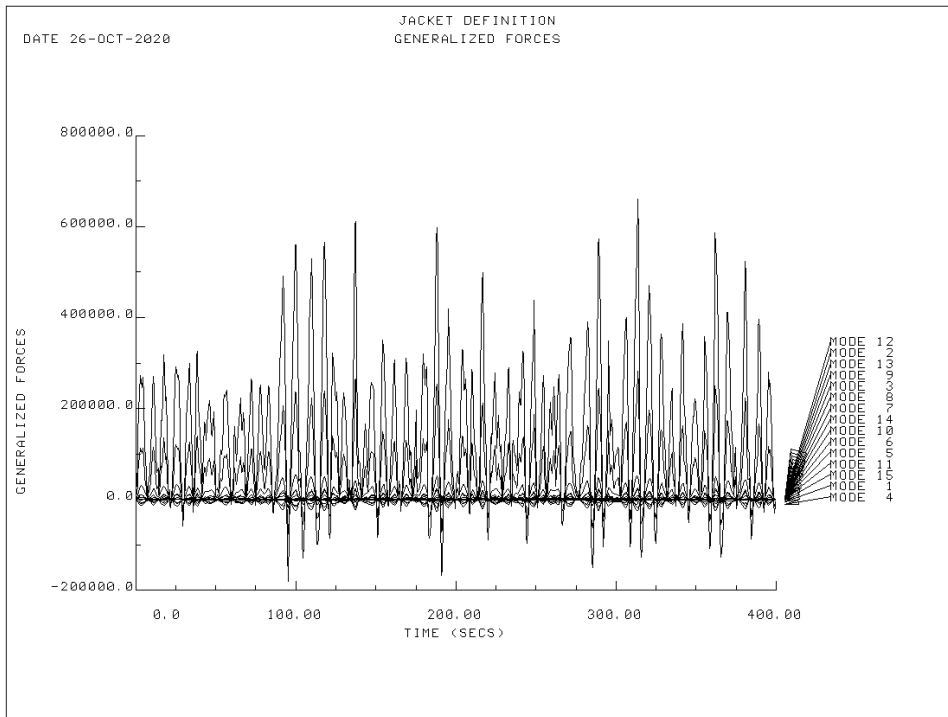
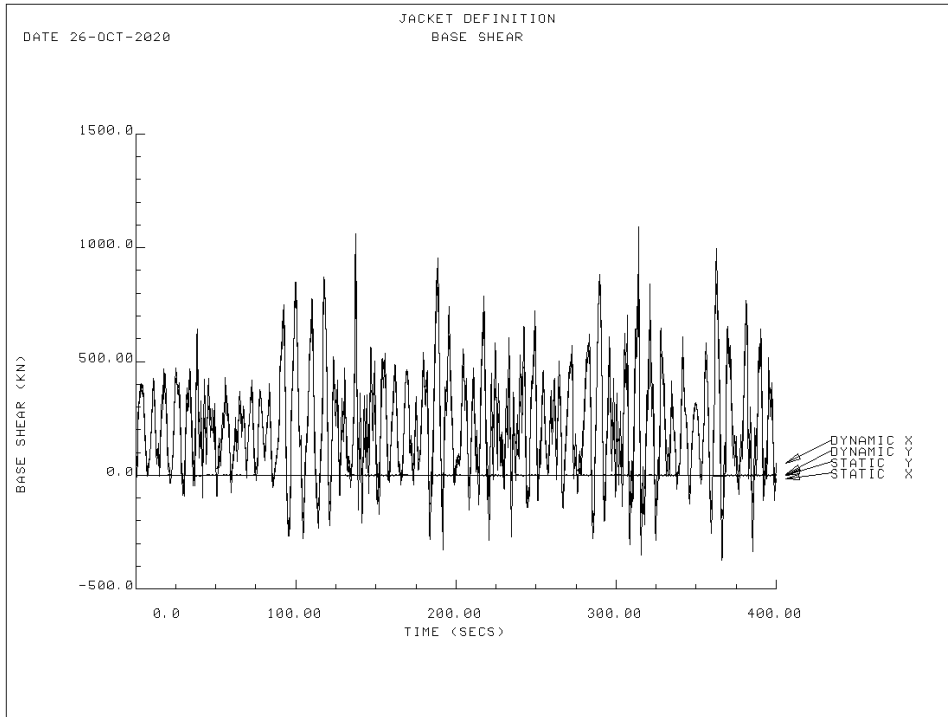
The comparison between the theoretical Pierson-Moskowitz spectrum and the spectrum of the surface profile generated by the Wave Response program follows:

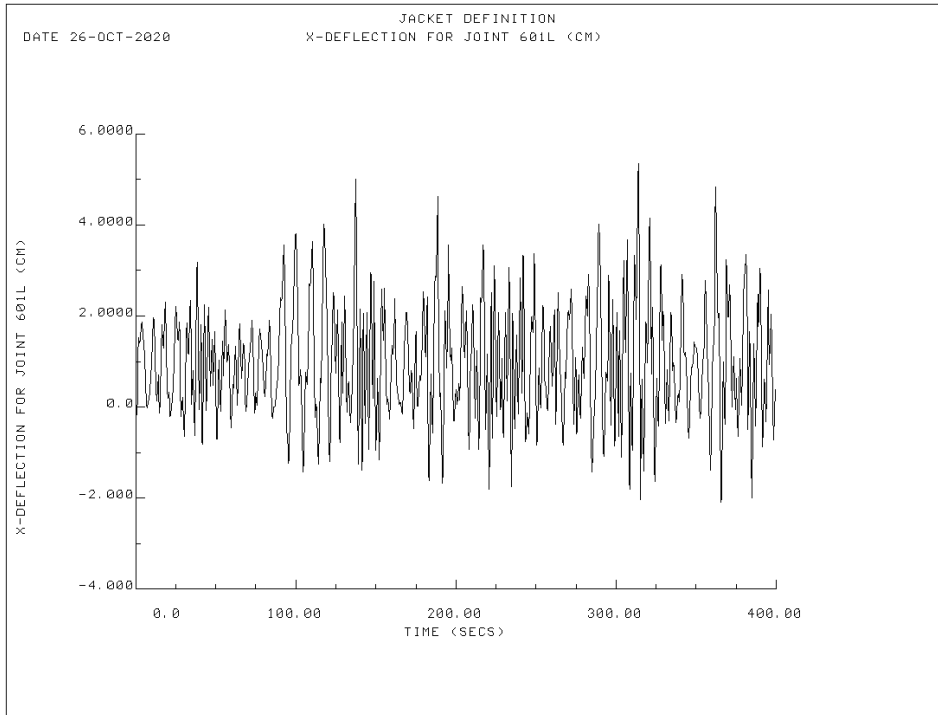
Note: Notice the slight discrepancy between the theoretical and calculated wave spectra. This may occur when the time duration of the surface time history is not long enough to represent the statistical properties sufficiently.



Additional time history plots including overturning moment, base shear, generalized forces and joint displacement were created and are shown below:







The equivalent static loading was saved to the output data file . The output data file contains the model and one load case containing the static and inertia loading corresponding to the time point causing the maximum base shear. The output data file may then be solved statically and combined with other static loading such as self weight, live load, etc. in order to perform code checks.

4.5 SAMPLE PROBLEM 5 – RANDOM WAVE ANALYSIS PLOT AND REPORT

Sample Problem 5 illustrates the random wave analysis surface profile plot and report capabilities. Surface profile plots are generated using a Ochi-Hubble wave spectrum to define the wave properties for 20 seeds using the time history integration approach.

The Seastate program is used to generate the hydrodynamic properties of each of the members in the model. Additionally, a dummy reference wave that will wet the parts of the structure that will be wetted during any time of the random wave analysis is used by the Seastate to create the member load segmentation.

Note: The Wave Response program calculates the water particle velocities from the Airy wave components. Therefore, the actual wave specified in the Seastate input file is of little importance as long as it provides an accurate representation of the wetted surface.

The dummy wave and hydrodynamic data was specified by the user in a Seastate input file as shown below. A detailed description of each of the applicable input lines ensues.

	1	2	3	4	5	6	7	8
1	123456789012345678901234567890123456789012345678901234567890							
1	* Surface Profile Plots							
2	LDOPT	+Z	1.03	7.85	-50.00	50.00	ME DYN	NPNP
3	FILE S							
4	CDM							
5	CDM AP							
6	MGROV							
7	MGROV	0.000	10.00	2.500				
8	MGROV	10.000	45.000	5.000				
9	GRPOV							
10	GRPOV	BL1 F 0.001	5.0 5.0 3.0 3.0				F	
11	GRPOV	BL2 F 0.001	3.0 3.0 2.0 2.0				F	
12	*****More Group overrides							
13	GRPOV	PL4NN	0.001	0.001	0.001			
14	GRPOV	W.BN	0.010	0.001	0.001	0.001		
15	LOAD							
16	LOADCN	1						
17	WAVE							
18	WAVE	AIRY	22.	15.	L	18	1MS	0
19	CURR							
20	END							

The following describes the input lines in the Seastate input file used for Sample Problem 5:

2. The LDOPT line specifies the Seastate options. The 'DYN' option in columns 56-58 designates that Seastate is to be used for dynamic modeling and wave data generation only.
3. The FILE line designates that the model data and Seastate data are specified in separate files. The 'S' option specifies that only loading data in the Seastate input file is to be considered.
18. The dummy wave is specified on the WAVE line of load case 1 as follows:
 - a. The wave type, height and period are designated in columns 9-12, 14-18 and 25-30, respectively.
 - b. The wave direction is 0.00 degrees as input in columns 39-44.

c. The wave is to be positioned so that the crest is at the center of the structure as designated by '1' in column 68.

Note: The number of crest position steps is of little importance. Typically, positioning the crest at the center of the structure ensures both sufficient wetted surface and the maximum number of member load segments.

The following is the Wave Response input file used to generate the surface profile plots. A detailed description of each of the applicable input lines ensues.

	1	2	3	4	5	6	7	8
1	1234567890123456789012345678901234567890123456789012345678901234567890							
2	WROPT	PSL			20	-1		TH
3	PSEL	SP		JO MF OM BS				
4	DAMP		2.0					
5	WAVTIM	+Z1402.0-1400.0.0	0.00		1.00		VS 1	2
6	WSPEC20	100H	11.80	16.00	1.0	4.0	5.00	60. SP 64.0 6.0 3.0
7	CURR							
8	CURR	0.0	0.6	0.0				
9	CURR	1150.	0.6					
10	CURR	1200.	1.9					
11	CURR	1390.	1.9					
12	CURR	1395.	3.7					
13	CURR	1400.	3.7					
13	END							

The following describes the input lines in the Wave Response input file used for Sample Problem 5:

1. The WROPT line specifies the Wave Response options as follows:
 - a. Default units are to be used.
 - b. Plots are to be generated as stipulated by 'PSL' in columns 10-12.
 - c. Twenty modes are to be considered as designated by '20' in columns 44-45.
 - d. A single pass analysis (no iterations) is to be performed as designated by '-1' in columns 49-50.
 - e. Random wave analysis time history integration is to be used by 'TH' in columns 68-69.
2. Surface profile plots requested by 'SP' in columns 7-9 on the PSEL line.
3. Three percent overall critical damping is to be used.
4. Wave time and position parameters are entered on the WAVTIM line as follows:
 - a. The water depth and mudline elevation are specified in columns 11-16 and 17-22, respectively.
 - b. The analysis time increment is input as 1.0 seconds.
 - c. Both Fast Fourier transformation options are selected by '1' and '2' in columns 61 and 67, respectively.
5. The wave height spectral density is defined using the WSPEC line as follows:
 - a. Ochi-Hubble spectrum is to be used as stipulated by 'OH'.

b. Twenty sequential seeds starting with a value of 10 are to be used to generate wave surface profiles.

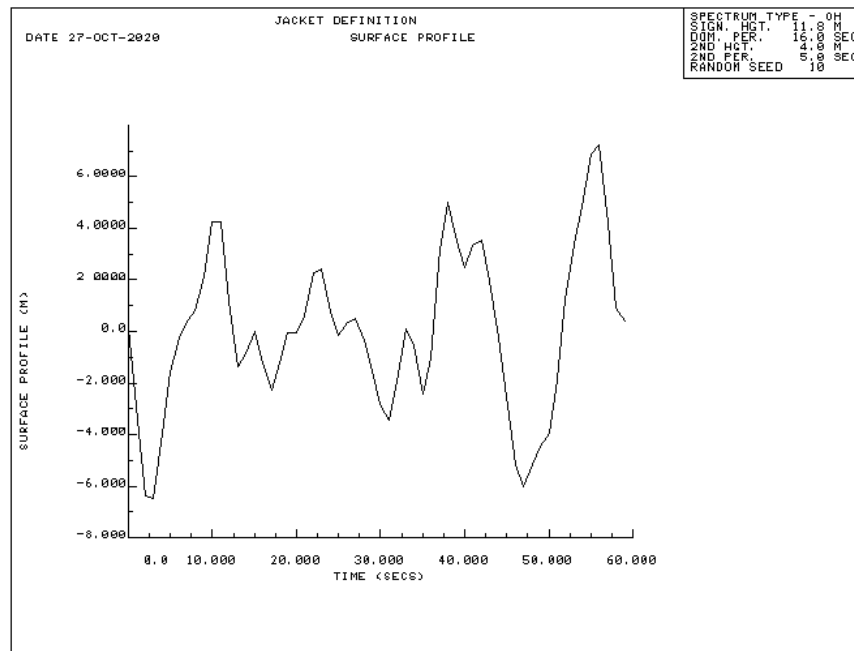
c. Only surface profile plots are to be generated as designated by 'SP' in columns 57-58.

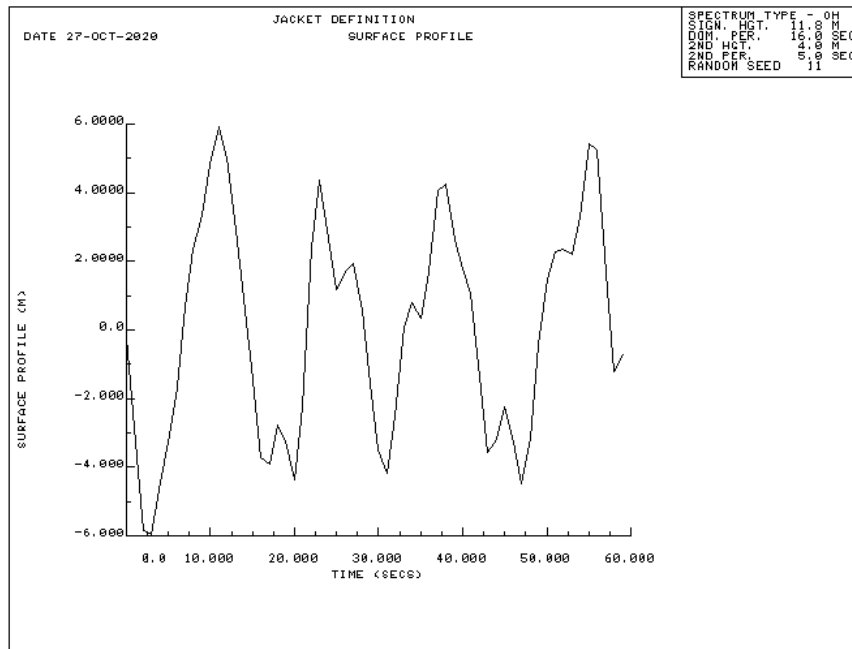
d. The time duration is to be 60.0 seconds as designated in columns 48-54.

F. The CURR lines are used to define the current.

Note: Current is ignored when generating surface profile plots.

The following are the surface profile plots for seeds 10 and 11.





A portion of the output listing file including the surface profile statistics follows:

** WAVE RESPONSE PARAMETERS **

NO. JOINTS	204
NO. MEMBERS	401
NO. PLATES	0
NO. SHELLS	0
NO. SOLIDS	0
MAX. ITERATIONS	-1
ITERATION CONVERGENCE	1.00 PERCENT
FOURIER SERIES CONVERGENCE .	1.00 PERCENT
OUTPUT LOAD OPTION	NO OUTPUT
WATER DEPTH	1402.0 M
MUDLINE ELEVATION	-1400.0 M
WAVE DIRECTION	0.00 DEGREES
WAVE INITIAL OFFSET	0.0 M
ANALYSIS TIME INCREMENT	1.000 SECONDS
WAVE KINEMATICS FACTOR	1.000
FOURIER SERIES TOLERANCE ...	1.000 PERCENT
WAVE APPLICATION TYPE	VERTICALLY STRETCHED
RANDOM WAVE ANALYSIS	TIME HISTORY INTEGRATION
REL. VELOCITY EFFECTS ...	INCLUDED
SPECTRAL WAVE SELECTION	OCHI-HUBBLE

SIGNIFICANT WAVE HEIGHT 11.80 M
 DOMINANT PERIOD 16.000 SECONDS
 WIND SIGN. WAVE HEIGHT 4.00 M
 WIND DOMINANT PERIOD 5.000 SECONDS
 SWELL LAMDA 6.00000
 WIND GENERATED LAMDA 3.00000

 WAVE COMPONENT ENERGY BAND . 1.000 PERCENT
 TIME HISTORY DURATION 60.00 SECONDS
 NUMBER OF RANDOM SEEDS 20
 SELECTED SEEDS 10 11 12 13 14
 SELECTED SEEDS 15 16 17 18 19
 SELECTED SEEDS 20 21 22 23 24
 SELECTED SEEDS 25 26 27 28 29
 CURRENT DIRECTION 0.00000 DEGREES
 CURRENT STRETCHINGVERTICAL
 CURRENT BLOCKING FACTOR 1.00000
 CURRENT DESCRIPTION

ELEVATION (M)	VELOCITY (M /SECS)
0.000	0.000
0.010	0.600
1150.000	0.600
1200.000	1.900
1390.000	1.900
1395.000	3.700
1400.000	3.700

 MAX. INTEGRATION TIME STEP.. 1.00000 SECS
 MIN. INTEGRATION TIME STEP..0.00000092 SECS
 INTEGRATION ERR. FACTOR 1.00000
 TRANSIENT REDUCTION ON

***** WAVE PARAMETERS *****

NO. OF WAVE TIME POINTS 60

MAXIMUM CREST HEIGHT 6.48 M

AT TIME 58.00 SECS

MINIMUM TROUGH DEPTH -6.55 M

AT TIME 51.00 SECS

MAXIMUM WAVE HEIGHT 11.93 M

AT TIME 48.00 SECS

ZERO UPCROSSING PERIOD 10.00 SECS

MEAN VALUE 0.00 M

STANDARD DEVIATION 3.11 M

SKEWNESS -0.0563

KURTOSIS 2.4398

RANDOM SEED 15

NO. PAIRS OF FOURIER SERIES. 6

NO. AIRY WAVES USED 6

***** AIRY WAVE PARAMETERS *****

NO.	HEIGHT M	PERIOD SECS	LENGTH M	PHASE RAD.
1	3.44	20.00	624.80	1.715
2	7.20	15.00	351.45	1.755
3	2.40	12.00	224.92	2.656
4	2.00	5.45	46.47	-1.743
5	1.61	4.62	33.27	-2.382
6	1.26	3.75	21.97	1.767

***** SURFACE PROFILE STATISTICS *****

SEED NO.	SEED VALUE	* MAXIMUM WAVE * HEIGHT	* TIME	** MAX. CREST ** HEIGHT	* TIME	** MIN. TROUGH ** HEIGHT	* TIME	ZERO UPCROSSING PERIOD	STANDARD DEVIATION
		M	SECS	M	SECS	M	SECS	SECS	
1	10	11.02	42.50	7.26	56.00	-6.50	3.00	8.57	3.11
2	11	10.28	15.50	5.90	11.00	-5.97	3.00	15.00	3.11
3	12	12.56	17.00	6.58	14.00	-5.98	20.00	12.00	3.11
4	13	10.75	12.50	5.74	0.00	-5.61	53.00	15.00	3.11
5	14	12.36	48.00	6.58	59.00	-6.28	52.00	12.00	3.11
6	15	11.93	48.00	6.48	58.00	-6.55	51.00	10.00	3.11
7	16	14.64	45.00	7.24	41.00	-7.40	49.00	10.00	3.11
8	17	12.84	50.50	7.38	54.00	-5.46	47.00	12.00	3.11
9	18	13.79	14.50	7.02	11.00	-6.78	18.00	12.00	3.11
10	19	11.60	35.00	6.12	31.00	-6.27	6.00	12.00	3.11
11	20	13.39	46.00	7.51	42.00	-5.88	50.00	12.00	3.11
12	21	11.27	34.00	5.32	38.00	-5.95	30.00	15.00	3.11
13	22	12.84	24.50	6.73	29.00	-6.11	20.00	12.00	3.11
14	23	11.96	13.50	7.17	2.00	-6.32	10.00	12.00	3.11
15	24	14.03	7.00	6.85	3.00	-7.18	11.00	12.00	3.11
16	25	12.37	9.00	6.47	6.00	-6.11	0.00	12.00	3.11
17	26	13.86	11.00	7.31	7.00	-8.26	0.00	12.00	3.11
18	27	10.98	49.00	6.47	53.00	-7.80	0.00	12.00	3.11
19	28	10.78	31.00	6.00	52.00	-7.66	0.00	12.00	3.11
20	29	12.20	11.50	7.46	7.00	-6.97	0.00	12.00	3.11

4.6 SAMPLE PROBLEM 6 – TIME HISTORY RESPONSE FILE

Sample Problem 6 uses time history integration to generate a response file used by the Global Loading program to select critical time points for numerous random wave seeds based on structural elevations designated by the user. It also demonstrates generating load cases at the critical time points by retrieving loads from the response file.

Modal coordinates, velocities and accelerations were generated for an Ochi-Hubble wave spectrum using seed values of 10, 20 and 21 and also including the effects of current. The time history integration approach was used with wave kinematics and fluid damping terms calculated at grid points in order to speed execution.

The Seastate program is used to generate the hydrodynamic properties of each of the members in the model. Additionally, a dummy reference wave that will wet the parts of the structure that will be wetted during any time of the random wave analysis is used by the Seastate to create the member load segmentation.

Note: The Wave Response program calculates the water particle velocities from the Airy wave components. Therefore, the actual wave specified in the Seastate input file is of little importance as long as it provides an accurate representation of the wetted surface.

The dummy wave and hydrodynamic data was specified by the user in a Seastate input file as shown below. A detailed description of each of the applicable input lines ensues.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	* Surface Profile Plots							
2	LDOPT	+Z	1.03	7.85	-50.00	50.00	ME DYN	NPNP
3	FILE S							
4	CDM							
5	CDM AP							
6	MGROV							
7	MGROV	0.000	10.00	2.500				
8	MGROV	10.000	45.000	5.000				
9	GRPOV							
10	GRPOV	BL1 F 0.001			5.0	5.0	3.0	3.0 F
11	GRPOV	BL2 F 0.001			3.0	3.0	2.0	2.0 F
12	*****More Group overrides							
13	GRPOV	PL4NN		0.001	0.001	0.001		
14	GRPOV	W.BN 0.010		0.001	0.001	0.001		
15	DUMMY BOATLANDDUMMY STRUCTURE DATA							
16	KEEP	803	903	807	907			
17	DELETE	1803	1903	1807	1907			
18	LOAD							
19	LOADCN	1						
20	WAVE							
21	WAVE	AIRY	22.	15.	L	18	1MS	0
22	CURR							
23	END							

The following describes the input lines in the Seastate input file used for Sample Problem 6:

2. The LDOPT line specifies the Seastate options. The 'DYN' option in columns 56-58 designates that Seastate is to be used for dynamic modeling and wave data generation only.
3. The FILE line designates that the model data and Seastate data are specified in separate files. The 'S' option specifies that only loading data in the Seastate input file is to be considered.

18. The dummy wave is specified on the WAVE line of load case 1 as follows:

- The wave type, height and period are designated in columns 9-12, 14-18 and 25-30, respectively.
- The wave direction is 0.00 degrees as input in columns 39-44.
- The wave is to be positioned so that the crest is at the center of the structure as designated by '1' in column 68.

Note: The number of crest position steps is of little importance. Typically, positioning the crest at the center of the structure ensures both sufficient wetted surface and the maximum number of member load segments.

The analysis options, current data, and wave spectra are input in the Wave Response input file. The following is the Wave Response input file used to generate the response file to be used by the Global Loading module. A detailed description of each of the applicable input lines ensues.

	1	2	3	4	5	6	7	8
	12345678901234567890123456789012345678901234567890123456789012345678901234567890							
1	WROPT	PSL			20	-1		TH
2	PSEL	SP	MCB	MVB	MAB			
3	DAMP		2.0					
4	WAVTIM	+Z1402.0-1400.0.0	0.00	1.00		VS 1	2	
5	WSPEC	OH 11.80 16.00	1.0	4.0	5.00	60.	6.0	3.0
6	THIST	5 10			1.0			
7	CURR							
8	CURR	0.0	0.6	0.0				
9	CURR	1150.	0.6					
10	CURR	1200.	1.9					
11	CURR	1390.	1.9					
12	CURR	1395.	3.7					
13	CURR	1400.	3.7					
14	RNSEED	10. 20. 21.						
15	END							

The following describes the input lines in the Wave Response input file used for Sample Problem 6:

1. The WROPT line specifies the Wave Response options as follows:

- Default units are to be used.
- Plots are to be generated as stipulated by 'PSL' in columns 10-12.
- Twenty modes are to be considered as designated by '20' in columns 44-45.
- A single pass analysis (no iterations) is to be performed as designated by '-1' in columns 49-50.
- Random wave analysis time history integration using a grid is to be used by 'TG' in columns 68-69.

2. Plot and response file data is specified on the PSEL line.

- Surface profile plots requested by 'SP' in columns 7-9.
- Modal coordinates, velocities and accelerations data is to be plotted and saved to the response file as designated by 'MCB', 'MVB' and 'MAB', respectively.

Note: Modal coordinates, velocities and accelerations must be saved to the response file in order to retrieve loads from the response file. This requires that the user specify these options unless the 'GLOB' load option is used. This data is automatically saved to the response file when the 'GLOB' load option is used.

3. Two percent overall critical damping is to be used.
4. Wave time and position parameters are entered on the WAVTIM line as follows:
 - a. The water depth and mudline elevation are specified in columns 11-16 and 17-22, respectively.
 - b. The analysis time increment is input as 1.0 seconds.
 - c. Both Fast Fourier transformation options are selected by '1' and '2' in columns 61 and 67, respectively.
5. The wave height spectral density is defined using the WSPEC line as follows:
 - a. Ochi-Hubble spectrum is to be used as stipulated by 'OH' in columns 11-12.
 - b. The swell period and significant wave height are 16 and 11.8, respectively.
 - c. The time duration is to be 60.0 seconds as designated in columns 48-54.
6. Wave kinematics are to be determined at 5 horizontal grid points and 10 vertical grid points rather than at each member segment end as input on the THIST line.
7. The CURR lines are used to define the current.
14. The seeds used to generate the surface profiles are 10, 20 and 21 as entered on the RNSEED line.

The response file generated is used by the Global Loading module to determine critical time points. Elevations 0.0, -50, -150, -250, -550, -800, -1050, -1200 and -1400 will be used to determine the critical positions based on maximum shear and moment. The Global loading input file along with a description follows:

	1	2	3	4	5	6	7	8
	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890	1234567890
1	GLAOPT	EC XY						
2	WAVRSP	1		1	0.0	Y	N	Y
3	PLOT MY	TIM						
4	PLOT FX	TIM						
5	PLOT MY	STA						
6	PLOT FX	STA						
7	LOCATE	0.0	-50.0	-150.0	-250.0	-550.0	-800.0	-1050.0
8	END							-1399.9

The following describes the input lines in the Global Loading input file:

1. The GLAOPT line specifies that the structure locations are in the XY plane by 'XY' in columns 13-14.:
2. The WAVRSP line is used to input response parameters and indicates the following:
 - a. One response file is to be used as input as designated by '1' in columns 7-9.

-
- b. Only one wave approach direction, namely '0.0' is to be considered.
 - c. Critical time points will be determined by checking the elevations designated for maximum shear and maximum moment. Minimum moment and shear will not be considered.
- 3. Plots will be generated for moment about the Y axis and shear along the X axis versus time.
 - 5. Plots will be generated for moment about the Y axis and shear along the X axis versus station or elevation.
 - 7. The locations to be checked are input on the LOCATE line as 0, -50, -150, -250, -550, -800, -1050, -1200 and -1399.9.

A portion of the output listing file including the time point selection data follows.

```
*** GLOBAL LOADING ANALYSIS PARAMETERS ***  
  
LOAD SUMMATION TYPE ..... INTERNAL  
NOMENCLATURE TYPE ..... STANDARD  
PLANE SELECTED ..... XY  
AXIS LOCATION - X ..... 0.00 M  
AXIS LOCATION - Y ..... 0.00 M  
NUMBER OF STATIONS ..... 9  
NUMBER OF MEMBER GROUPS .. 38  
NUMBER OF PLATE GROUPS ... 0  
NUMBER OF JOINTS ..... 204  
NUMBER OF MEMBERS ..... 401  
NUMBER OF PLATES ..... 0  
  
WAVE RESPONSE ANALYSIS ... ON  
NUMBER OF FILES ..... 1  
TRANSIENT TIME ..... 0.000 SECS
```

***** MAXIMUM AND MINIMUM MOMENT AND SHEAR VALUES FOR ANGLE OF 0.00 DEGREES *****																		
***** MAXIMUM MOMENT *****					***** MAXIMUM SHEAR *****					***** MINIMUM MOMENT *****					***** MINIMUM SHEAR *****			
STAT	DIST	MOMENT	TIME	TIME RUN	SEED	SHEAR	TIME	TIME RUN	SEED	MOMENT	TIME	TIME RUN	SEED	SHEAR	TIME	TIME RUN	SEED	SEED
	M	KN-M	SECS	PT	NO. NO.	KN	SECS	PT	NO. NO.	KN-M	SECS	PT	NO. NO.	KN	SECS	PT	NO. NO.	NO.
1-1399.9		0.	0.0	1	1 1	0.0	0.0	1	1 1	0.	0.0	1	1 1	0.0	0.0	1	1 1	1
2-1200.0		0.	0.0	1	1 1	0.0	0.0	1	1 1	0.	0.0	1	1 1	0.0	0.0	1	1 1	1
3-1050.0		0.	0.0	1	1 1	0.0	0.0	1	1 1	0.	0.0	1	1 1	0.0	0.0	1	1 1	1
4 -800.0		0.	0.0	1	1 1	0.0	0.0	1	1 1	0.	0.0	1	1 1	0.0	0.0	1	1 1	1
5 -550.0		0.	0.0	1	1 1	0.0	0.0	1	1 1	0.	0.0	1	1 1	0.0	0.0	1	1 1	1
6 -250.0		0.	0.0	1	1 1	0.0	0.0	1	1 1	0.	0.0	1	1 1	0.0	0.0	1	1 1	1
7 -150.0		0.	0.0	1	1 1	0.0	0.0	1	1 1	0.	0.0	1	1 1	0.0	0.0	1	1 1	1
8 -50.0	370864.	17.0	18	1	1	0.0	0.0	1	1 1	0.	0.0	1	1 1	-3994.8	17.0	18	1	1
9 0.0	8439.	17.0	18	1	1	2494.8	17.0	18	1 1	-1472.	1.0	2	1 1	0.0	0.0	1	1 1	1
***** CRITICAL CONDITIONS *****																		
BASED ON : MAXIMUM MOMENT..YES MINIMUM MOMENT..NO																		
MAXIMUM SHEAR...YES MINIMUM SHEAR...NO																		
RUN SEED TIME TIME																		
NO. NO. PT SECS																		
1 1 1 0.00																		
1 1 18 17.00																		
*** MOMENT AND SHEAR REPORT ***																		
FILE NO. 1 ANGLE = 0.0 DEGREES																		
***** MAXIMUM MOMENTS ***** MAXIMUM SHEARS *****																		
STAT	DIST	NO. SEEDS	* MAXIMUM * MAXIMUM MOMENTS				* MINIMUM * MINIMUM MOMENTS				* MAXIMUM * MAXIMUM SHEARS				* MINIMUM * MINIMUM SHEARS			
			MOMENT SEED	MOMENT SEED	MOMENT SEED	MOMENT SEED	MOMENT SEED	MOMENT SEED	MOMENT SEED	SHEAR SEED	SHEAR SEED	SHEAR SEED	SHEAR SEED	MOMENT SEED	MOMENT SEED	MOMENT SEED	MOMENT SEED	MOMENT SEED
	M		KN-M	KN-M	KN-M	KN-M	KN-M	KN-M	KN-M	KN	KN	KN	KN	KN	KN	KN	KN	KN
1-1399.9		1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1
2-1200.0		1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1
3-1050.0		1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1
4 -800.0		1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1
5 -550.0		1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1
6 -250.0		1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1
7 -150.0		1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0. 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1	0.0 1
8 -50.0		1	370864. 1	0. 1	370864. 1	0. 1	370864. 1	0. 1	370864. 1	0.0 1	-3994.8 1	0.0 1	-3994.8 1	0.0 1	-3994.8 1	0.0 1	-3994.8 1	-3994.8 1
9 0.0		1	8439. 1	-1472. 1	8439. 1	-1472. 1	8439. 1	-1472. 1	8439. 1	2494.8 1	0.0 1	0.0 1	2494.8 1	0.0 1	2494.8 1	0.0 1	2494.8 1	0.0 1

From the Global Loading output, only three time points are required to define load cases representing the maximum shear and moment cases at the desired elevations, namely Seed 1 time point 60, seed 2 time point 58 and seed 2 time point 59. A load case consisting of equivalent static loads will be retrieved from the response file for each of the critical cases.

Note: The critical time point information is saved in the time selection file created by the Global Loading program. This file may be specified when creating the retrieve load run file or the time points may be specified by the user.

The Wave Response input file was modified in order to create the loading and is shown below along with a description of each line.

	1								2								3								4								5								6								7								8							
	12345678901234567890123456789012345678901234567890123456789012345678901234567890																																																															
1	WROPT	RLALL ES																20								-1								TG																														
2	DAMP	2.0																																																														
3	WAVTIM	+Z1402.0-1400.0.0								0.00								1.00								VS 1								2																														
4	WSPEC	OH 11.80 16.00								1.0								4.0 5.00								60.								6.0								3.0																						
5	THIST	5																																																														
6	CURR																																																															
7	CURR	0.0								0.6								0.0																																														
8	CURR	1150.								0.6																																																						
9	CURR	1200.								1.9																																																						
10	CURR	1390.								1.9																																																						
11	CURR	1395.								3.7																																																						
12	CURR	1400.								3.7																																																						
13	PTSEED	60. 1								58. 2								59. 2																																														
14	END																																																															

1. The WROPT line specifies the Wave Response options as follows:

- 'RL' in columns 13-14 designates that loading is to be retrieved from the response file.
- Equivalent static loads are to be created for all time points designated by 'ES' in columns 19-20.

2. The PTSEED line designates the time points for each seed that correspond to critical points for load case creation. Time point 60 from seed 1 along with time points 58 and 59 from seed 2 are designated.

Note: The PTSEED line is optional. If no TIMSEL, PTSEL or PTSEED line is specified, a time selection file may be used when creating the runfile.

5 INPUT LINES

INDIVIDUAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE. IF A SINGLE PASS ANALYSIS IS BEING USED, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING VALUES IN THESE MODAL VALUES. THIS LINE IS REQUIRED ONLY IF DIFFERENT MODES HAVE DIFFERENT DAMPING VALUES. EACH MODE SHOULD HAVE A NON-ZERO POSITIVE VALUE FOR THE PERCENT CRITICAL DAMPING. IF MORE THAN 10 MODES ARE BEING USED IN THE ANALYSIS, USE ADDITIONAL LINES TO SPECIFY ALL THE MODAL DAMPING VALUES.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(16-60)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES.

[illegible]

OVERALL MODAL DAMPING CONTROL

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS AUTOMATICALLY ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE. IF A SINGLE PASS ANALYSIS IS BEING USED, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING VALUES IN THESE MODAL VALUES. IF THE DAMPING VALUES ARE THE SAME FOR ALL MODES, THIS VALUE CAN BE INPUT USING THIS LINE AND SKIPPING THE INDIVIDUAL MODAL DAMPING LINE. OTHERWISE LEAVE THE DAMPING VALUE BLANK IN THIS LINE AND ENTER EACH MODAL DAMPING VALUE ON THE INDIVIDUAL MODAL DAMPING LINE.
(11-15)	ENTER THE OVERALL MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES HAVE THE SAME VALUE. OTHERWISE LEAVE THIS FIELD BLANK.

LINE LABEL	OVERALL MODAL DAMPING VALUE	LEAVE BLANK
DAMP		
1-- 4	11<--15	16-----80
DEFAULT		
ENGLISH	PERCENT	
METRIC	PERCENT	

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS THE LAST LINE OF THE INPUT FILE.
(1- 3)	ENTER 'END'.

LINE LABEL	LEAVE THIS FIELD BLANK
END	
1-- 3	4-----80

MODE SELECTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT MODES TO BE USED IN THE ANALYSIS. THE NUMBER OF MODES SELECTED MUST AGREE WITH THE NUMBER ENTERED IN COLUMNS 41-45 OF THE 'WROPT' LINE.
(7- 9)	ENTER THE FIRST MODE SELECTION (THIS VALUE MUST BE GREATER THAN ZERO AND LESS THAN OR EQUAL TO THE HIGHEST MODE SELECTED IN THE DYNPAC ANALYSIS.
(10-12)	ENTER THE SECOND MODE.
(13-78)	ENTER THE REMAINING MODES DESIRED. IF THE NUMBER OF MODES DESIRED IS GREATER THAN 24, CONTINUE ON A SECOND 'MODSEL' LINE.

LINE LABEL	MODE SELECTIONS																							
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH	11TH	12TH	13TH	14TH	15TH	16TH	17TH	18TH	19TH	20TH	21TH	22TH	23TH	24TH
MODSEL																								
1-- 6	7--> 9	10-->12	13-->15	16-->18	19-->21	22-->24	25-->27	28-->30	31-->33	34-->36	37-->39	40-->42	43-->45	46-->48	49-->51	52-->54	55-->57	58-->60	61-->63	64-->66	67-->69	70-->72	73-->75	76-->78

PLOT SELECTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE 'PSL' OPTION WAS SELECTED ON THE 'WROPT' LINE. ENTRIES ON THIS LINE ALLOW THE USER TO SELECT VARIOUS VARIABLES TO BE PLOTTED AND/OR OUTPUT ON A FORMATTED DATA FILE. IN THE CASE OF THE PLOTTED OUTPUT, A NEUTRAL PICTURE FILE IS CREATED FOR SUBSEQUENT PROCESSING FOR PARTICULAR PLOTTERS.
(7- 8)	ENTER ANY ONE OF THE FOLLOWING PLOT/DATA SELECTION OPTIONS: 'SP' - RANDOM WAVE SURFACE PROFILE (AVAILABLE FOR RANDOM WAVE OPTION ONLY) 'GF' - GENERALIZED FORCES 'MC' - MODAL COORDINATES 'MV' - MODAL VELOCITIES 'MA' - MODAL ACCELERATIONS 'JO' - JOINT DISPLACEMENTS (REQUIRES ADDITIONAL 'PSJO' LINE) 'MF' - MEMBER FORCES (REQUIRES ADDITIONAL 'PSMF' LINE) 'OM' - OVERTURNING MOMENT 'BS' - BASE SHEAR 'HF' - TOTAL HYDRODYNAMIC FORCES 'FE' - HYDRODYNAMIC FORCES AT SPECIFIED ELEVATIONS (SEE 'ELVSEL' LINE) 'WS' - WIND SPECTRUM 'WV' - WIND VELOCITIES 'CS' - PRINT OPTION FOR COMPARISON OF MODAL STATIC BASE SHEAR TO TOTAL HYDRODYNAMIC FORCES. 'IF' - INPUT FORCE SUMMATION (TIME HISTORY ONLY) NOTE: THE 'CS' OPTION DOES NOT PRODUCE EITHER PLOTS OR A DATA FILE.

COLUMNS	COMMENTARY
(9)	ENTER THE DESIRED TYPE OF OUTPUT FOR THE SELECTED PARAMETER: LEAVE BLANK FOR PLOTS ONLY ENTER 'D' FOR OUTPUT ON FORMATTED DATA FILE ONLY ENTER 'B' FOR BOTH PLOTTER OUTPUT AND FORMATTED FILE OUTPUT
(10-51)	ENTER ANY OR ALL OF THE AVAILABLE PLOT/DATA SELECTIONS FROM THE ABOVE LIST.
(52-56)	ENTER THE PLOT SIZE FOR THE X-AXIS (ABSCISSA).
(57-61)	ENTER THE PLOT SIZE FOR THE Y-AXIS (ORDINATE).
(62-66)	ENTER THE CHARACTER SIZE TO BE USED IN THE PLOT LABELS.
(67-68)	ENTER THE NUMBER OF PENS AVAILABLE FOR THE PLOTTER.
(69)	ENTER 'S' IF EACH DATA POINT ON A TRANSFER FUNCTION PLOT IS TO BE MARKED WITH A SYMBOL.

LINE LABEL	PLOT SELECTIONS																														PLOT PARAMETERS				
	SEL 1		SEL 2		SEL 3		SEL 4		SEL 5		SEL 6		SEL 7		SEL 8		SEL 9		SEL 10		SEL 11		SEL 12		SEL 13		SEL 14		SEL 15		X SIZE	Y SIZE	CHAR. SIZE	NUMB. PENS	DATA SYMBOL OPTION
	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T					
PSEL																																			
1-- 4	7-- 8	9	10--11	12	13--14	15	16--17	18	19--20	21	22--23	24	25--26	27	28--29	30	31--32	33	34--35	36	37--38	39	40--41	42	43--44	45	46--47	48	49--50	51	52<--56	57<--61	62<--66	67-->68	69
DEFAULT																															11.0 ENGL	8.5 ENGL	0.10 ENGL	1	
ENGLISH																															IN	IN	IN		
METRIC																															CM	CM	CM		

JOINT DOF SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE 'JO' OPTION WAS SELECTED ON THE 'PSEL' LINE. ENTRIES ON THIS LINE ALLOW THE USER TO SELECT VARIOUS DEGREES OF FREEDOM OF JOINTS TO BE PLOTTED AND/OR OUTPUT ON A FORMATTED DATA FILE. THE USER CAN SELECT SEVERAL DEGREES OF FREEDOM FROM THE SAME JOINT IF DESIRED. A MAXIMUM OF 10 SELECTIONS IS ALLOWED.
(7-10)	ENTER THE JOINT NAME FOR THIS SELECTION.
(11-12)	ENTER ANY ONE OF THE FOLLOWING DEGREE OF FREEDOM OPTIONS: 'DX' - DISPLACEMENT IN THE GLOBAL X-DIRECTION 'DY' - DISPLACEMENT IN THE GLOBAL Y-DIRECTION 'DZ' - DISPLACEMENT IN THE GLOBAL Z-DIRECTION 'RX' - ROTATION ABOUT THE GLOBAL X-AXIS 'RY' - ROTATION ABOUT THE GLOBAL Y-AXIS 'RZ' - ROTATION ABOUT THE GLOBAL Z-AXIS 'AX' - ACCELERATION IN THE GLOBAL X-AXIS 'AY' - ACCELERATION IN THE GLOBAL Y-AXIS 'AZ' - ACCELERATION IN THE GLOBAL Z-AZIS 'VX' - VELOCITY IN THE GLOBAL X-AXIS 'VY' - VELOCITY IN THE GLOBAL Y-AYSIS 'VZ' - VELOCITY IN THE GLOBAL Z-AZIS
(13-66)	ENTER ANY REMAINING JOINT DEGREE OF FREEDOM SELECTIONS.

LINE LABEL	JOINT DISPLACEMENT SELECTIONS																			
	1ST		2ND		3RD		4TH		5TH		6TH		7TH		8TH		9TH		10TH	
	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F
PSJO																				
1-- 4	7-->10	11--12	13-->16	17--18	19-->22	23--24	25-->28	29--30	31-->34	35--36	37-->40	41--42	43-->46	47--48	49-->52	53--54	55-->58	59--60	61-->64	65--66

MEMBER FORCE SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE 'MF' OPTION WAS SELECTED ON THE 'PSEL' LINE. ENTRIES ON THIS LINE ALLOW THE USER TO SELECT VARIOUS INTERNAL LOADS OF MEMBERS TO BE PLOTTED AND/OR OUTPUT ON A FORMATTED DATA FILE. THE USER CAN SELECT SEVERAL INTERNAL LOADS FROM THE SAME MEMBER IF DESIRED. A MAXIMUM OF NINETY NINE SELECTIONS IS ALLOWED USING MULTIPLE LINES.
(7-10)	ENTER THE FIRST JOINT NAME FOR THIS MEMBER.
(11-14)	ENTER THE SECOND JOINT NAME FOR THIS MEMBER.
(15-17)	ENTER ANY ONE OF THE FOLLOWING INTERNAL LOAD OPTIONS: 'FXA' - AXIAL LOAD AT END A 'FYA' - SHEAR IN THE Y-DIRECTION AT END A 'FZA' - SHEAR IN THE Z-DIRECTION AT END A 'MXA' - TORSION AT END A 'MYA' - MOMENT ABOUT LOCAL Y-AXIS AT END A 'MZA' - MOMENT ABOUT LOCAL Z-AXIS AT END A 'FXB' - AXIAL LOAD AT END B 'FYB' - SHEAR IN THE Y-DIRECTION AT END B 'FZB' - SHEAR IN THE Z-DIRECTION AT END B 'MXB' - TORSION AT END B 'MYB' - MOMENT ABOUT LOCAL Y-AXIS AT END B 'MZB' - MOMENT ABOUT LOCAL Z-AXIS AT END B
(18-72)	ENTER ANY REMAINING MEMBER INTERNAL LOAD SELECTIONS. IF MORE THAN SIX INTERNAL LOADS ARE NEEDED, USE ADDITIONAL 'PSMF' LINES. UP TO NINETY NINE LOAD SELECTIONS MAY BE SPECIFIED.

LINE LABEL	MEMBER FORCE SELECTIONS																	
	1ST			2ND			3RD			4TH			5TH			6TH		
	JOINT A	JOINT B	INT. LOAD	JOINT A	JOINT B	INT. LOAD	JOINT A	JOINT B	INT. LOAD	JOINT A	JOINT B	INT. LOAD	JOINT A	JOINT B	INT. LOAD	JOINT A	JOINT B	INT. LOAD
PSMF																		
1-- 4	7-->10	11-->14	15--17	18-->21	22-->25	26--28	29-->32	33-->36	37--39	40-->43	44-->47	48--50	51-->54	55-->58	59--61	62-->65	66-->69	70--72

WAVE RESPONSE OPTIONS

COLUMNS	SYNOPSIS
GENERAL	THIS LINE WHICH SPECIFIES ANALYSIS PARAMETERS AND THE OUTPUT LOADS DESIRED, IS REQUIRED IN ANY WAVE RESPONSE RUN.
(7)	ENTER 'S' TO COMPUTE DYNAMIC SUPERELEMENT FORCES.
(8 - 9)	SELECT THE UNITS AS: 'EN' - ENGLISH UNITS 'MN' - METRIC WITH KILONEWTION FORCE 'ME' - METRIC WITH KILOGRAM FORCE
(10-12)	ENTER 'PSL' TO CREATE PLOTS. IF THIS OPTION IS SELECTED, A 'PSEL' OR 'PLTTF' LINE WILL IMMEDIATELY FOLLOW.
(13-14)	ENTER 'RI' TO CREATE LOADING FROM TIME POINTS SPECIFIED ON 'PTSEL' LINES. THIS OPTION REQUIRES A RESPONSE FILE AS INPUT.
(15-18)	ENTER THE CRITERIA FOR OUTPUT LOAD CASE SELECTION: 'ALL' - ALL TIME POINTS. 'GLOB' - TIME POINTS DETERMINED AUTOMATICALLY BY GLOBAL LOADING. 'MAXM' - MAXIMUM OVERTURNING MOMENT. 'MAXS' - MAXIMUM BASE SHEAR. 'M-NM' - MAX. - MIN. OVERTURNING MOMENT. 'M-NS' - MAX. - MIN. BASE SHEAR. 'MM*M' - * POINTS AT MAX. AND MIN. OVERTURNING MOMENT POSITIONS WHERE * IS AN INTEGER FROM 1 TO 9. 'MM*S' - * POINTS AT MAX. AND MIN. BASE SHEAR POSITIONS. 'MK*M' - SKIP EVERY * POINTS INCLUDING THE MAX. OTM. 'MK*S' - SKIP EVERY * POINTS INCLUDING THE MAX. BS. 'WAVE' - USE WAVE CREST POSITION OPTION SPECIFIED ON THE WAVE DEFINITION. 'MST' - MAX & MIN OTM, BS, & TORSION. NOTE: SELECTED FROM ESL GENERATED USING 'ALL' OPTION.
(19-20)	ENTER THE EQUIVALENT STATIC LOAD OPTION. THESE INCLUDE INERTIA LOADS AND HYDRODYNAMIC LOADS FOR THE TIME POINTS SELECTED AND CAN BE SOLVED IN A STATIC ANALYSIS. 'ES' - EQUIVALENT STATIC LOADS ARE TO BE GENERATED (USED WITH DYN OPTION IN SEASTATE). 'EX' - EXTREME WAVE EQUIVALENT STATIC LOADS (USED WITH DWY OPTION IN SEASTATE). 'EI' - EQUIVALENT STATIC INERTIA LOADS ONLY. 'US' - FATIGUE ANALYSIS EQUIVALENT STATIC LOADS. 'MR' - CREATE COMBINE INPUT WITH MODAL PARTICIPATION FACTORS.
NOTE:	THE 'US' OPTION ELIMINATES SORTING LOADS INTO LOAD CASES WHICH CAN SIGNIFICANTLY REDUCE ANALYSIS RUNTIME.
(21-30)	ENTER THE PERCENT RMS ERROR ALLOWED ON GENERALIZED FORCES BETWEEN ITERATIONS.
(31-40)	ENTER THE FOURIER SERIES CONVERGENCE TOLERANCE. THIS TOLERANCE IS USED TO DETERMINE THE NUMBER OF TERMS IN THE FOURIER SERIES REPRESENTATION OF THE GENERALIZED FORCES.
(41-45)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. DEFAULT IS THE NUMBER OF MODES IN THE MODAL SOLUTION FILE.

COLUMNS	COMMENTARY
(46-50)	ENTER THE NUMBER OF ALLOWABLE ITERATIONS TO INCLUDE RELATIVE VELOCITY AND ACCELERATION BETWEEN FLUID AND STRUCTURE. ENTER '-1' IF NO RELATIVE VELOCITY OR ACCELERATION IS TO BE INCLUDED IN THIS ANALYSIS (FLUID VERSUS STRUCTURE). THIS OPTION IS NORMALLY SELECTED FOR FATIGUE ANALYSIS.
(52-55)	ENTER 'STAT' IF MODAL STATIC LOADS ARE TO BE PRODUCED. THIS OPTION IS GENERALLY NOT USED EXCEPT IN SPECIAL APPLICATIONS.
(56-57)	ENTER 'MR' IF MODAL RESPONSES ARE TO BE INCLUDED IN THE OUTPUT.
(58-59)	ENTER 'GP' IF GAP ELEMENTS ARE TO BE INCLUDED IN THIS ANALYSIS.
(60)	ENTER 'N' IF NO RELATIVE VELOCITY EFFECTS ARE TO BE INCLUDED. APPLICABLE FOR TIME HISTORY INTEGRATION ONLY.
(61-64)	ENTER 'BUOY' IF THE EFFECTS OF CHANGES IN BUOYANCY ARE TO BE INCLUDED IN THIS ANALYSIS. THESE EFFECTS ARE NOT NORMALLY IMPORTANT EXCEPT IN CASES OF FLOATING STRUCTURES.
(65)	ENTER 'A' IF THE CUTOFF ELEVATION FOR RANDOM WAVES IS TO BE DETERMINED AUTOMATICALLY USING THE RATIO ENTERED IN COLUMNS 70-75.
(66-67)	ENTER 'AO' TO EXCLUDE THE CONSTANT TERM IN THE FOURIER ANALYSIS. THIS IS USUALLY ONLY REQUIRED FOR FLOATING STRUCTURES.
(68-69)	LEAVE BLANK FOR DETERMINISTIC WAVES OR ENTER 'RW' FOR RANDOM WAVE ANALYSIS, 'TH' FOR TIME HISTORY INTEGRATION OR 'TG' FOR TIME HISTORY INTEGRATION WITH WAVE KINEMATICS CALCULATED AT GRID POINTS INSTEAD OF AT EACH END OF EACH SEGMENT, 'WM' FOR WINDMILL OR 'WL' FOR WIND LOAD OPTION.
NOTE:	'WAVTIM' LINE REQUIRED WHEN USING 'RW', 'TH', 'TG', 'WM' OR 'WL' OPTION. 'WM' OPTION SHOULD ONLY BE USED FOR GENERATING GENERALIZED FORCES FOR SACS-FAST ANALYSIS, WHILE 'WL' OPTION SHOULD ONLY BE USED FOR INCLUDING LOADS FROM SACS-FAST RUNS IN WAVE RESPONSE ANALYSIS.
(70-75)	ENTER THE ELEVATION CUTOFF BELOW WHICH NO WAVE KINEMATICS WILL BE CALCULATED. IF LEFT BLANK OR ZERO WAVE KINEMATICS WILL BE CALCULATED FROM THE MUDLINE TO THE SURFACE ELEVATION. IF THE AUTOMATIC CUTOFF LIMIT OPTION IS USED ('A' IN COLUMN 65), ENTER THE VELOCITY RATIO USED TO DETERMINE THE CUTOFF ELEVATION. THIS IS THE RATIO OF HORIZONTAL VELOCITY OVER THE HORIZONTAL VELOCITY AT THE WATER SURFACE. DEFAULT IS 0.05 WHICH YIELDS A DRAG RATIO OF 0.0025.
(76-80)	ENTER INERTIA LOAD FACTOR FOR EQUIVALENT STATIC LOADS. ONLY EFFECTS OUTPUT INERTIA LOADS AND NORMALLY APPLIES TO LRFD CODES.
(81)	ENTER 'C' TO USE CATEGORY SORT FOR LOAD LINES.

[illegible]

INDIVIDUAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE. IF A SINGLE PASS ANALYSIS IS BEING USED, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING VALUES IN THESE MODAL VALUES. THIS LINE IS REQUIRED ONLY IF DIFFERENT MODES HAVE DIFFERENT DAMPING VALUES. EACH MODE SHOULD HAVE A NON-ZERO POSITIVE VALUE FOR THE PERCENT CRITICAL DAMPING. IF MORE THAN 10 MODES ARE BEING USED IN THE ANALYSIS, USE ADDITIONAL LINES TO SPECIFY ALL THE MODAL DAMPING VALUES.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(16-60)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES.

[illegible]

OVERALL MODAL DAMPING CONTROL

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS AUTOMATICALLY ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE. IF A SINGLE PASS ANALYSIS IS BEING USED, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING VALUES IN THESE MODAL VALUES. IF THE DAMPING VALUES ARE THE SAME FOR ALL MODES, THIS VALUE CAN BE INPUT USING THIS LINE AND SKIPPING THE INDIVIDUAL MODAL DAMPING LINE. OTHERWISE LEAVE THE DAMPING VALUE BLANK IN THIS LINE AND ENTER EACH MODAL DAMPING VALUE ON THE INDIVIDUAL MODAL DAMPING LINE.
(11-15)	ENTER THE OVERALL MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES HAVE THE SAME VALUE. OTHERWISE LEAVE THIS FIELD BLANK.

LINE LABEL	OVERALL MODAL DAMPING VALUE	LEAVE BLANK
DAMP		
1-- 4	11<--15	16-----80
DEFAULT		
ENGLISH	PERCENT	
METRIC	PERCENT	

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS THE LAST LINE OF THE INPUT FILE.
(1- 3)	ENTER 'END'.

LINE LABEL	LEAVE THIS FIELD BLANK
END	
1-- 3	4-----80

MODE SELECTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT MODES TO BE USED IN THE ANALYSIS. THE NUMBER OF MODES SELECTED MUST AGREE WITH THE NUMBER ENTERED IN COLUMNS 41-45 OF THE 'WROPT' LINE.
(7- 9)	ENTER THE FIRST MODE SELECTION (THIS VALUE MUST BE GREATER THAN ZERO AND LESS THAN OR EQUAL TO THE HIGHEST MODE SELECTED IN THE DYNPAC ANALYSIS.
(10-12)	ENTER THE SECOND MODE.
(13-78)	ENTER THE REMAINING MODES DESIRED. IF THE NUMBER OF MODES DESIRED IS GREATER THAN 24, CONTINUE ON A SECOND 'MODSEL' LINE.

LINE LABEL	MODE SELECTIONS																							
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH	11TH	12TH	13TH	14TH	15TH	16TH	17TH	18TH	19TH	20TH	21TH	22TH	23TH	24TH
MODSEL																								
1-- 6	7--> 9	10-->12	13-->15	16-->18	19-->21	22-->24	25-->27	28-->30	31-->33	34-->36	37-->39	40-->42	43-->45	46-->48	49-->51	52-->54	55-->57	58-->60	61-->63	64-->66	67-->69	70-->72	73-->75	76-->78

TRANSFER AND RESPONSE FUNCTION PLOT SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS OPTIONAL AND IS USED ONLY FOR THE DETERMINISTIC OPTION TO CREATE PLOTS OF TRANSFER AND RESPONSE FUNCTIONS DEFINED BY A SERIES OF WAVES OF DECREASING PERIODS IN THE SEASTATE DATA. THIS LINE IS USED IN CONJUNCTION WITH THE 'TFLCAS' LINE(S) TO DEFINE WHICH SEASTATE LOAD CASES DEFINE THE TRANSFER FUNCTIONS AND 'WSPEC' LINE TO DESIGNATE WHICH WAVE SPECTRUM IS TO BE USED FOR THE RESPONSE FUNCTION.
(6)	ENTER 'F' IF THE TRANSFER AND/OR RESPONSE PLOTS DATA IS TO BE INCLUDED IN THE OUTPUT DATA FILE.
(7 - 8)	ENTER ANY ONE OF THE FOLLOWING PLOT/DATA SELECTION OPTIONS: 'GF' - GENERALIZED FORCES 'MC' - MODAL COORDINATES 'MV' - MODAL VELOCITIES 'MA' - MODAL ACCELERATIONS 'JO' - JOINT DISPLACEMENTS (REQUIRES ADDITIONAL 'PSJO' LINE) 'MF' - MEMBER FORCES (REQUIRES ADDITIONAL 'PSMF' LINE) 'OM' - OVERTURNING MOMENT 'BS' - BASE SHEAR
(9)	ENTER THE DESIRED TYPE OF OUTPUT FOR THE SELECTED PARAMETER: LEAVE BLANK FOR TRANSFER FUNCTION ONLY ENTER 'D' FOR RESPONSE FUNCTION ONLY ENTER 'B' FOR BOTH
(10-51)	ENTER ANY OR ALL OF THE AVAILABLE PLOT/DATA SELECTIONS FROM THE ABOVE LIST.
(52-53)	ENTER THE INDEPENDENT VARIABLE FOR THE TRANSFER FUNCTION: 'FQ' - FREQUENCY 'PR' - PERIOD 'PF' - FREQUENCY AND PERIOD (2 PLOTS FOR EACH FUNCTION)
(54)	ENTER 'S' IF SYMBOLS ARE TO BE PLOTTED AT EACH POINT ON THE TRANSFER FUNCTION CURVE.

[illegible]

PLOT SELECTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE 'PSL' OPTION WAS SELECTED ON THE 'WROPT' LINE. ENTRIES ON THIS LINE ALLOW THE USER TO SELECT VARIOUS VARIABLES TO BE PLOTTED AND/OR OUTPUT ON A FORMATTED DATA FILE. IN THE CASE OF THE PLOTTED OUTPUT, A NEUTRAL PICTURE FILE IS CREATED FOR SUBSEQUENT PROCESSING FOR PARTICULAR PLOTTERS.
(7- 8)	ENTER ANY ONE OF THE FOLLOWING PLOT/DATA SELECTION OPTIONS: 'SP' - RANDOM WAVE SURFACE PROFILE (AVAILABLE FOR RANDOM WAVE OPTION ONLY) 'GF' - GENERALIZED FORCES 'MC' - MODAL COORDINATES 'MV' - MODAL VELOCITIES 'MA' - MODAL ACCELERATIONS 'JO' - JOINT DISPLACEMENTS (REQUIRES ADDITIONAL 'PSJO' LINE) 'MF' - MEMBER FORCES (REQUIRES ADDITIONAL 'PSMF' LINE) 'OM' - OVERTURNING MOMENT 'BS' - BASE SHEAR 'HF' - TOTAL HYDRODYNAMIC FORCES 'FE' - HYDRODYNAMIC FORCES AT SPECIFIED ELEVATIONS (SEE 'ELVSEL' LINE) 'WS' - WIND SPECTRUM 'WV' - WIND VELOCITIES 'CS' - PRINT OPTION FOR COMPARISON OF MODAL STATIC BASE SHEAR TO TOTAL HYDRODYNAMIC FORCES. 'IF' - INPUT FORCE SUMMATION (TIME HISTORY ONLY) NOTE: THE 'CS' OPTION DOES NOT PRODUCE EITHER PLOTS OR A DATA FILE.

COLUMNS	COMMENTARY
(9)	ENTER THE DESIRED TYPE OF OUTPUT FOR THE SELECTED PARAMETER: LEAVE BLANK FOR PLOTS ONLY ENTER 'D' FOR OUTPUT ON FORMATTED DATA FILE ONLY ENTER 'B' FOR BOTH PLOTTER OUTPUT AND FORMATTED FILE OUTPUT
(10-51)	ENTER ANY OR ALL OF THE AVAILABLE PLOT/DATA SELECTIONS FROM THE ABOVE LIST.
(52-56)	ENTER THE PLOT SIZE FOR THE X-AXIS (ABSCISSA).
(57-61)	ENTER THE PLOT SIZE FOR THE Y-AXIS (ORDINATE).
(62-66)	ENTER THE CHARACTER SIZE TO BE USED IN THE PLOT LABELS.
(67-68)	ENTER THE NUMBER OF PENS AVAILABLE FOR THE PLOTTER.
(69)	ENTER 'S' IF EACH DATA POINT ON A TRANSFER FUNCTION PLOT IS TO BE MARKED WITH A SYMBOL.

LINE LABEL	PLOT SELECTIONS																														PLOT PARAMETERS				
	SEL 1		SEL 2		SEL 3		SEL 4		SEL 5		SEL 6		SEL 7		SEL 8		SEL 9		SEL 10		SEL 11		SEL 12		SEL 13		SEL 14		SEL 15		X SIZE	Y SIZE	CHAR. SIZE	NUMB. PENS	DATA SYMBOL OPTION
	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T					
PSEL																																			
1-- 4	7-- 8	9	10--11	12	13--14	15	16--17	18	19--20	21	22--23	24	25--26	27	28--29	30	31--32	33	34--35	36	37--38	39	40--41	42	43--44	45	46--47	48	49--50	51	52<--56	57<--61	62<--66	67-->68	69
DEFAULT																															11.0 ENGL	8.5 ENGL	0.10 ENGL	1	
ENGLISH																															IN	IN	IN		
METRIC																															CM	CM	CM		

TRANSFER FUNCTION WAVE SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT THE SEASTATE WAVES THAT ARE TO BE USED TO CREATE TRANSFER FUNCTIONS PLOTS. DO NOT USE THIS LINE IF RESPONSE FUNCTION PLOTS ARE GENERATED. NOTE: TRANSFER FUNCTIONS AND RESPONSE FUNCTIONS PLOTS ARE GENERATED USING THE 'WSPEC' LINE.
(7-14)	ENTER THE FIRST WAVE NUMBER AND THE LAST WAVE NUMBER TO BE USED TO CREATE THE 1ST TRANSFER FUNCTION. THESE SEASTATE WAVES MUST HAVE PERIODS IN DESCENDING ORDER AND MUST BE IN THE SAME DIRECTION.
(17-64)	ENTER THE FIRST WAVE NUMBER AND THE LAST WAVE NUMBER FOR ADDITIONAL TRANSFER FUNCTIONS. REPEAT THIS RECORD FOR A MAXIMUM OF 20 TRANSFER FUNCTIONS.

LINE LABEL	1ST TRANSFER FUNCTION		2ND TRANSFER FUNCTION		3RD TRANSFER FUNCTION		4TH TRANSFER FUNCTION		5TH TRANSFER FUNCTION		6TH TRANSFER FUNCTION		LEAVE BLANK
	FIRST WAVE NUMBER	LAST WAVE NUMBER	FIRST WAVE NUMBER	LAST WAVE NUMBER	FIRST WAVE NUMBER	LAST WAVE NUMBER	FIRST WAVE NUMBER	LAST WAVE NUMBER	FIRST WAVE NUMBER	LAST WAVE NUMBER	FIRST WAVE NUMBER	LAST WAVE NUMBER	
TFLCAS													
1-- 6	7-->10	11-->14	17-->20	21-->24	27-->30	31-->34	37-->40	41-->44	47-->50	51-->54	57-->60	61-->64	65--80

WAVE RESPONSE OPTIONS

COLUMNS	COMMENTS
GENERAL	THIS LINE WHICH SPECIFIES ANALYSIS PARAMETERS AND THE OUTPUT LOADS DESIRED, IS REQUIRED IN ANY WAVE RESPONSE RUN.
(7)	ENTER 'S' TO COMPUTE DYNAMIC SUPERELEMENT FORCES.
(8 - 9)	SELECT THE UNITS AS: 'EN' - ENGLISH UNITS 'MN' - METRIC WITH KILONEWTON FORCE 'ME' - METRIC WITH KILOGRAM FORCE
(10-12)	ENTER 'PSL' TO CREATE PLOTS. IF THIS OPTION IS SELECTED, A 'PSEL' OR 'PLTTF' LINE WILL IMMEDIATELY FOLLOW.
(13-14)	ENTER 'RL' TO CREATE LOADING FROM TIME POINTS SPECIFIED ON 'PTSEL' LINES. THIS OPTION REQUIRES A RESPONSE FILE AS INPUT.
(15-18)	ENTER THE CRITERIA FOR OUTPUT LOAD CASE SELECTION: 'ALL' - ALL TIME POINTS. 'GLOB' - TIME POINTS DETERMINED AUTOMATICALLY BY GLOBAL LOADING. 'MAXM' - MAXIMUM OVERTURNING MOMENT. 'MAXS' - MAXIMUM BASE SHEAR. 'M-NM' - MAX. - MIN. OVERTURNING MOMENT. 'M-NS' - MAX. - MIN. BASE SHEAR. 'MM*M' - * POINTS AT MAX. AND MIN. OVERTURNING MOMENT POSITIONS WHERE * IS AN INTEGER FROM 1 TO 9. 'MM*S' - * POINTS AT MAX. AND MIN. BASE SHEAR POSITIONS. 'MK*M' - SKIP EVERY * POINTS INCLUDING THE MAX. OTM. 'MK*S' - SKIP EVERY * POINTS INCLUDING THE MAX. BS. 'WAVE' - USE WAVE CREST POSITION OPTION SPECIFIED ON THE WAVE DEFINITION. 'MST' - MAX & MIN OTM, BS, & TORSION. NOTE: SELECTED FROM ESL GENERATED USING 'ALL' OPTION.
(19-20)	ENTER THE EQUIVALENT STATIC LOAD OPTION. THESE INCLUDE INERTIA LOADS AND HYDRODYNAMIC LOADS FOR THE TIME POINTS SELECTED AND CAN BE SOLVED IN A STATIC ANALYSIS. 'ES' - EQUIVALENT STATIC LOADS ARE TO BE GENERATED (USED WITH DYN OPTION IN SEASTATE). 'EX' - EXTREME WAVE EQUIVALENT STATIC LOADS (USED WITH DYN OPTION IN SEASTATE). 'EI' - EQUIVALENT STATIC INERTIA LOADS ONLY. 'US' - FATIGUE ANALYSIS EQUIVALENT STATIC LOADS. 'MR' - CREATE COMBINE INPUT WITH MODAL PARTICIPATION FACTORS.
NOTE:	THE 'US' OPTION ELIMINATES SORTING LOADS INTO LOAD CASES WHICH CAN SIGNIFICANTLY REDUCE ANALYSIS RUNTIME.
(21-30)	ENTER THE PERCENT RMS ERROR ALLOWED ON GENERALIZED FORCES BETWEEN ITERATIONS.
(31-40)	ENTER THE FOURIER SERIES CONVERGENCE TOLERANCE. THIS TOLERANCE IS USED TO DETERMINE THE NUMBER OF TERMS IN THE FOURIER SERIES REPRESENTATION OF THE GENERALIZED FORCES.
(41-45)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. DEFAULT IS THE NUMBER OF MODES IN THE MODAL SOLUTION FILE.

COLUMNS	COMMENTARY
(46-50)	ENTER THE NUMBER OF ALLOWABLE ITERATIONS TO INCLUDE RELATIVE VELOCITY AND ACCELERATION BETWEEN FLUID AND STRUCTURE. ENTER '-1' IF NO RELATIVE VELOCITY OR ACCELERATION IS TO BE INCLUDED IN THIS ANALYSIS (FLUID VERSUS STRUCTURE). THIS OPTION IS NORMALLY SELECTED FOR FATIGUE ANALYSIS.
(52-55)	ENTER 'STAT' IF MODAL STATIC LOADS ARE TO BE PRODUCED. THIS OPTION IS GENERALLY NOT USED EXCEPT IN SPECIAL APPLICATIONS.
(56-57)	ENTER 'MR' IF MODAL RESPONSES ARE TO BE INCLUDED IN THE OUTPUT.
(58-59)	ENTER 'GP' IF GAP ELEMENTS ARE TO BE INCLUDED IN THIS ANALYSIS.
(60)	ENTER 'N' IF NO RELATIVE VELOCITY EFFECTS ARE TO BE INCLUDED. APPLICABLE FOR TIME HISTORY INTEGRATION ONLY.
(61-64)	ENTER 'BUOY' IF THE EFFECTS OF CHANGES IN BUOYANCY ARE TO BE INCLUDED IN THIS ANALYSIS. THESE EFFECTS ARE NOT NORMALLY IMPORTANT EXCEPT IN CASES OF FLOATING STRUCTURES.
(65)	ENTER 'A' IF THE CUTOFF ELEVATION FOR RANDOM WAVES IS TO BE DETERMINED AUTOMATICALLY USING THE RATIO ENTERED IN COLUMNS 70-75.
(66-67)	ENTER 'AO' TO EXCLUDE THE CONSTANT TERM IN THE FOURIER ANALYSIS. THIS IS USUALLY ONLY REQUIRED FOR FLOATING STRUCTURES.
(68-69)	LEAVE BLANK FOR DETERMINISTIC WAVES OR ENTER 'RW' FOR RANDOM WAVE ANALYSIS, 'TH' FOR TIME HISTORY INTEGRATION OR 'TG' FOR TIME HISTORY INTEGRATION WITH WAVE KINEMATICS CALCULATED AT GRID POINTS INSTEAD OF AT EACH END OF EACH SEGMENT, 'WM' FOR WINDMILL OR 'WL' FOR WIND LOAD OPTION.
NOTE:	'WAVTIM' LINE REQUIRED WHEN USING 'RW', 'TH', 'TG', 'WM' OR 'WL' OPTION. 'WM' OPTION SHOULD ONLY BE USED FOR GENERATING GENERALIZED FORCES FOR SACS-FAST ANALYSIS, WHILE 'WL' OPTION SHOULD ONLY USED FOR INCLUDING LOADS FROM SACS-FAST RUNS IN WAVE RESPONSE ANALYSIS.
(70-75)	ENTER THE ELEVATION CUTOFF BELOW WHICH NO WAVE KINEMATICS WILL BE CALCULATED. IF LEFT BLANK OR ZERO WAVE KINEMATICS WILL BE CALCULATED FROM THE MUDLINE TO THE SURFACE ELEVATION. IF THE AUTOMATIC CUTOFF LIMIT OPTION IS USED ('A' IN COLUMN 65), ENTER THE VELOCITY RATIO USED TO DETERMINE THE CUTOFF ELEVATION. THIS IS THE RATIO OF HORIZONTAL VELOCITY OVER THE HORIZONTAL VELOCITY AT THE WATER SURFACE. DEFAULT IS 0.05 WHICH YIELDS A DRAG RATIO OF 0.0025.
(76-80)	ENTER INERTIA LOAD FACTOR FOR EQUIVALENT STATIC LOADS. ONLY EFFECTS OUTPUT INERTIA LOADS AND NORMALLY APPLIES TO LRFD CODES.
(81)	ENTER 'C' TO USE CATEGORY SORT FOR LOAD LINES.

[illegible]

PIERSON-MOSKOWITZ/JONSWAP WAVE SPECTRAL DENSITY

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED IF AND ONLY IF A SPECTRAL WAVE RESPONSE ANALYSIS IS BEING DONE. IT IS USED TO DESIGNATE THE FORM OF THE WAVE HEIGHT SPECTRAL DENSITY FUNCTION. THIS LINE FOLLOWS IMMEDIATELY AFTER THE 'WAVTIM' LINE.
(1- 5)	ENTER 'WSPEC'.
(6- 7)	IF NUMEROUS SURFACE PROFILES USING DIFFERENT SEEDS ARE TO BE GENERATED, ENTER THE TOTAL NUMBER OF SEEDS. THE SEEDS WILL BE CALCULATED BASED OF THE FIRST SEED ENTERED.
(8-10)	THIS IS THE SEED FOR A BUILT-IN PSEUDO-RANDOM NUMBER GENERATOR WHICH IS USED TO CALCULATE THE PHASE ANGLES FOR THE VARIOUS AIRY WAVE COMPONENTS. IF THE SAME ANALYSIS IS RUN ON THE SAME COMPUTER THEN THE RESULTS WILL BE THE SAME IF THE SAME SEED IS USED. IF A DIFFERENT SEED IS USED, A DIFFERENT SET OF RANDOM PHASE ANGLES WILL BE COMPUTED. THE PRIMARY USE FOR THIS PARAMETER IS TO CHECK THAT THE DURATION OF THE SPECTRUM IS OF SUFFICIENT LENGTH TO GUARANTEE THAT THE RESULTS HAVE REPEATABLE STATISTICAL PROPERTIES. NORMALLY, THIS FIELD IS LEFT BLANK. IF A SEED IS DESIRED, ENTER AN INTEGER FROM 1 TO 999.
(11-12)	ENTER THE TYPE OF SPECTRUM TO BE USED FOR THE WAVE HEIGHT SPECTRAL DENSITY FUNCTION. CHOOSE FROM BETWEEN THE FOLLOWING: . 'PM'...PIERSON-MOSKOWITZ SPECTRUM. THIS IS THE DEFAULT. 'JS'...JONSWAP SPECTRUM. .
(13-19)	ENTER THE "SIGNIFICANT WAVE HEIGHT" FOR THIS SPECTRUM.
(20-26)	ENTER THE "DOMINANT PERIOD" FOR THIS SPECTRUM.

COLUMNS	COMMENTARY
(27-33)	THE SPECTRUM IS REPRESENTED BY A SET OF AIRY WAVE COMPONENTS WHICH ARE COMBINED USING RANDOM PHASE ANGLES. THE FINENESS OF THE DIVISION IS CONTROLLED BY THIS PARAMETER. EACH WAVE COMPONENT WILL REPRESENT AT LEAST THIS SPECIFIED PERCENTAGE OF THE AREA UNDER THE SPECTRUM.
(34-47)	ENTER THE VALUES OF THE PARAMETERS "GAMMA" AND "C" REQUIRED TO FULLY DEFINE THE JONSWAP SPECTRUM IF 'JS' IS IN COLUMNS 11-12.
(48-54)	ENTER THE TIME DURATION TO BE GENERATED BY THIS SPECTRUM. A TIME HISTORY WAVE WILL BE GENERATED HAVING THE STATISTICAL PROPERTIES AS SPECIFIED BY THE SPECTRUM. THE DEFAULT IS 50 TIMES THE DOMINANT PERIOD.
(55-56)	ENTER 'RA' IF RANDOM AMPLITUDE IS TO BE USED IN ADDITION TO RANDOM PHASE ANGLES IN THE AIRY WAVE COMPONENT GENERATION.
(57-58)	ENTER 'SP' IF ONLY THE SURFACE PROFILE IS TO BE GENERATED. IF THIS OPTION IS SELECTED, THE PROGRAM WILL STOP AFTER SURFACE PROFILE IS GENERATED. THIS OPTION IS USEFUL TO CHECK IF THE TIME DURATION IS SUFFICIENT TO CREATE A SURFACE PROFILE THAT REPRESENTS THE SPECTRUM TO THE DESIRED DEGREE OF ACCURACY.
(59-62)	ENTER THE 1ST SEASTATE THAT WILL BE USED TO DEFINE TRANSFER AND/OR RESPONSE FUNCTION.
(63-66)	ENTER THE LAST SEASTATE THAT WILL BE USED TO DEFINE THE TRANSFER AND/OF RESPONSE FUNCTION.

LINE LABEL	NUMBER OF RANDOM NUMBER SEEDS	FIRST RANDOM NUMBER SEED	WAVE SPECTRUM TYPE	WAVE SPECTRUM PARAMETERS		MINIMUM PERCENT FOR EACH WAVE COMPONENT	JONSWAP PARAMETERS		TIME DURATION	RANDOM AMPLITUDE OPTION	SURFACE PROFILE GENERATION ONLY OPTION	BEGIN LOAD CASE	END LOAD CASE	LEAVE THIS FIELD BLANK
				SIGNIFICANT WAVE HEIGHT	DOMINANT PERIOD (DP)		"GAMMA"	"C"						
WSPEC														
1-- 5	6--> 7	8-->10	11--12	13<--19	20<--26	27<--33	34<--40	41<--47	48<--54	55--56	57--58	59-->62	63-->66	67--80
DEFAULT	1		'PM'			1	3.3	1.525	50 * DP					
ENGLISH				FT	SEC	PERCENT			SEC					
METRIC				M	SEC	PERCENT			SEC					

CURRENT INPUT DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO INCLUDE THE EFFECT OF A STEADY HORIZONTAL CURRENT ON THE STRUCTURAL LOADING. THIS LINE IS OPTIONAL AND ONLY APPLICABLE FOR RANDOM WAVE ANALYSIS.
(9-16)	ENTER THE ELEVATION ABOVE THE MUDLINE WHERE THIS CURRENT VALUE APPLIES. A MAXIMUM OF 10 ELEVATIONS MAY BE INPUT. THE LINES MUST BE INPUT IN ORDER OF INCREASING ELEVATION.
(17-24)	ENTER THE CURRENT VELOCITY FOR THIS ELEVATION. A LINEAR VARIATION IS ASSUMED BETWEEN ELEVATIONS. A CONSTANT CURRENT IS ASSUMED ABOVE THE HIGHEST ELEVATION.
(25-32)	ENTER THE DIRECTION OF THE CURRENT ONLY ON THE FIRST CURRENT DATA LINE. THE CURRENT DIRECTION IS THE SAME FOR ALL ELEVATIONS. THE DIRECTION IS MEASURED FROM THE X-AXIS TOWARD THE Y-AXIS FOR A VERTICAL POSITIVE Z-AXIS. FOR OTHER VERTICALS THE DIRECTION IS SIMILAR USING THE RIGHT-HAND RULE.
(41-48)	ENTER THE CURRENT BLOCKING FACTOR.

LINE LABEL	ELEVATION ABOVE MUDLINE	CURRENT VELOCITY	DIRECTION	BLOCKING FACTOR	LEAVE BLANK
CURR					
1-- 4	9<--16	17<--24	25<--32	41<--48	49-----80
DEFAULT				1	
ENGLISH	FT	KNOT	DEG		
METRIC	M	M/SEC	DEG		

INDIVIDUAL MODAL DAMPING

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE. IF A SINGLE PASS ANALYSIS IS BEING USED, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING VALUES IN THESE MODAL VALUES. THIS LINE IS REQUIRED ONLY IF DIFFERENT MODES HAVE DIFFERENT DAMPING VALUES. EACH MODE SHOULD HAVE A NON-ZERO POSITIVE VALUE FOR THE PERCENT CRITICAL DAMPING. IF MORE THAN 10 MODES ARE BEING USED IN THE ANALYSIS, USE ADDITIONAL LINES TO SPECIFY ALL THE MODAL DAMPING VALUES.
(5-5)	ENTER 'R' IF RADIATION DAMPING NEEDS TO BE ADDED TO MODAL DAMPING. NOTE: THIS WILL BE APPLIED ONLY TO THE FIRST MODE.
(11-15)	ENTER THE MODAL DAMPING AS A PERCENT OF CRITICAL FOR THE FIRST MODE.
(16-60)	ENTER MODAL DAMPING VALUES FOR THE REMAINING MODES.

[illegible]

OVERALL MODAL DAMPING CONTROL

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED TO SPECIFY THE MODAL DAMPING VALUES. THESE VALUES SHOULD INCLUDE ALL SOURCES OF DAMPING OTHER THAN THE FLUID DAMPING IF AN ITERATIVE SOLUTION IS BEING USED. IN THAT CASE THE FLUID DAMPING IS AUTOMATICALLY ACCOUNTED FOR BY CALCULATING THE ACTUAL FLUID FORCES DUE TO MOTION OF THE STRUCTURE. IF A SINGLE PASS ANALYSIS IS BEING USED, THEN THE USER SHOULD INCLUDE THE FLUID DAMPING VALUES IN THESE MODAL VALUES. IF THE DAMPING VALUES ARE THE SAME FOR ALL MODES, THIS VALUE CAN BE INPUT USING THIS LINE AND SKIPPING THE INDIVIDUAL MODAL DAMPING LINE. OTHERWISE LEAVE THE DAMPING VALUE BLANK IN THIS LINE AND ENTER EACH MODAL DAMPING VALUE ON THE INDIVIDUAL MODAL DAMPING LINE.
(5-5)	ENTER 'R' IF RADIATION DAMPING NEEDS TO BE ADDED TO MODAL DAMPING. NOTE: THIS WILL BE APPLIED ONLY TO THE FIRST MODE.
(11-15)	ENTER THE OVERALL MODAL DAMPING AS A PERCENT OF CRITICAL IF ALL MODES HAVE THE SAME VALUE. OTHERWISE LEAVE THIS FIELD BLANK.

LINE LABEL	RADIATION DAMPING	OVERALL MODAL DAMPING VALUE	LEAVE BLANK
DAMP			
1-- 4	5	11<--15	16-----80
DEFAULT			
ENGLISH		PERCENT	
METRIC		PERCENT	

ELEVATION SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY ELEVATIONS FOR HYDRODYNAMIC FORCE SUMMATIONS. THE PROGRAM WILL COLLECT ALL APPLIED FORCES BETWEEN THESE VALUES AND PRINT OUT THE SUMMATIONS. THIS LINE IS OPTIONAL AND IS AVAILABLE ONLY FOR THE RANDOM WAVE ANALYSIS. THIS DATA WILL BE INCLUDED IN THE PLOTTED OUTPUT AND, IF SELECTED ON THE 'PSEL' LINE, IT WILL BE INCLUDED ON THE FORMATTED DATA FILE.
(7-12)	ENTER THE FIRST ELEVATION RELATIVE TO THE MUDLINE THAT THE FORCES ARE TO BE SUMMED. ALL APPLIED FORCES ABOVE THE MUDLINE AND HALFWAY TO THE NEXT ELEVATION WILL BE INCLUDED IN THE SUMMATION.
(13-18)	ENTER THE SECOND ELEVATION FOR FORCE SUMMATION. ALL FORCES HALFWAY FROM THE FIRST ELEVATION TO HALFWAY TO THE NEXT ELEVATION WILL BE INCLUDED IN THE SUMMATION.
(19-78)	ENTER THE REMAINING ELEVATIONS. THE LAST ELEVATION ENTERED WILL INCLUDE ALL APPLIED FORCES ABOVE THAT LEVEL.

[illegible]

END LINE

<u>COLUMNS</u>	<u>COMMENTARY</u>
GENERAL	THIS LINE IS THE LAST LINE OF THE INPUT FILE.
(1- 3)	ENTER 'END'.

LINE LABEL	LEAVE THIS FIELD BLANK
END	
1-- 3	4-----80

JOINT DISPLACEMENT PART 1

COLUMNS	COMMENTARY
GENERAL	THIS LINE SET ENABLES THE TIME HISTORY APPLICATION OF TRANSLATIONS AND ROTATIONS ON THE JOINTS.
(1- 4)	ENTER 'JDIS' ON EACH LINE IN THIS SET.
(5- 8)	ENTER THE JOINT NAME TO WHICH DISPLACEMENTS/VELOCITIES/ACCELERATIONS ON THIS LINE ARE TO BE APPLIED.
(9-20)	TRANSLATIONAL DISPLACEMENTS IN GLOBAL X DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE X DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(21-32)	TRANSLATIONAL DISPLACEMENTS IN GLOBAL Y DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Y DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(33-44)	TRANSLATIONAL DISPLACEMENTS IN GLOBAL Z DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Z DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(45-56)	ROTATIONAL DISPLACEMENTS IN GLOBAL X DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(57-68)	ROTATIONAL DISPLACEMENTS IN GLOBAL Y DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(69-80)	ROTATIONAL DISPLACEMENTS IN GLOBAL Z DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(81-224)	SEE JLOD PART 2.

LINE LABEL	JOINT NAME	JOINT TRANSLATION AND ROTATION DATA IN GLOBAL COORDINATES						SEE JLOD LINE PART 2
		TRANSLATIONS			ROTATIONS			
		Dx	Dy	Dz	RDx	RDy	RDz	
JDIS								
1-- 4	5--> 8	9<--20	21<--32	33<--44	45<--56	57<--68	69<--80	81<--224
DEFAULT								
ENGLISH		FT	FT	FT	RADIANS	RADIANS	RADIANS	
METRIC (KN)		M	M	M	RADIANS	RADIANS	RADIANS	
METRIC (KG)		M	M	M	RADIANS	RADIANS	RADIANS	

JOINT DISPLACEMENT PART 2

COLUMNS	COMMENTARY
GENERAL	THIS LINE SET ENABLES THE TIME HISTORY APPLICATION OF TRANSLATIONS AND ROTATIONS ON THE JOINTS.
(1- 80)	SEE JLOD PART 1
(81-92)	TRANSLATIONAL VELOCITIES IN GLOBAL X DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE X DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(93-104)	TRANSLATIONAL VELOCITIES IN GLOBAL Y DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Y DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(105-116)	TRANSLATIONAL VELOCITIES IN GLOBAL Z DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Z DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(117-128)	ROTATIONALVELOCITIES IN GLOBAL X DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(129-140)	ROTATIONAL VELOCITIES IN GLOBAL Y DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(141-152)	ROTATIONAL VELOCITIES IN GLOBAL Z DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(153-224)	SEE JLOD PART 3.

LINE LABEL	SEE JLOD LINE PART 1	JOINT TRANSLATION AND ROTATION DATA IN GLOBAL COORDINATES						SEE JLOD LINE PART 3
		TRANSLATIONS			ROTATIONS			
		Vx	Vy	Vz	RVx	RVy	RVz	
JDIS								
1-- 4	5--> 80	81<--92	93<--104	105<--116	117<--128	129<--140	141<--152	153<--224
DEFAULT								
ENGLISH		FT	FT	FT	RADIANS	RADIANS	RADIANS	
METRIC (KN)		M	M	M	RADIANS	RADIANS	RADIANS	
METRIC (KG)		M	M	M	RADIANS	RADIANS	RADIANS	

JOINT DISPLACEMENT PART 3

COLUMNS	COMMENTARY
GENERAL	THIS LINE SET ENABLES THE TIME HISTORY APPLICATION OF TRANSLATIONS AND ROTATIONS ON THE JOINTS.
(1- 152)	SEE JLOD PART 2.
(153-164)	TRANSLATIONAL DISP/VEL/ACC IN GLOBAL X DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE X DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(165-176)	TRANSLATIONAL DISP/VEL/ACC IN GLOBAL Y DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Y DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(177-188)	TRANSLATIONAL DISP/VEL/ACC IN GLOBAL Z DIRECTION ON THIS JOINT. A POSITIVE DISPLACEMENT IS IN THE POSITIVE Z DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(189-200)	ROTATIONAL DISP/VEL/ACC IN GLOBAL X DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(201-212)	ROTATIONAL DISP/VEL/ACC IN GLOBAL Y DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.
(213-224)	ROTATIONAL DISP/VEL/ACC IN GLOBAL Z DIRECTION ON THIS JOINT. THE SIGN OF THE ROTATION IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THDISP' LINE.

LINE LABEL	SEE JLOD LINE PART 2	JOINT TRANSLATION AND ROTATION DATA IN GLOBAL COORDINATES					
		TRANSLATIONS			ROTATIONS		
		Ax	Ay	Az	RAx	RAy	RAz
JDIS							
1-- 4	5--> 152	153<--164	165<--176	177<--188	189<--200	201<--212	213<--224
DEFAULT							
ENGLISH		FT	FT	FT	RADIANS	RADIANS	RADIANS
METRIC (KN)		M	M	M	RADIANS	RADIANS	RADIANS
METRIC (KG)		M	M	M	RADIANS	RADIANS	RADIANS

JOINT LOAD

COLUMNS	COMMENTARY
GENERAL	THIS LINE SET ENABLES THE TIME HISTORY APPLICATION OF FORCES AND MOMENTS ON THE JOINTS.
(1- 4)	ENTER 'JLOD' ON EACH LINE IN THIS SET.
(5- 8)	ENTER THE JOINT NAME TO WHICH LOADS ON THIS LINE ARE TO BE APPLIED.
(9-20)	FORCE IN GLOBAL X DIRECTION ON THIS JOINT. A POSITIVE FORCE IS IN THE POSITIVE X DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THLOAD' LINE.
(21-32)	FORCE IN GLOBAL Y DIRECTION ON THIS JOINT. A POSITIVE FORCE IS IN THE POSITIVE Y DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THLOAD' LINE.
(33-44)	FORCE IN GLOBAL Z DIRECTION ON THIS JOINT. A POSITIVE FORCE IS IN THE POSITIVE Z DIRECTION. UNITS ARE DETERMINED FROM PREVIOUS 'THLOAD' LINE.
(45-56)	MOMENT ACTING IN GLOBAL X DIRECTION ON THIS JOINT. THE SIGN OF THE MOMENT IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THLOAD' LINE.
(57-68)	MOMENT ACTING IN GLOBAL Y DIRECTION ON THIS JOINT. THE SIGN OF THE MOMENT IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THLOAD' LINE.
(69-80)	MOMENT ACTING IN GLOBAL Z DIRECTION ON THIS JOINT. THE SIGN OF THE MOMENT IS DETERMINED USING THE RIGHT-HAND RULE. UNITS ARE DETERMINED FROM PREVIOUS 'THLOAD' LINE.

LINE LABEL	JOINT NAME	JOINT FORCE AND MOMENT DATA IN GLOBAL COORDINATES					
		FORCES			MOMENTS		
		Fx	Fy	Fz	Mx	My	Mz
JLOD							
1-- 4	5--> 8	9<--20	21<--32	33<--44	45<--56	57<--68	69<--80
DEFAULT							
ENGLISH		KIP	KIP	KIP	KIP-IN	KIP-IN	KIP-IN
METRIC (KN)		KN	KN	KN	KN-M	KN-M	KN-M
METRIC (KG)		KG	KG	KG	KG-CM	KG-CM	KG-CM

MODE SELECTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT MODES TO BE USED IN THE ANALYSIS. THE NUMBER OF MODES SELECTED MUST AGREE WITH THE NUMBER ENTERED IN COLUMNS 41-45 OF THE 'WROPT' LINE.
(7- 9)	ENTER THE FIRST MODE SELECTION (THIS VALUE MUST BE GREATER THAN ZERO AND LESS THAN OR EQUAL TO THE HIGHEST MODE SELECTED IN THE DYNPAC ANALYSIS.
(10-12)	ENTER THE SECOND MODE.
(13-78)	ENTER THE REMAINING MODES DESIRED. IF THE NUMBER OF MODES DESIRED IS GREATER THAN 24, CONTINUE ON A SECOND 'MODSEL' LINE.

LINE LABEL	MODE SELECTIONS																							
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH	11TH	12TH	13TH	14TH	15TH	16TH	17TH	18TH	19TH	20TH	21TH	22TH	23TH	24TH
MODSEL																								
1-- 6	7--> 9	10-->12	13-->15	16-->18	19-->21	22-->24	25-->27	28-->30	31-->33	34-->36	37-->39	40-->42	43-->45	46-->48	49-->51	52-->54	55-->57	58-->60	61-->63	64-->66	67-->69	70-->72	73-->75	76-->78

NONLINEAR FOUNDATION OPTIONS (TECHNOLOGY PREVIEW)

COLUMNS	COMMENTARY
GENERAL	THIS OPTIONAL LINE IS USED TO SET OPTIONS TO INCLUDE NONLINEAR FOUNDATIONS LIKE NONLINEAR PILE SOIL INTERACTION. NONLINEAR FOUNDATION IS ONLY AVAILABLE IN TIME-HISTORY WAVE RESPONSE ANALYSIS. IF THIS LINE IS ENTERED IN THE WAVE RESPONSE INPUT FILE, THE ANALYSIS REQUIRES TWO NEW INPUTS: PILE-SOIL-INTERACTION (PSI) INPUT FILE AND THE PILEHEAD SUPERELEMENT FILE.
(8-10)	ENTER 'PSI' TO SELECT SACS PILE INTERACTION AS THE NONLINEAR FOUNDATION FOR WAVE RESPONSE ANALYSIS.

LINE LABEL	FOUNDATION TYPE	LEAVE THIS FIELD BLANK
NLFOPT		
1-- 6	8--10	11-----80

PERIOD RANGE SPECIFICATION

COLUMNS

COMMENTARY

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GENERAL      THIS LINE IS USED TO REFIN THE DIVISION OF THE SPECTRUM INTO
              AIRY WAVE COMPONENTS. THIS LINE IS USED ONLY FOR SPECTRAL
              ANALYSIS. THIS LINE ALLOWS THE USER TO DIVIDE THE SPECTRUM
              WITH DIFFERENT PERCENTAGES IN DIFFERENT AREAS OF THE
              SPECTRUM. THIS IS USED INSTEAD OF THE CONSTANT VALUE
              SPECIFIED IN COLUMNS 27-33 OF THE 'WSPEC' LINE. THE VALUE ON
              THE 'SPEC' LINE WILL BE USED UP TO THE FIRST ENTERED PERIOD
              ON THIS PERIOD RANGE LINE. THIS LINE IS OPTIONAL.

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(7-11) ENTER THE PERIOD OF THE SPECTRUM. THE FOLLOWING PERCENTAGE
WILL BE USED UP TO THE NEXT ENTERED PERIOD.

(12-16) ENTER THE MINIMUM PERCENT OF THE SPECTRUM THAT EACH WAVE
COMPONENT WILL CONTAIN.

(17-76) ENTER THE REMAINING PERIOD RANGES TO COMPLETE THE SPECTRUM.
THE PERIODS MUST BE ENTERED IN DECENDING ORDER.

[illegible]

WAVE VELOCITY VERSUS DEPTH PLOTS

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO CREATE WAVE VELOCITY VERSUS DEPTH PLOTS.
(8)	ENTER 'Q' IF PROGRAM IS TO BE TERMINATED AFTER PLOTS ARE CREATED.
(9-12)	ENTER NUMBER OF EQUALLY SPACED DEPTH STEPS WHERE VELOCITIES ARE CALCULATED.
(13-20)	ENTER THE BEGINNING ELEVATION FROM MUDLINE. THE REMAINING ELEVATIONS ARE CALCULATED FROM THIS POINT TO THE WAVE SURFACE ELEVATION.

LINE LABEL	EXECUTION TERMINATION OPTION	NUMBER OF DEPTH STEPS	BEGINNING ELEVATION ABOVE MUDLINE	LEAVE BLANK
PLTWV				
1-- 5	8	9--12	13--20	21-----80
DEFAULT		100		
ENGLISH			FT	
METRIC			M	

PLOT SELECTION DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE 'PSL' OPTION WAS SELECTED ON THE 'WROPT' LINE. ENTRIES ON THIS LINE ALLOW THE USER TO SELECT VARIOUS VARIABLES TO BE PLOTTED AND/OR OUTPUT ON A FORMATTED DATA FILE. IN THE CASE OF THE PLOTTED OUTPUT, A NEUTRAL PICTURE FILE IS CREATED FOR SUBSEQUENT PROCESSING FOR PARTICULAR PLOTTERS.
(7- 8)	ENTER ANY ONE OF THE FOLLOWING PLOT/DATA SELECTION OPTIONS: 'SP' - RANDOM WAVE SURFACE PROFILE (AVAILABLE FOR RANDOM WAVE OPTION ONLY) 'GF' - GENERALIZED FORCES 'MC' - MODAL COORDINATES 'MV' - MODAL VELOCITIES 'MA' - MODAL ACCELERATIONS 'JO' - JOINT DISPLACEMENTS (REQUIRES ADDITIONAL 'PSJO' LINE) 'MF' - MEMBER FORCES (REQUIRES ADDITIONAL 'PSMF' LINE) 'OM' - OVERTURNING MOMENT 'BS' - BASE SHEAR 'HF' - TOTAL HYDRODYNAMIC FORCES 'FE' - HYDRODYNAMIC FORCES AT SPECIFIED ELEVATIONS (SEE 'ELVSEL' LINE) 'WS' - WIND SPECTRUM 'WV' - WIND VELOCITIES 'CS' - PRINT OPTION FOR COMPARISON OF MODAL STATIC BASE SHEAR TO TOTAL HYDRODYNAMIC FORCES. 'IF' - INPUT FORCE SUMMATION (TIME HISTORY ONLY)
NOTE:	THE 'CS' OPTION DOES NOT PRODUCE EITHER PLOTS OR A DATA FILE.

COLUMNS	COMMENTARY
(9)	ENTER THE DESIRED TYPE OF OUTPUT FOR THE SELECTED PARAMETER: LEAVE BLANK FOR PLOTS ONLY ENTER 'D' FOR OUTPUT ON FORMATTED DATA FILE ONLY ENTER 'B' FOR BOTH PLOTTER OUTPUT AND FORMATTED FILE OUTPUT
(10-51)	ENTER ANY OR ALL OF THE AVAILABLE PLOT/DATA SELECTIONS FROM THE ABOVE LIST.
(52-56)	ENTER THE PLOT SIZE FOR THE X-AXIS (ABSCISSA).
(57-61)	ENTER THE PLOT SIZE FOR THE Y-AXIS (ORDINATE).
(62-66)	ENTER THE CHARACTER SIZE TO BE USED IN THE PLOT LABELS.
(67-68)	ENTER THE NUMBER OF PENS AVAILABLE FOR THE PLOTTER.
(69)	ENTER 'S' IF EACH DATA POINT ON A TRANSFER FUNCTION PLOT IS TO BE MARKED WITH A SYMBOL.
(70-71)	ENTER 'SF' TO ADD CONSTRAINED WAVE'S FLOW FIELDS TO PLOTTER.

LINE LABEL	PLOT SELECTIONS																																PLOT PARAMETERS				
	SEL 1		SEL 2		SEL 3		SEL 4		SEL 5		SEL 6		SEL 7		SEL 8		SEL 9		SEL 10		SEL 11		SEL 12		SEL 13		SEL 14		SEL 15		SEL 16	X SIZE	Y SIZE	CHAR. SIZE	NUMB. PENS	DATA SYMBOL OPTION	
	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T	VAR	O P T							
PSEL																																					
1-- 4	7-- 8	9	10--11	12	13--14	15	16--17	18	19--20	21	22--23	24	25--26	27	28--29	30	31--32	33	34--35	36	37--38	39	40--41	42	43--44	45	46--47	48	49--50	51	70-71	52<--56	57<--61	62<--66	67-->68	69	
DEFAULT																																11.0 ENGL	8.5 ENGL	0.10 ENGL	1		
ENGLISH																																IN	IN	IN			
METRIC																																CM	CM	CM			

JOINT DOF SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE 'JO' OPTION WAS SELECTED ON THE 'PSEL' LINE. ENTRIES ON THIS LINE ALLOW THE USER TO SELECT VARIOUS DEGREES OF FREEDOM OF JOINTS TO BE PLOTTED AND/OR OUTPUT ON A FORMATTED DATA FILE. THE USER CAN SELECT SEVERAL DEGREES OF FREEDOM FROM THE SAME JOINT IF DESIRED. A MAXIMUM OF 10 SELECTIONS IS ALLOWED.
(7-10)	ENTER THE JOINT NAME FOR THIS SELECTION.
(11-12)	ENTER ANY ONE OF THE FOLLOWING DEGREE OF FREEDOM OPTIONS: 'DX' - DISPLACEMENT IN THE GLOBAL X-DIRECTION 'DY' - DISPLACEMENT IN THE GLOBAL Y-DIRECTION 'DZ' - DISPLACEMENT IN THE GLOBAL Z-DIRECTION 'RX' - ROTATION ABOUT THE GLOBAL X-AXIS 'RY' - ROTATION ABOUT THE GLOBAL Y-AXIS 'RZ' - ROTATION ABOUT THE GLOBAL Z-AXIS 'AX' - ACCELERATION IN THE GLOBAL X-AXIS 'AY' - ACCELERATION IN THE GLOBAL Y-AXIS 'AZ' - ACCELERATION IN THE GLOBAL Z-AZIS 'VX' - VELOCITY IN THE GLOBAL X-AXIS 'VY' - VELOCITY IN THE GLOBAL Y-AYSIS 'VZ' - VELOCITY IN THE GLOBAL Z-AZIS
(13-66)	ENTER ANY REMAINING JOINT DEGREE OF FREEDOM SELECTIONS.

LINE LABEL	JOINT DISPLACEMENT SELECTIONS																			
	1ST		2ND		3RD		4TH		5TH		6TH		7TH		8TH		9TH		10TH	
	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F	JOINT NAME	D O F
PSJO																				
1-- 4	7-->10	11--12	13-->16	17--18	19-->22	23--24	25-->28	29--30	31-->34	35--36	37-->40	41--42	43-->46	47--48	49-->52	53--54	55-->58	59--60	61-->64	65--66

MEMBER FORCE SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE 'MF' OPTION WAS SELECTED ON THE 'PSEL' LINE. ENTRIES ON THIS LINE ALLOW THE USER TO SELECT VARIOUS INTERNAL LOADS OF MEMBERS TO BE PLOTTED AND/OR OUTPUT ON A FORMATTED DATA FILE. THE USER CAN SELECT SEVERAL INTERNAL LOADS FROM THE SAME MEMBER IF DESIRED. A MAXIMUM OF NINETY NINE SELECTIONS IS ALLOWED USING MULTIPLE LINES.
(7-10)	ENTER THE FIRST JOINT NAME FOR THIS MEMBER.
(11-14)	ENTER THE SECOND JOINT NAME FOR THIS MEMBER.
(15-17)	ENTER ANY ONE OF THE FOLLOWING INTERNAL LOAD OPTIONS: 'FXA' - AXIAL LOAD AT END A 'FYA' - SHEAR IN THE Y-DIRECTION AT END A 'FZA' - SHEAR IN THE Z-DIRECTION AT END A 'MXA' - TORSION AT END A 'MYA' - MOMENT ABOUT LOCAL Y-AXIS AT END A 'MZA' - MOMENT ABOUT LOCAL Z-AXIS AT END A 'FXB' - AXIAL LOAD AT END B 'FYB' - SHEAR IN THE Y-DIRECTION AT END B 'FZB' - SHEAR IN THE Z-DIRECTION AT END B 'MXB' - TORSION AT END B 'MYB' - MOMENT ABOUT LOCAL Y-AXIS AT END B 'MZB' - MOMENT ABOUT LOCAL Z-AXIS AT END B
(18-72)	ENTER ANY REMAINING MEMBER INTERNAL LOAD SELECTIONS. IF MORE THAN SIX INTERNAL LOADS ARE NEEDED, USE ADDITIONAL 'PSMF' LINES. UP TO NINETY NINE LOAD SELECTIONS MAY BE SPECIFIED.

LINE LABEL	MEMBER FORCE SELECTIONS																	
	1ST			2ND			3RD			4TH			5TH			6TH		
	JOINT A	JOINT B	INT. LOAD	JOINT A	JOINT B	INT. LOAD	JOINT A	JOINT B	INT. LOAD	JOINT A	JOINT B	INT. LOAD	JOINT A	JOINT B	INT. LOAD	JOINT A	JOINT B	INT. LOAD
PSMF																		
1-- 4	7-->10	11-->14	15--17	18-->21	22-->25	26--28	29-->32	33-->36	37--39	40-->43	44-->47	48--50	51-->54	55-->58	59--61	62-->65	66-->69	70--72

TIME INCREMENT AND SEED SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT TIME POINTS FOR OUTPUT LOADS WHEN MULTIPLE SEEDS ARE USED IN THE RANDOM WAVE ANALYSIS.
(9-14)	ENTER THE FIRST TIME INCREMENT TO PRODUCE LOADS. THE OUTPUT LOADS WILL BE MODAL RESPONSE FACTORS AND/OR THE EQUIVALENT STATIC LOADS AS SPECIFIED ON THE 'WROPT' LINE. THIS VALUE IS AN INTEGER VALUE CORRESPONDING TO THE ANALYSIS TIME INTERVAL. FOR EXAMPLE, TIME ZERO WOULD BE INCREMENT NUMBER 1 AND THE 20TH INCREMENT WOULD CORRESPOND TO TIME = 19*DT WHERE DT IS THE TIME INCREMENT BETWEEN ANALYSIS TIME POINTS.
(15-17)	ENTER THE RANDOM SEED NUMBER. THIS IS NOT THE ACTUAL SEED VALUE, BUT THE ORDER IN WHICH THE SEEDS ARE ANALYZED. THE FIRST SEED IS NUMBER 1, ETC.
(18-80)	ENTER THE REMAINING TIME POINTS. ALL TIME POINTS IN ASCENDING ORDER FOR THE FIRST SEED ARE FOLLOWED BY ALL TIME POINTS FOR THE SECOND SEED, ETC. A LIMIT OF 100 TIME POINTS IS ALLOWED FOR ALL SEEDS.

LINE LABEL	SELECTED TIME INCREMENT POINTS															
	1ST POINT		2ND POINT		3RD POINT		4TH POINT		5TH POINT		6TH POINT		7TH POINT		8TH POINT	
	TIME INCR	SEED NUMBER	TIME INCR	SEED NUMBER	TIME INCR	SEED NUMBER	TIME INCR	SEED NUMBER	TIME INCR	SEED NUMBER	TIME INCR	SEED NUMBER	TIME INCR	SEED NUMBER	TIME INCR	SEED NUMBER
PTSEED																
1-- 6	9-->14	15-->17	18-->23	24-->26	27-->32	33-->35	36-->41	42-->44	45-->50	51-->53	54-->59	60-->62	63-->68	69-->71	72-->77	78-->80

TIME INCREMENT SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT TIME POINTS FOR OUTPUT LOADS. THIS LINE HAS BEEN REPLACED BY THE 'TMSL' LINE AND IS ONLY MAINTAINED FOR BACKWARD COMPATIBILITY.
(6-10)	ENTER THE FIRST TIME INCREMENT TO PRODUCE LOADS. THE OUTPUT LOADS WILL BE MODAL RESPONSE FACTORS AND/OR THE EQUIVALENT STATIC LOADS AS SPECIFIED ON THE 'WROPT' LINE. THIS VALUE IS AN INTEGER VALUE CORRESPONDING TO THE ANALYSIS TIME INTERVAL. FOR EXAMPLE, TIME ZERO WOULD BE INCREMENT NUMBER 1 AND THE 20TH INCREMENT WOULD CORRESPOND TO TIME = 19*DT WHERE DT IS THE TIME INCREMENT BETWEEN ANALYSIS TIME POINTS.
(11-80)	ENTER THE REMAINING TIME POINTS IN ASCENDING ORDER. A LIMIT OF 12 TIME POINTS IS ALLOWED. IF A NEGATIVE VALUE IS ENTERED IN COLUMNS 76-80, THE VALUES IN COLUMNS 6-75 WILL BE USED AS 7 PAIRS OF RANGES SUCH THAT ALL INCREMENTS FALLING WITHIN THE VALUES WILL BE INCLUDED AS OUTPUT LOAD CASES. IN THIS CASE THE RANGES MUST BE ENTERED IN ASCENDING ORDER.

LINE LABEL	SELECTED TIME INCREMENT POINTS OR RANGES (ASCENDING ORDER)														
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH	11TH	12TH	13TH	14TH	15TH
PTSEL															
1-- 5	6-->10	11-->15	16-->20	21-->25	26-->30	31-->35	36-->40	41-->45	46-->50	51-->55	56-->60	61-->65	66-->70	71-->75	76-->80

RANDOM SEED SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY SEEDS FOR RANDOM WAVE ANALYSIS.
(12-14)	ENTER THE FIRST RANDOM SEED.
(16-70)	ENTER ADDITIONAL SEEDS. THIS LINE CAN BE REPEATED FOR UP TO 99 SEEDS.

LINE LABEL	SELECTED 5 DIGIT SEEDS FOR RANDOM WAVE ANALYSIS									
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH
RNSEE5										
1-- 6	12-->16	18-->22	24-->28	30-->34	36-->40	42-->46	48-->52	54-->58	60-->64	66-->70

RANDOM SEED SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY SEEDS FOR RANDOM WAVE ANALYSIS.
(12-14)	ENTER THE FIRST RANDOM SEED.
(16-70)	ENTER ADDITIONAL SEEDS. THIS LINE CAN BE REPEATED FOR UP TO 99 SEEDS.

LINE LABEL	SELECTED SEEDS FOR RANDOM WAVE ANALYSIS														
	1ST	2ND	3RD	4TH	5TH	6TH	7TH	8TH	9TH	10TH	11TH	12TH	13TH	14TH	15TH
RNSEED															
1-- 6	12-->14	16-->18	20-->22	24-->26	28-->30	32-->34	36-->38	40-->42	44-->46	48-->50	52-->54	56-->58	60-->62	64-->66	68-->70

USER-DEFINED SPECTRUM

COLUMNS	COMMENTARY
GENERAL	THIS LINE DEFINES THE USER SPECTRUM. FOR THOSE CASES WHERE A PIERSON-MOSKOWITZ OR JONSWAP SPECTRUM DEFINITION IS NOT SATISFACTORY, THE USER CAN DEFINE A PARTICULAR SPECTRUM WITH UP TO 100 POINTS USING 4 VALUES PER LINE AND UP TO 25 RECORDS. THESE LINES FOLLOW THE 'WSPEC' LINE WITH THE 'US' OPTION.
(1- 4)	ENTER 'SPEC'.
(5)	SELECT EITHER PERIOD ('P') OR FREQUENCY ('F') IF THE SPECTRUM IS DEFINED BY PERIOD OR FREQUENCY.
(9-16)	ENTER THE PERIOD OR FREQUENCY OF THE FIRST POINT OF THE WAVE SPECTRUM. FREQUENCIES ARE ENTERED IN ASCENDING ORDER. PERIODS MUST BE ENTERED IN DESCENDING ORDER. A PERIOD OF ZERO SHOULD NEVER BE ENTERED. IF THE PERIOD/FREQUENCY FIELD IS BLANK, THEN THIS POINT IS IGNORED.
(17-24)	ENTER THE WAVE SPECTRUM VALUE IN TERMS OF WAVE HEIGHT SQUARED OVER HERTZ.
(25-40)	ENTER THE SECOND POINT. IF THE PERIOD/FREQUENCY FIELD IS BLANK, THEN THIS POINT IS IGNORED.
(41-56)	ENTER THE THIRD POINT. IF THE PERIOD/FREQUENCY FIELD IS BLANK, THEN THIS POINT IS IGNORED.
(57-72)	ENTER THE FOURTH POINT. IF THE PERIOD/FREQUENCY FIELD IS BLANK, THEN THIS POINT IS IGNORED.

LINE LABEL	FREQ OR PERIOD OPTION	USER DEFINED SPECTRAL DATA								LEAVE BLANK
		1ST POINT		2ND POINT		3RD POINT		4TH POINT		
		PERIOD OR FREQUENCY	SPECTRA VALUE	PERIOD OR FREQUENCY	SPECTRA VALUE	PERIOD OR FREQUENCY	SPECTRA VALUE	PERIOD OR FREQUENCY	SPECTRA VALUE	
SPEC										
1-- 4	5	9<--16	17<--24	25<--32	33<--40	41<--48	49<--56	57<--64	65<--72	73--80
DEFAULT	'P'									
ENGLISH		SEC	FT**2/HZ	SEC	FT**2/HZ	SEC	FT**2/HZ	SEC	FT**2/HZ	
METRIC		SEC	M**2/HZ	SEC	M**2/HZ	SEC	M**2/HZ	SEC	M**2/HZ	

DISPLACEMENT TIME HISTORY DATA HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE DISPLACEMENT TIME HISTORY IS INCLUDED IN THE WAVE RESPONSE ANALYSIS. THE OVERALL PARAMETERS OF THE DISPLACEMENT TIME HISTORY INPUT ARE DESCRIBED HERE.
(12-13)	SELECT INPUT DISPLACEMENT UNITS FROM THE FOLLOWING: 'EN' - ENGLISH 'MN' - METRIC (KILONEWTON FORCE) 'ME' - METRIC (KILOGRAM FORCE)
(14-20)	ENTER TRANSLATION FACTOR. THIS FACTOR WILL APPLY TO ALL TRANSLATIONS.
(21-27)	ENTER ROTATION FACTOR. THIS FACTOR WILL APPLY TO ALL ROTATIONS.

LINE LABEL	DISPLACEMENT INPUT UNITS	FORCE FACTOR	MOMENT FACTOR	LEAVE BLANK	LEAVE BLANK
THDISP					
1-- 6	12--13	14<--20	21<--27	28-----50	51-----80
DEFAULT	'EN'	1	1		

TIME HISTORY INTEGRATION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY TIME HISTORY INTEGRATION PARAMETERS.
(7-10)	IF A GRID IS TO BE USED, ENTER THE NUMBER OF HORIZONTAL POINTS IN THE GRID. THE SPACING OF THESE GRID POINTS IS DETERMINED BY THE PROGRAM.
(11-14)	IF A GRID IS TO BE USED, ENTER THE NUMBER OF VERTICAL POINTS IN THE GRID. IF LEFT BLANK, THE PROGRAM WILL DETERMINE THE NUMBER OF VERTICAL GRID POINTS. THE SPACING OF THE GRID POINTS IS DETERMINED BY THE PROGRAM.
(31-40)	ENTER THE MAXIMUM INTEGRATION STEP SIZE. IF LEFT BLANK, THE ANALYSIS TIME INCREMENT WILL BE USED.
(41-50)	ENTER THE MINIMUM INTEGRATION STEP SIZE. IF LEFT BLANK, THE PROGRAM WILL DETERMINE THE MINIMUM STEP SIZE.
(51-60)	ENTER THE INTEGRATION ERROR CONTROL FACTOR. FOR MORE ACCURATE INTEGRATION, ENTER A NUMBER LESS THAN 1.0.
(61-62)	ENTER 'TR' TO INCLUDE A REDUCTION IN THE INITIAL TRANSIENTS FOR TIME HISTORY (DEFAULT). ENTER 'NR' IF NO TRANSIENT REDUCTION IS DESIRED. ENTER 'OR' TO OMIT INITIAL TRANSIENTS. (ENTER VALUE IN COLUMNS 63-72).
(63-72)	ENTER THE INITIAL TRANIENT TIME TO OMIT.

LINE LABEL	GRID POINTS		INTEGRATION STEP SIZES		INTEGRATION ERROR CONTROL FACTOR	TRANSIENT REDUCTION OPTION	TRANSIENT TIME TO OMIT	LEAVE BLANK
	HORIZONTAL	VERTICAL	MAXIMUM	MINIMUM				
THIST								
1-- 5	7-->10	11-->14	31<--40	41<--50	51<--60	61--62	63-72	73--80
DEFAULT	10				1	TR		
ENGLISH			SEC	SEC				
METRIC			SEC	SEC			SEC	

LOAD TIME HISTORY DATA HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE LOAD TIME HISTORY IS INCLUDED IN THE WAVE RESPONSE ANALYSIS. THE OVERALL PARAMETERS OF THE LOAD TIME HISTORY INPUT ARE DESCRIBED HERE.
(12-13)	SELECT INPUT LOAD UNITS FROM THE FOLLOWING: 'EN' - ENGLISH 'MN' - METRIC (KILONEWTON FORCE) 'ME' - METRIC (KILOGRAM FORCE)
(14-20)	ENTER FORCE FACTOR. THIS FACTOR WILL APPLY TO ALL FORCES.
(21-27)	ENTER MOMENT FACTOR. THIS FACTOR WILL APPLY TO ALL MOMENTS.
(28)	SELECT LOAD INPUT FILE TYPE: ' ' - SACS 'F' - FLEX5 OUTPUT FILE ('N' & 'N.M' UNITS ONLY FOR FORCE AND MOMENT) 'B' - BHAWC PUTPUT FILE ('KN' & 'KN.M' UNITS ONLY FOR FORCE AND MOMENT)
(29-32)	ENTER THE JOINT NAME TO WHICH LOADS FROM FLEX5 / BHAWC ARE TO BE APPLIED.
(33-35)	ENTER COLUMN NUMBER FOR FX IN BHAWC FILE
(36-38)	ENTER COLUMN NUMBER FOR FY IN BHAWC FILE
(39-41)	ENTER COLUMN NUMBER FOR FZ IN BHAWC FILE
(42-44)	ENTER COLUMN NUMBER FOR MX IN BHAWC FILE
(45-47)	ENTER COLUMN NUMBER FOR MY IN BHAWC FILE
(48-50)	ENTER COLUMN NUMBER FOR MZ IN BHAWC FILE

[illegible]

WAVE TIME HISTORY DATA HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE WAVE TIME HISTORY IS INCLUDED IN THE WAVE RESPONSE ANALYSIS. THE OVERALL PARAMETERS OF THE WAVE TIME HISTORY INPUT ARE DESCRIBED HERE.
(8-10)	ENTER THE INPUT FORMAT FOR THE SUBSEQUENT DATA INPUT. THERE ARE TWO OPTIONS, 'NON' FOR NONUNIFORM TIME INCREMENTS AND 'UNI' FOR UNIFORM TIME INCREMENTS.
(12-12)	SELECT INPUT WAVE HEIGHT UNITS FROM THE FOLLOWING: 'F' - FEET 'M' - METERS
(15-22)	ENTER THE TIME INTERVAL BETWEEN INPUT TIME POINTS FOR THE UNIFORM INPUT FORMAT. LEAVE BLANK FOR NONUNIFORM INPUT.
(23-29)	ENTER THE WAVE HEIGHT FACTOR. EACH SURFACE PROFILE ENTRY WILL BE MULTIPLIED BY THIS FACTOR.

LINE LABEL	INPUT FORM	WAVE SURFACE UNITS	TIME INTERVAL 'UNI' ONLY	WAVE HEIGHT FACTOR	LEAVE BLANK
THWAVE					
1-- 6	8--10	12	15<--22	23<--29	30-----80
DEFAULT	'NON'	'F'		1	
ENGLISH			SEC		
METRIC			SEC		

WIND TIME HISTORY DATA HEADER

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IF THE WIND TIME HISTORY IS SPECIFIED ON THE 'WINDSP' LINE. THE OVERALL PARAMETERS OF THE WIND TIME HISTORY INPUT ARE DESCRIBED HERE.
(8-10)	ENTER THE INPUT FORMAT FOR THE SUBSEQUENT DATA INPUT. THERE ARE TWO OPTIONS, 'NON' FOR NONUNIFORM TIME INCREMENTS AND 'UNI' FOR UNIFORM TIME INCREMENTS.
(12-14)	SELECT INPUT WAVE VELOCITY UNITS FROM THE FOLLOWING: 'KTS' - KNOTS (DEFAULT). 'MPS' - METERS PER SECOND. 'MPH' - MILES PER HOUR. 'FPS' - FEET PER SECOND.
(15-22)	ENTER THE TIME INTERVAL BETWEEN INPUT TIME POINTS FOR THE UNIFORM INPUT FORMAT. LEAVE BLANK FOR NONUNIFORM INPUT.
(23-29)	ENTER THE WIND VELOCITY FACTOR. EACH WIND VELOCITY ENTRY WILL BE MULTIPLIED BY THIS VALUE.

LINE LABEL	INPUT FORM	WIND VELOCITY UNITS	TIME INTERVAL 'UNI' ONLY	WIND VELOCITY FACTOR	LEAVE BLANK
THWIND					
1-- 6	8--10	12--14	15<--22	23<--29	30-----80
DEFAULT	'NON'	'KTS'		1	
ENGLISH			SEC		
METRIC			SEC		

LOAD TIME DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS REQUIRED IN THE LOAD TIME HISTORY TO DENOTE THAT THE FOLLOWING 'JLOD' RECORD OCCUR AT THIS SPECIFIED TIME.
(6-17)	ENTER THE TIME IN SECONDS.

LINE LABEL	LOAD TIME	LEAVE BLANK
TIME		
1-- 4	6<--17	18-----80
DEFAULT		
ENGLISH	SEC	
METRIC	SEC	

TIME SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT TIME FOR OUTPUT LOADS. THIS LINE IS OPTIONAL.
(7-12)	ENTER THE FIRST TIME POINT TO PRODUCE LOADS. THE OUTPUT LOADS WILL BE MODAL RESPONSE FACTORS AND/OR THE EQUIVALENT STATIC LOADS AS SPECIFIED ON THE 'WROPT' LINE.
(13-78)	ENTER THE REMAINING TIME POINTS IN ASCENDING ORDER. A LIMIT OF 12 TIME POINTS IS ALLOWED.
(79)	ENTER 'R' IF THE INPUT VALUES ARE TO BE TAKEN IN PAIRS AND USED AS RANGES. EACH SUCCESSIVE RANGE MUST COVER A TIME PERIOD GREATER THAN THE PREVIOUS RANGE.

[illegible]

WAVE VELOCITY PLOT TIME SELECTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SELECT TIME FOR OUTPUT WAVE VELOCITY VERSUS DEPTH PLOTS. THIS LINE IS OPTIONAL.
(7-12)	ENTER THE FIRST TIME POINT.
(13-78)	ENTER THE REMAINING TIME POINTS. A LIMIT OF 12 TIME POINTS IS ALLOWED.

[illegible]

NONUNIFORM WAVE TIME HISTORY DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED WHEN THE NONUNIFORM OPTION IS SPECIFIED ON THE 'THWAVE' LINE. THE WAVE HEIGHT VALUES ARE SPECIFIED HERE USING THE UNITS AS SPECIFIED ON THE 'THWAVE' LINE. REPEAT THIS LINE FOR ALL TIME HISTORY VALUES.
(9-80)	ENTER THE TIME AND WAVE HEIGHT VALUES. TIME VALUES MUST BE ENTERED IN ASCENDING ORDER.

LINE LABEL	1ST WAVE POINT		2ND WAVE POINT		3RD WAVE POINT	
	TIME	HEIGHT	TIME	HEIGHT	TIME	HEIGHT
WAVENON						
1-- 7	9<--20	21<--32	33<--44	45<--56	57<--68	69<--80
DEFAULT						
ENGLISH	SEC	'THWAVE'	SEC	'THWAVE'	SEC	'THWAVE'
METRIC	SEC	'THWAVE'	SEC	'THWAVE'	SEC	'THWAVE'

UNIFORM WAVE TIME HISTORY DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED WHEN THE UNIFORM OPTION IS SPECIFIED ON THE 'THWAVE' LINE. THE WAVE HEIGHT VALUES ARE SPECIFIED HERE USING THE UNITS AS SPECIFIED ON THE 'THWAVE' LINE. REPEAT THIS LINE FOR ALL TIME HISTORY VALUES.
(9-80)	ENTER THE WAVE HEIGHT VALUES.

LINE LABEL	1ST WAVE HEIGHT	2ND WAVE HEIGHT	3RD WAVE HEIGHT	4TH WAVE HEIGHT	5TH WAVE HEIGHT	6TH WAVE HEIGHT
WAVEUNI						
1-- 7	9<--20	21<--32	33<--44	45<--56	57<--68	69<--80
DEFAULT						
ENGLISH	'THWAVE'	'THWAVE'	'THWAVE'	'THWAVE'	'THWAVE'	'THWAVE'
METRIC	'THWAVE'	'THWAVE'	'THWAVE'	'THWAVE'	'THWAVE'	'THWAVE'

WAVE TIME AND POSITION PARAMETERS PART 1

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS NECESSARY ONLY FOR A RANDOM WAVE ANALYSIS. IT IS USED TO SPECIFY THE TIME SERIES WAVE PARAMETERS AND OVERALL RANDOM ANALYSIS PARAMETERS.
(7- 8)	THE WAVE KINEMATIC CALCULATIONS CAN EITHER BE BASED ON THE POSITION OF THE STRUCTURE IN THE ORIGINAL LOCATION OR IN THE DISPLACED POSITION FOR SUCCESSIVE ITERATIONS. IF THE MOTION OF THE STRUCTURE IS SMALL RELATIVE TO THE SIGNIFICANT WAVE LENGTHS OF THE VARIOUS AIRY WAVE COMPONENTS, THEN THE WAVE KINEMATICS CALCULATED IN THE FIRST PASS CAN BE USED IN SUBSEQUENT ITERATIONS WITH NO SIGNIFICANT LOSS IN ACCURACY. IN THIS CASE LEAVE THIS FIELD BLANK. IF THE WAVES ARE TO BE RECALCULATED FOR EACH ITERATION, ENTER 'PD' IN THIS FIELD.
(9-10)	ENTER THE VERTICAL COORDINATE OF THE STRUCTURE (+ UP) .
(11-16)	ENTER THE WATER DEPTH FOR THIS ANALYSIS.
(17-22)	ENTER THE VALUE FOR THE VERTICAL COORDINATE OF THE MUDLINE.
(23-28)	ENTER THE DIRECTION THAT THE WAVE IS TRAVELING, MEASURED IN DEGREES FROM THE GLOBAL REFERENCE AXIS. IF THE VERTICAL AXIS IS +Z THEN THIS VALUE IS MEASURED FROM X TOWARD Y.
(29-34)	ENTER THE STARTING POSITION OF THE WAVE RELATIVE TO THE GLOBAL VERTICAL COORDINATE. NORMALLY THIS FIELD IS LEFT BLANK SINCE THE STARTING POSITION IN A RANDOM ANALYSIS IS IMMATERIAL.
(35-41)	FOR A WAVE SPECTRUM OR WIND SPECTRUM INPUT, ENTER THE TIME DURATION OF THE SPECTRUM. IF A USER-SUPPLIED SURFACE PROFILE IS BEING USED, ENTER THE TIME INCREMENT FOR THAT PROFILE.
(42-48)	THIS VALUE IS THE TIME INCREMENT USED FOR THE ANALYSIS. THE WAVE AND/OR WIND KINEMATICS ARE EVALUATED AT EACH OF THESE TIME POINTS.
(49-52)	ENTER THE MAXIMUM NUMBER OF WAVE COMPONENTS USED TO REPRESENT THE RANDOM WAVE TIME HISTORY SAMPLE.
(53-88)	SEE WAVTIM LINE PART 2 .

LINE LABEL	POSITION DEPENDENT WAVE OPTION	VERTICAL COORD.	WAVE PARAMETERS				TIME DURATION OR INPUT WAVE TIME INCR.	ANALYSIS TIME INCR.	MAXIMUM WAVE COMPONENTS	SEE WAVTIM LINE PART 2
			WATER DEPTH	MUDLINE ELEV.	WAVE DIRECTION	INITIAL OFFSET				
WAVTIM										
1-- 6	7-- 8	9--10	11<--16	17<--22	23<--28	29<--34	35<--41	42<--48	49-->52	53--88
DEFAULT		' +Z '							30	
ENGLISH			FT	FT	DEG	FT	SEC	SEC		
METRIC			M	M	DEG	M	SEC	SEC		SEC

WAVE TIME AND POSITION PARAMETERS PART 2

COLUMNS

COMMENTARY

- (49-52) ENTER THE MAXIMUM NUMBER OF WAVE COMPONENTS USED TO REPRESENT THE RANDOM WAVE TIME HISTORY SAMPLE.
- (53-57) ENTER THE FOURIER SERIES CONVERGENCE TOLERANCE. THIS VALUE IS USED TO DETERMINE THE NUMBER OF AIRY WAVE COMPONENTS USED TO REPRESENT A USER DEFINED SURFACE PROFILE.
- (58-59) SELECT THE AIRY WAVE CALCULATION FROM THE FOLLOWING OPTIONS:
 'ST' - STANDARD AIRY WAVE USING THE WAVE KINEMATICS AS CALCULATED IN THE CREST AND TROUGH.
 'VS' - USE THE WAVE KINEMATICS IN THE CREST AS CALCULATED AT THE MEAN WATER LINE
 'WH' - USE THE KINEMATICS AS CALCULATED AT THE MEAN WATER LINE AT THE SURFACE AND THEN DECAY THESE VALUES USING NORMAL AIRY EXPONENTIAL DECAY FUNCTIONS TO VALUES BELOW THE SURFACE.
 'CA' - USE "CLASSICAL" AIRY WAVE THEORY WHERE THERE IS NO CREST OR TROUGH AND THE WAVE KINEMATICS ARE CALCULATED FROM THE MEAN WATER LINE.
 'LS' - USE LINEAR STRETCHING (SEE 'WH') ABOVE MEAN WATER LINE AND NORMAL AIRY WAVE BELOW.
- (62-63) ENTER 'LN' TO USE LINEAR CURRENT STRETCHING. THIS OPTION USES THE LAST CURRENT (I.E. HIGHEST ELEVATION) TO STRETCH THE CURRENT FIELD TO THE INSTANTANEOUS WAVE SURFACE LOCATION.
- (68-71) ENTER WAVE KINEMATICS FACTOR USED TO ACCOUNT FOR SPREADING & WAVE PROFILE REGULARITY.
- (72-74) IF MULTI-SEED USER-SUPPLIED WAVE SURFACE PROFILE IS BEING USED, ENTER THE NUMBER OF SEEDS.
- (75) ENTER 'S' IF SECOND ORDER SURFACE WAVE THEORY IS TO BE USED. (ONLY TIME HISTORY ANALYSIS).
- (76) ENTER 'D' TO INCLUDE DIFFRACTION CALCULATIONS.
- (77-78) ENTER 'TR' TO INCLUDE A REDUCTION IN THE INITIAL TRANSIENTS FOR TIME HISTORY. ENTER 'NR' IF NO TRANSIENT REDUCTION IS DESIRED. ENTER 'OR' TO OMIT INITIAL TRANSIENTS. (ENTER VALUE IN COLUMNS 63-72).
- (79-88) ENTER THE INITIAL TRANIENT TIME TO OMIT.

LINE LABEL	SEE WAVTIM LINE PART 1	FOURIER COEFF CONVERGENCE	AIRY WAVE OPTION	CURRENT STRETCH OPTION	WAVE KINEMATICS FACTOR	NUMBER OF RANDOM SEEDS FOR USER DEFINED SURFACE	USE SECOND ORDER SURFACE WAVE THEORY	INCLUDE WAVE DIFFRACTION	TRANSIENT REDUCTION OPTION	TRANSIENT TIME TO OMIT
WAVTIM										
1-- 6	7-- 52	53<--57	58--59	62--63	68<--71	72-->74	75	76	77--78	79--88
DEFAULT		1			1				TR	
ENGLISH		%								
METRIC		%								SEC

NONUNIFORM WIND TIME HISTORY DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED WHEN THE NONUNIFORM OPTION IS SPECIFIED ON THE 'THWIND' LINE. THE WIND VELOCITY VALUES ARE SPECIFIED HERE USING THE UNITS AS SPECIFIED ON THE 'THWIND' LINE. REPEAT THIS LINE FOR ALL TIME HISTORY VALUES.
(9-80)	ENTER THE TIME AND WIND VELOCITY VALUES. TIME VALUES MUST BE ENTERED IN ASCENDING ORDER.

LINE LABEL	1ST WIND POINT		2ND WIND POINT		3RD WIND POINT	
	TIME	VELOCITY	TIME	VELOCITY	TIME	VELOCITY
WINDNON						
1-- 7	9<--20	21<--32	33<--44	45<--56	57<--68	69<--80
DEFAULT						
ENGLISH	SEC	'THWIND'	SEC	'THWIND'	SEC	'THWIND'
METRIC	SEC	'THWIND'	SEC	'THWIND'	SEC	'THWIND'

WIND SPECTRUM PARAMETERS

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS NECESSARY ONLY FOR A RANDOM WIND ANALYSIS. IT IS USED TO SPECIFY THE FROYA/API/NORSOK/ISO19902 WIND SPECTRUM PARAMETERS.
(7- 8)	SELECT THE WIND SPECTRUM TYPE: 'FR' - FROYA/API/NORSOK/ISO19902
(10)	SELECT THE WIND VELOCITY UNITS FOR THE ENGLISH SYSTEM. ENTER 'M' FOR MILES PER HOUR, 'F' FOR FEET PER SECOND, OR LEAVE BLANK FOR KNOTS.
(11-17)	ENTER THE WIND VELOCITY. NOTE: 1 MILE PER HOUR = 0.8690 KNOTS. THIS IS THE WIND SPEED AT 10 METERS ABOVE THE WATER LINE. THIS WIND SPEED IS USED TO DEFINE THE HARRIS WIND SPECTRUM. THE WIND SPEED USED IN SEASTATE INPUT WILL HAVE NO EFFECT ON THE SPECTRUM.
(67-71)	THIS IS THE SEED FOR A BUILT-IN PSEUDO-RANDOM NUMBER GENERATOR WHICH IS USED TO CALCULATE THE PHASE ANGLES FOR THE VARIOUS WIND FOURIER COEFFICIENTS. IF THE SAME ANALYSIS IS RUN ON THE SAME COMPUTER THEN THE RESULTS WILL BE THE SAME IF THE SAME SEED IS USED. IF A DIFFERENT SEED IS USED, A DIFFERENT SET OF RANDOM PHASE ANGLES WILL BE COMPUTED. NORMALLY, THIS FIELD IS LEFT BLANK. IF A SEED IS DESIRED, ENTER AN INTEGER FROM 1 TO 99999.

LINE LABEL	WIND SPECTRUM TYPE	WIND VELOCITY UNITS OPTION	WIND VELOCITY AT 10 METERS	LEAVE BLANK	WIND RANDOM SEED	LEAVE BLANK
WINDSP						
1-- 6	7-- 8	10	11<--17	39-----66	67<--71	72-----80
DEFAULT		KNOTS			0	
ENGLISH			SEE COLUMN 10			
METRIC			M/SEC			

WIND SPECTRUM PARAMETERS

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS NECESSARY ONLY FOR A RANDOM WIND ANALYSIS. IT IS USED TO SPECIFY THE KAIMAL WIND SPECTRUM PARAMETERS.
(7- 8)	SELECT THE WIND SPECTRUM TYPE: 'KM' - KAIMAL.
(10)	SELECT THE WIND VELOCITY UNITS FOR THE ENGLISH SYSTEM. ENTER 'M' FOR MILES PER HOUR, 'F' FOR FEET PER SECOND, OR LEAVE BLANK FOR KNOTS.
(11-17)	ENTER THE WIND VELOCITY. NOTE: 1 MILE PER HOUR = 0.8690 KNOTS. THIS IS THE WIND SPEED AT 10 METERS ABOVE THE WATER LINE. THIS WIND SPEED IS USED TO DEFINE THE HARRIS WIND SPECTRUM. THE WIND SPEED USED IN SEASTATE INPUT WILL HAVE NO EFFECT ON THE SPECTRUM.
(39-45)	ENTER THE TURBULENCE INTENSITY FACTOR.
(67-71)	THIS IS THE SEED FOR A BUILT-IN PSEUDO-RANDOM NUMBER GENERATOR WHICH IS USED TO CALCULATE THE PHASE ANGLES FOR THE VARIOUS WIND FOURIER COEFFICIENTS. IF THE SAME ANALYSIS IS RUN ON THE SAME COMPUTER THEN THE RESULTS WILL BE THE SAME IF THE SAME SEED IS USED. IF A DIFFERENT SEED IS USED, A DIFFERENT SET OF RANDOM PHASE ANGLES WILL BE COMPUTED. NORMALLY, THIS FIELD IS LEFT BLANK. IF A SEED IS DESIRED, ENTER AN INTEGER FROM 1 TO 99999.

LINE LABEL	WIND SPECTRUM TYPE	WIND VELOCITY UNITS OPTION	WIND VELOCITY AT 10 METERS	TURBULENCE INTENSITY FACTOR	LEAVE BLANK	WIND RANDOM SEED	LEAVE BLANK
WINDSP							
1-- 6	7-- 8	10	11--17	39<--45	46-----66	67<--71	72-----80
DEFAULT		KNOTS				0	
ENGLISH			SEE COLUMN 10				
METRIC			M/SEC				

WIND SPECTRUM PARAMETERS

COLUMNS	COMMENTARY	COLUMNS	COMMENTARY
GENERAL	THIS LINE IS NECESSARY ONLY FOR A RANDOM WIND ANALYSIS. IT IS USED TO SPECIFY THE HARRIS WIND SPECTRUM PARAMETERS AND LATERAL COHERENCE PARAMETERS.	(67-71)	THIS IS THE SEED FOR A BUILT-IN PSEUDO-RANDOM NUMBER GENERATOR WHICH IS USED TO CALCULATE THE PHASE ANGLES FOR THE VARIOUS WIND FOURIER COEFFICIENTS. IF THE SAME ANALYSIS IS RUN ON THE SAME COMPUTER THEN THE RESULTS WILL BE THE SAME IF THE SAME SEED IS USED. IF A DIFFERENT SEED IS USED, A DIFFERENT SET OF RANDOM PHASE ANGLES WILL BE COMPUTED. NORMALLY, THIS FIELD IS LEFT BLANK. IF A SEED IS DESIRED, ENTER AN INTEGER FROM 1 TO 99999.
(7- 8)	SELECT THE WIND SPECTRUM TYPE: 'HR' - HARRIS. 'VK' - VON KARMAN. 'TH' - TIME HISTORY INPUT BY USER.		
(10)	SELECT THE WIND VELOCITY UNITS FOR THE ENGLISH SYSTEM. ENTER 'M' FOR MILES PER HOUR, 'F' FOR FEET PER SECOND, OR LEAVE BLANK FOR KNOTS.		
(11-17)	ENTER THE WIND VELOCITY. NOTE: 1 MILE PER HOUR = 0.8690 KNOTS. THIS IS THE WIND SPEED AT 10 METERS ABOVE THE WATER LINE. THIS WIND SPEED IS USED TO DEFINE THE HARRIS WIND SPECTRUM. THE WIND SPEED USED IN SEASTATE INPUT WILL HAVE NO EFFECT ON THE SPECTRUM.		
(18-24)	ENTER THE WIDTH OF THE MAIN DECK. THIS DIMENSION IS USED TO CALCULATE THE WIND COHERENCE FUNCTION.		
(25-31)	ENTER THE LATERAL COHERENCE PARAMETER. THIS IS ALSO USED TO CALCULATE THE WIND COHERENCE FUNCTION.		
(32-38)	ENTER THE HARRIS SPECTRUM LENGTH.		
(39-45)	ENTER THE HARRIS SPECTRUM ROUGHNESS.		
(46-52)	'A' PARAMETER IN THE VON KARMON TURBULENCE INTENSITY FACTOR		
(53-59)	'B' PARAMETER IN THE VON KARMON TURBULENCE INTENSITY FACTOR WHERE $TIF = A + B * V_{mean}$ THE UNITS FOR THE 'B' PARAMETER ARE ONE OVER VELOCITY UNITS.		

LINE LABEL	WIND SPECTRUM TYPE	WIND VELOCITY UNITS OPTION	WIND VELOCITY AT 10 METERS	WIDTH OF MAIN DECK	LATERAL COHERENCE PARAMETER	HARRIS SPECTRUM LENGTH	HARRIS SPECTRUM ROUGHNESS	VON KARMON 'A' PARAMETER	VON KARMON 'B' PARAMETER	WIND RANDOM SEED
WINDSP										
1-- 6	7-- 8	10	11<--17	18<--24	25<--31	32<--38	39<--45	46<--52	53<--59	67<--71
DEFAULT	'HR'	KNOTS			16	1800. METRIC	0.0025			0
ENGLISH			SEE COLUMN 10	FT		FT			SEE COLUMN 10	
METRIC			M/SEC	M		M			SEC/M	

UNIFORM WIND TIME HISTORY DATA

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED WHEN THE UNIFORM OPTION IS SPECIFIED ON THE 'THWIND' LINE. THE WIND VELOCITY VALUES ARE SPECIFIED HERE USING THE UNITS AS SPECIFIED ON THE 'THWIND' LINE. REPEAT THIS LINE FOR ALL TIME HISTORY VALUES.
(9-80)	ENTER THE WIND VELOCITY VALUES.

LINE LABEL	1ST WIND VELOCITY	2ND WIND VELOCITY	3RD WIND VELOCITY	4TH WIND VELOCITY	5TH WIND VELOCITY	6TH WIND VELOCITY
WINDUNI						
1-- 7	9<--20	21<--32	33<--44	45<--56	57<--68	69<--80
DEFAULT						
ENGLISH	'THWIND'	'THWIND'	'THWIND'	'THWIND'	'THWIND'	'THWIND'
METRIC	'THWIND'	'THWIND'	'THWIND'	'THWIND'	'THWIND'	'THWIND'

WAVE RESPONSE OPTIONS

COLUMNS	COMMENTARY
GENERAL	THIS LINE WHICH SPECIFIES ANALYSIS PARAMETERS AND THE OUTPUT LOADS DESIRED, IS REQUIRED IN ANY WAVE RESPONSE RUN.
(7)	ENTER 'S' TO COMPUTE DYNAMIC SUPERELEMENT FORCES.
(8 - 9)	SELECT THE UNITS AS: 'EN' - ENGLISH UNITS 'MN' - METRIC WITH KILONEWTON FORCE 'ME' - METRIC WITH KILOGRAM FORCE
(10-12)	ENTER 'PSL' TO CREATE PLOTS. IF THIS OPTION IS SELECTED, A 'PSEL' OR 'PLTTF' LINE WILL IMMEDIATELY FOLLOW.
(13-14)	ENTER 'RL' TO CREATE LOADING FROM TIME POINTS SPECIFIED ON 'PTSEL' LINES. THIS OPTION REQUIRES A RESPONSE FILE AS INPUT.
(15-18)	ENTER THE CRITERIA FOR OUTPUT LOAD CASE SELECTION: 'ALL' - ALL TIME POINTS. 'GLOB' - TIME POINTS DETERMINED AUTOMATICALLY BY GLOBAL LOADING. 'MAXM' - MAXIMUM OVERTURNING MOMENT. 'MAXS' - MAXIMUM BASE SHEAR. 'M-NM' - MAX. - MIN. OVERTURNING MOMENT. 'M-NS' - MAX. - MIN. BASE SHEAR. 'MM*M' - * POINTS AT MAX. AND MIN. OVERTURNING MOMENT POSITIONS WHERE * IS AN INTEGER FROM 1 TO 9. 'MM*S' - * POINTS AT MAX. AND MIN. BASE SHEAR POSITIONS. 'MK*M' - SKIP EVERY * POINTS INCLUDING THE MAX. OTM. 'MK*S' - SKIP EVERY * POINTS INCLUDING THE MAX. BS. 'WAVE' - USE WAVE CREST POSITION OPTION SPECIFIED ON THE WAVE DEFINITION. 'MST' - MAX & MIN OTM, BS, & TORSION. NOTE: SELECTED FROM ESL GENERATED USING 'ALL' OPTION.
(19-20)	ENTER THE EQUIVALENT STATIC LOAD OPTION. THESE INCLUDE INERTIA LOADS AND HYDRODYNAMIC LOADS FOR THE TIME POINTS SELECTED AND CAN BE SOLVED IN A STATIC ANALYSIS. 'ES' - EQUIVALENT STATIC LOADS ARE TO BE GENERATED (USED WITH DYN OPTION IN SEASTATE). 'EX' - EXTREME WAVE EQUIVALENT STATIC LOADS (USED WITH DYN OPTION IN SEASTATE). 'EI' - EQUIVALENT STATIC INERTIA LOADS ONLY. 'US' - FATIGUE ANALYSIS EQUIVALENT STATIC LOADS. 'MR' - CREATE COMBINE INPUT WITH MODAL PARTICIPATION FACTORS.
NOTE:	THE 'US' OPTION ELIMINATES SORTING LOADS INTO LOAD CASES WHICH CAN SIGNIFICANTLY REDUCE ANALYSIS RUNTIME.
(21-30)	ENTER THE PERCENT RMS ERROR ALLOWED ON GENERALIZED FORCES BETWEEN ITERATIONS.
(31-40)	ENTER THE FOURIER SERIES CONVERGENCE TOLERANCE. THIS TOLERANCE IS USED TO DETERMINE THE NUMBER OF TERMS IN THE FOURIER SERIES REPRESENTATION OF THE GENERALIZED FORCES.
(41-45)	ENTER THE NUMBER OF MODES TO BE USED IN THE ANALYSIS. DEFAULT IS THE NUMBER OF MODES IN THE MODAL SOLUTION FILE.

COLUMNS	COMMENTARY
(46-50)	ENTER THE NUMBER OF ALLOWABLE ITERATIONS TO INCLUDE RELATIVE VELOCITY AND ACCELERATION BETWEEN FLUID AND STRUCTURE. ENTER '-1' IF NO RELATIVE VELOCITY OR ACCELERATION IS TO BE INCLUDED IN THIS ANALYSIS (FLUID VERSUS STRUCTURE). THIS OPTION IS NORMALLY SELECTED FOR FATIGUE ANALYSIS.
(52-55)	ENTER 'STAT' IF MODAL STATIC LOADS ARE TO BE PRODUCED. THIS OPTION IS GENERALLY NOT USED EXCEPT IN SPECIAL APPLICATIONS.
(56-57)	ENTER 'MR' IF MODAL RESPONSES ARE TO BE INCLUDED IN THE OUTPUT.
(58-59)	ENTER 'GP' IF GAP ELEMENTS ARE TO BE INCLUDED IN THIS ANALYSIS.
(60)	ENTER 'N' IF NO RELATIVE VELOCITY EFFECTS ARE TO BE INCLUDED. APPLICABLE FOR TIME HISTORY INTEGRATION ONLY.
(61-64)	ENTER 'BUOY' IF THE EFFECTS OF CHANGES IN BUOYANCY ARE TO BE INCLUDED IN THIS ANALYSIS. THESE EFFECTS ARE NOT NORMALLY IMPORTANT EXCEPT IN CASES OF FLOATING STRUCTURES.
(65)	ENTER 'A' IF THE CUTOFF ELEVATION FOR RANDOM WAVES IS TO BE DETERMINED AUTOMATICALLY USING THE RATIO ENTERED IN COLUMNS 70-75.
(66-67)	ENTER 'AO' TO EXCLUDE THE CONSTANT TERM IN THE FOURIER ANALYSIS. THIS IS USUALLY ONLY REQUIRED FOR FLOATING STRUCTURES.
(68-69)	LEAVE BLANK FOR DETERMINISTIC WAVES OR ENTER 'RW' FOR RANDOM WAVE ANALYSIS, 'TH' FOR TIME HISTORY INTEGRATION OR 'TG' FOR TIME HISTORY INTEGRATION WITH WAVE KINEMATICS CALCULATED AT GRID POINTS INSTEAD OF AT EACH END OF EACH SEGMENT, 'WM' FOR WINDMILL OR 'WL' FOR WIND LOAD OPTION.
NOTE:	'WAVTIM' LINE REQUIRED WHEN USING 'RW', 'TH', 'TG', 'WM' OR 'WL' OPTION. 'WM' OPTION SHOULD ONLY BE USED FOR GENERATING GENERALIZED FORCES FOR SACS-FAST ANALYSIS, WHILE 'WL' OPTION SHOULD ONLY BE USED FOR INCLUDING LOADS FROM SACS-FAST RUNS IN WAVE RESPONSE ANALYSIS.
(70-75)	ENTER THE ELEVATION CUTOFF BELOW WHICH NO WAVE KINEMATICS WILL BE CALCULATED. IF LEFT BLANK OR ZERO WAVE KINEMATICS WILL BE CALCULATED FROM THE MUDLINE TO THE SURFACE ELEVATION. IF THE AUTOMATIC CUTOFF LIMIT OPTION IS USED ('A' IN COLUMN 65), ENTER THE VELOCITY RATIO USED TO DETERMINE THE CUTOFF ELEVATION. THIS IS THE RATIO OF HORIZONTAL VELOCITY OVER THE HORIZONTAL VELOCITY AT THE WATER SURFACE. DEFAULT IS 0.05 WHICH YIELDS A DRAG RATIO OF 0.0025.
(76-80)	ENTER INERTIA LOAD FACTOR FOR EQUIVALENT STATIC LOADS. ONLY EFFECTS OUTPUT INERTIA LOADS AND NORMALLY APPLIES TO LRFD CODES.
(81)	ENTER 'C' TO USE CATEGORY SORT FOR LOAD LINES.

[illegible]

OCHI-HUBBLE WAVE SPECTRAL DENSITY

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED IF AND ONLY IF A SPECTRAL WAVE RESPONSE ANALYSIS IS BEING DONE AND THE WAVE SPECTRUM IS OF THE OCHI-HUBBLE TYPE. THIS LINE FOLLOWS IMMEDIATELY AFTER THE 'WAVTIM' LINE.
(1- 5)	ENTER 'WSPEC'.
(6- 7)	IF NUMEROUS SURFACE PROFILES USING DIFFERENT SEEDS IS TO BE GENERATED, ENTER THE TOTAL NUMBER OF SEEDS. THE SEEDS WILL BE CALCULATED BASED OF THE FIRST SEED ENTERED.
(8-10)	THIS IS THE SEED FOR A BUILT-IN PSEUDO-RANDOM NUMBER GENERATOR WHICH IS USED TO CALCULATE THE PHASE ANGLES FOR THE VARIOUS AIRY WAVE COMPONENTS. IF THE SAME ANALYSIS IS RUN ON THE SAME COMPUTER THEN THE RESULTS WILL BE THE SAME IF THE SAME SEED IS USED. IF A DIFFERENT SEED IS USED, A DIFFERENT SET OF RANDOM PHASE ANGLES WILL BE COMPUTED. THE PRIMARY USE FOR THIS PARAMETER IS TO CHECK THAT THE DURATION OF THE SPECTRUM IS OF SUFFICIENT LENGTH TO GUARANTEE THAT THE RESULTS HAVE REPEATABLE STATISTICAL PROPERTIES. NORMALLY, THIS FIELD IS LEFT BLANK. IF A SEED IS DESIRED, ENTER AN INTEGER FROM 1 TO 999.
(11-12)	ENTER 'OH' TO DESIGNATE THAT THIS SPECTRUM IS AN OCHI-HUBBLE TYPE.
(13-19)	ENTER THE "SIGNIFICANT WAVE HEIGHT" FOR THE SWELL PORTION OF THE SPECTRUM. THIS NUMBER MUST BE GREATER THAN ZERO.
(20-26)	ENTER THE "DOMINANT PERIOD" FOR THE SWELL PORTION OF THE SPECTRUM. THIS NUMBER MUST BE GREATER THAN ZERO.

COLUMNS	COMMENTARY
(27-33)	THE SPECTRUM IS REPRESENTED BY A SET OF AIRY WAVE COMPONENTS WHICH ARE COMBINED USING RANDOM PHASE ANGLES. THE FINENESS OF THE DIVISION IS CONTROLLED BY THIS PARAMETER. EACH WAVE COMPONENT WILL REPRESENT AT LEAST THIS SPECIFIED PERCENTAGE OF THE AREA UNDER THE SPECTRUM.
(34-40)	ENTER THE "SIGNIFICANT WAVE HEIGHT" FOR THE WIND PORTION OF THE SPECTRUM.
(41-47)	ENTER THE "DOMINANT PERIOD" FOR THE WIND PORTION OF THE SPECTRUM.
(48-54)	ENTER THE TIME DURATION TO BE GENERATED BY THIS SPECTRUM. A TIME HISTORY WAVE WILL BE GENERATED HAVING THE STATISTICAL PROPERTIES AS SPECIFIED BY THE SPECTRUM. THE DEFAULT IS 50 TIMES THE DOMINANT PERIOD.
(55-56)	ENTER 'RA' IF RANDOM AMPLITUDE IS TO BE USED IN ADDITION TO RANDOM PHASE ANGLES IN THE AIRY WAVE COMPONENT GENERATION.
(57-58)	ENTER 'SP' IF ONLY THE SURFACE PROFILE IS TO BE GENERATED. IF THIS OPTION IS SELECTED, THE PROGRAM WILL STOP AFTER SURFACE PROFILE IS GENERATED. THIS OPTION IS USEFUL TO CHECK IF THE TIME DURATION IS SUFFICIENT TO CREATE A SURFACE PROFILE THAT REPRESENTS THE SPECTRUM TO THE DESIRED DEGREE OF ACCURACY.
(59-62)	ENTER THE 1ST SEASTATE THAT WILL BE USED TO DEFINE TRANSFER AND/OR RESPONSE FUNCTION.
(63-66)	ENTER THE LAST SEASTATE THAT WILL BE USED TO DEFINE THE TRANSFER AND/OF RESPONSE FUNCTION.
(67-73)	ENTER THE SWELL LAMDA PARAMETER.
(74-80)	ENTER THE WIND LAMDA PARAMETER.

LINE LABEL	NUMBER OF RANDOM NUMBER SEEDS	FIRST RANDOM NUMBER SEED	WAVE SPECTRUM TYPE	SWELL PARAMETERS		MINIMUM PERCENT FOR EACH WAVE COMPONENT	WIND PARAMETERS		TIME DURATION	RANDOM AMPLITUDE OPTION	SURFACE PROFILE GENERATION ONLY OPTION	BEGIN LOAD CASE	END LOAD CASE	LAMDA PARAMETERS	
				SIGNIF-ICANT WAVE HEIGHT	DOMINANT PERIOD (DP)		SIGNIF-ICANT WAVE HEIGHT	DOMINANT PERIOD (DP)						SWELL GENERATED	WIND GENERATED
WSPEC															
1-- 5	6--> 7	8-->10	11--12	13<--19	20<--26	27<--33	34<--40	41<--47	48<--54	55--56	57--58	59-->62	63-->66	67<--73	74<--80
DEFAULT	1	0				1			50 * DP					2.72	CALCULATED
ENGLISH				FT	SEC	PERCENT	FT	SEC	SEC						
METRIC				M	SEC	PERCENT	M	SEC	SEC						

PIERSON-MOSKOWITZ/JONSWAP WAVE SPECTRAL DENSITY

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED IF AND ONLY IF A SPECTRAL WAVE RESPONSE ANALYSIS IS BEING DONE. IT IS USED TO DESIGNATE THE FORM OF THE WAVE HEIGHT SPECTRAL DENSITY FUNCTION. THIS LINE FOLLOWS IMMEDIATELY AFTER THE 'WAVTIM' LINE.
(1- 5)	ENTER 'WSPEC'.
(6- 7)	IF NUMEROUS SURFACE PROFILES USING DIFFERENT SEEDS ARE TO BE GENERATED, ENTER THE TOTAL NUMBER OF SEEDS. THE SEEDS WILL BE CALCULATED BASED OF THE FIRST SEED ENTERED.
(8-10)	THIS IS THE SEED FOR A BUILT-IN PSEUDO-RANDOM NUMBER GENERATOR WHICH IS USED TO CALCULATE THE PHASE ANGLES FOR THE VARIOUS AIRY WAVE COMPONENTS. IF THE SAME ANALYSIS IS RUN ON THE SAME COMPUTER THEN THE RESULTS WILL BE THE SAME IF THE SAME SEED IS USED. IF A DIFFERENT SEED IS USED, A DIFFERENT SET OF RANDOM PHASE ANGLES WILL BE COMPUTED. THE PRIMARY USE FOR THIS PARAMETER IS TO CHECK THAT THE DURATION OF THE SPECTRUM IS OF SUFFICIENT LENGTH TO GUARANTEE THAT THE RESULTS HAVE REPEATABLE STATISTICAL PROPERTIES. NORMALLY, THIS FIELD IS LEFT BLANK. IF A SEED IS DESIRED, ENTER AN INTEGER FROM 1 TO 999.
(11-12)	ENTER THE TYPE OF SPECTRUM TO BE USED FOR THE WAVE HEIGHT SPECTRAL DENSITY FUNCTION. CHOOSE FROM BETWEEN THE FOLLOWING: . 'PM'...PIERSON-MOSKOWITZ SPECTRUM. THIS IS THE DEFAULT. 'JS'...JONSWAP SPECTRUM. .
(13-19)	ENTER THE "SIGNIFICANT WAVE HEIGHT" FOR THIS SPECTRUM.
(20-26)	ENTER THE "DOMINANT PERIOD" FOR THIS SPECTRUM.

COLUMNS	COMMENTARY
(27-33)	THE SPECTRUM IS REPRESENTED BY A SET OF AIRY WAVE COMPONENTS WHICH ARE COMBINED USING RANDOM PHASE ANGLES. THE FINENESS OF THE DIVISION IS CONTROLLED BY THIS PARAMETER. EACH WAVE COMPONENT WILL REPRESENT AT LEAST THIS SPECIFIED PERCENTAGE OF THE AREA UNDER THE SPECTRUM.
(34-47)	ENTER THE VALUES OF THE PARAMETERS "GAMMA" AND "C" REQUIRED TO FULLY DEFINE THE JONSWAP SPECTRUM IF 'JS' IS IN COLUMNS 11-12.
(48-54)	ENTER THE TIME DURATION TO BE GENERATED BY THIS SPECTRUM. A TIME HISTORY WAVE WILL BE GENERATED HAVING THE STATISTICAL PROPERTIES AS SPECIFIED BY THE SPECTRUM. THE DEFAULT IS 50 TIMES THE DOMINANT PERIOD.
(55-56)	ENTER 'RA' IF RANDOM AMPLITUDE IS TO BE USED IN ADDITION TO RANDOM PHASE ANGLES IN THE AIRY WAVE COMPONENT GENERATION.
(57-58)	ENTER 'SP' IF ONLY THE SURFACE PROFILE IS TO BE GENERATED. IF THIS OPTION IS SELECTED, THE PROGRAM WILL STOP AFTER SURFACE PROFILE IS GENERATED. THIS OPTION IS USEFUL TO CHECK IF THE TIME DURATION IS SUFFICIENT TO CREATE A SURFACE PROFILE THAT REPRESENTS THE SPECTRUM TO THE DESIRED DEGREE OF ACCURACY.
(59-62)	ENTER THE 1ST SEASTATE THAT WILL BE USED TO DEFINE TRANSFER AND/OR RESPONSE FUNCTION.
(63-66)	ENTER THE LAST SEASTATE THAT WILL BE USED TO DEFINE THE TRANSFER AND/OF RESPONSE FUNCTION.

LINE LABEL	NUMBER OF RANDOM NUMBER SEEDS	FIRST RANDOM NUMBER SEED	WAVE SPECTRUM TYPE	WAVE SPECTRUM PARAMETERS		MINIMUM PERCENT FOR EACH WAVE COMPONENT	JONSWAP PARAMETERS		TIME DURATION	RANDOM AMPLITUDE OPTION	SURFACE PROFILE GENERATION ONLY OPTION	BEGIN LOAD CASE	END LOAD CASE	LEAVE THIS FIELD BLANK
				SIGNIFICANT WAVE HEIGHT	DOMINANT PERIOD (DP)		"GAMMA"	"C"						
WSPEC														
1-- 5	6--> 7	8-->10	11--12	13<--19	20<--26	27<--33	34<--40	41<--47	48<--54	55--56	57--58	59-->62	63-->66	67--80
DEFAULT	1	0	'PM'			1	3.3	1.525	50 * DP					
ENGLISH				FT	SEC	PERCENT			SEC					
METRIC				M	SEC	PERCENT			SEC					

USER DEFINED WAVE SPECTRAL DENSITY

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED IF AND ONLY IF A SPECTRAL WAVE RESPONSE ANALYSIS IS BEING DONE. IT IS USED TO DESIGNATE THE FORM OF THE WAVE HEIGHT SPECTRAL DENSITY FUNCTION. THIS LINE FOLLOWS IMMEDIATELY AFTER THE 'WAVTIM' LINE.
(1- 5)	ENTER 'WSPEC'.
(6- 7)	IF NUMEROUS SURFACE PROFILES USING DIFFERENT SEEDS ARE TO BE GENERATED, ENTER THE TOTAL NUMBER OF SEEDS. THE SEEDS WILL BE CALCULATED BASED ON THE FIRST SEED ENTERED.
(8-10)	THIS IS THE SEED FOR A BUILT-IN PSEUDO-RANDOM NUMBER GENERATOR WHICH IS USED TO CALCULATE THE PHASE ANGLES FOR THE VARIOUS AIRY WAVE COMPONENTS. IF THE SAME ANALYSIS IS RUN ON THE SAME COMPUTER THEN THE RESULTS WILL BE THE SAME IF THE SAME SEED IS USED. IF A DIFFERENT SEED IS USED, A DIFFERENT SET OF RANDOM PHASE ANGLES WILL BE COMPUTED. THE PRIMARY USE FOR THIS PARAMETER IS TO CHECK THAT THE DURATION OF THE SPECTRUM IS OF SUFFICIENT LENGTH TO GUARANTEE THAT THE RESULTS HAVE REPEATABLE STATISTICAL PROPERTIES. NORMALLY, THIS FIELD IS LEFT BLANK. IF A SEED IS DESIRED, ENTER AN INTEGER FROM 1 TO 999.
(11-12)	ENTER 'US' TO DESIGNATE THAT THIS SPECTRUM IS A USER-DEFINED WAVE SPECTRUM. THE 'SPEC' DATA LINES DEFINING THE SPECTRUM IMMEDIATELY FOLLOW THIS LINE.

COLUMNS	COMMENTARY
(27-33)	THE SPECTRUM IS REPRESENTED BY A SET OF AIRY WAVE COMPONENTS WHICH ARE COMBINED USING RANDOM PHASE ANGLES. THE FINENESS OF THE DIVISION IS CONTROLLED BY THIS PARAMETER. EACH WAVE COMPONENT WILL REPRESENT AT LEAST THIS SPECIFIED PERCENTAGE OF THE AREA UNDER THE SPECTRUM.
(48-54)	ENTER THE TIME DURATION TO BE GENERATED BY THIS SPECTRUM. A TIME HISTORY WAVE WILL BE GENERATED HAVING THE STATISTICAL PROPERTIES AS SPECIFIED BY THE SPECTRUM. THE DEFAULT IS 50 TIMES THE DOMINANT PERIOD.
(55-56)	ENTER 'RA' IF RANDOM AMPLITUDE IS TO BE USED IN ADDITION TO RANDOM PHASE ANGLES IN THE AIRY WAVE COMPONENT GENERATION.
(57-58)	ENTER 'SP' IF ONLY THE SURFACE PROFILE IS TO BE GENERATED. IF THIS OPTION IS SELECTED, THE PROGRAM WILL STOP AFTER THE SURFACE PROFILE IS GENERATED. THIS OPTION IS USEFUL TO CHECK IF THE TIME DURATION IS SUFFICIENT TO CREATE A SURFACE PROFILE THAT REPRESENTS THE SPECTRUM TO THE DESIRED DEGREE OF ACCURACY.
(59-62)	ENTER THE FIRST SEASTATE THAT WILL BE USED TO DEFINE THE TRANSFER AND/OR RESPONSE FUNCTIONS.
(63-66)	ENTER THE LAST SEASTATE THAT WILL BE USED TO DEFINE THE TRANSFER AND/OR RESPONSE FUNCTION.

LINE LABEL	NUMBER OF RANDOM NUMBER SEEDS	FIRST RANDOM NUMBER SEED	WAVE SPECTRUM TYPE	MINIMUM PERCENT FOR EACH WAVE COMPONENT	TIME DURATION	RANDOM AMPLITUDE OPTION	SURFACE PROFILE GENERATION ONLY OPTION	BEGIN LOAD CASE	END LOAD CASE	LEAVE THIS FIELD BLANK
WSPEC										
1-- 5	6--> 7	8-->10	11--12	27<--33	48<--54	55--56	57--58	59-->62	63-->66	67--80
DEFAULT	1	0		1	50 * DP					
ENGLISH				PERCENT	SEC					
METRIC				PERCENT	SEC					

WAVE INSERT DESCRIPTION

COLUMNS	COMMENTARY
GENERAL	THIS LINE IS USED TO SPECIFY A EXTREME WAVE TO BE INSERTED INTO A RANDOM WAVE TIME HISTORY.
(12)	ENTER 'P' IF THE MODIFIED SURFACE PROFILE IS TO BE PRINTED.
(13-20)	ENTER THE MAXIMUM CREST ELEVATION ABOVE MEAN WATER LINE.
(24-31)	ENTER THE PEAK PERIOD FOR THE WAVE INSERT.
(32-39)	ENTER THE TIME THAT THESE WAVE IS TO BE INSERTED INTO THE WAVE TIME HISTORY.
(40)	ENTER 'S' IF STREAM FUNCTION THEORY IS TO BE USED FOR THE CONSTRAINED WAVE.
(41-42)	IF 'S' IS IN COLUMN 40, ENTER THE DESIRED ORDER OF THE GENERATED STREAM FUNCTION WAVE. ODD VALUES SHOULD BE USED WITH A MAXIMUM OF 21. IF LEFT BLANK THE ORDER WILL BE SELECTED BASED ON ATKINS.
(43-44)	IF 'S' IS IN COLUMN 40, ENTER THE MAX. ITERATIONS OF THE GENERATED STREAM FUNCTION WAVE.
(45-50)	ENTER THE NORMAL DRAG COEFFICIENT OVERRIDE FOR CONSTRAINED WAVE (OPTIONAL) .
(51-56)	ENTER THE TANGENTIAL DRAG COEFFICIENT OVERRIDE FOR CONSTRAINED WAVE (OPTIONAL) .
(64)	IF 'S' IS IN COLUMN 40, ENTER E IF THE INSERT WINDOW NEEDS TO BE EXTENDED
(71)	IF 'S' IS IN COLUMN 40, ENTER 'A' TO REDUCE THE WAVEHEIGHT TO BELOW THE BREAKING LIMIT IF IT EXCEEDS IT.

LINE LABEL	OUTPUT MODIFIED SURFACE PROFILE OPTION	MAX. CREST ELEV.	PEAK PERIOD	INSERT TIME	USE STREAM FUNCTION THEORY	STREAM FUNCTION WAVE ORDER	STREAM FUNCTION MAX. ITERATIONS	DRAG COEFFICIENT - NORMAL	DRAG COEFFICIENT - TANGENTIAL	EXTEND INSERTION WINDOW	REDUCE BREAKING WAVE'S HEIGHT	LEAVE BLANK
WVINS												
1-- 5	12	13--20	24--31	32--39	40	41-42	43-44	45-50	50-56	64	71	57-----80
DEFAULT							24					
ENGLISH		FT	SECS	SECS								
METRIC		M	SECS	SECS								